Model Analysis

Table of contents

1 Data analysis /Dataframe import and filters

1.1 DataFrame import and filters

The data has been imported with the following(s) function(s):

```
def build_df():
    if flag_IR:
        df, features_engineered = import_data_IR()
        if flag_IR_maskP:
            df = mask_period_IR(df)
        if flag_IR_balR:
            df = mask_balanceHC_random_IR(df, n_samples=None)
    if flag_ER:
        df, features_engineered = import_data_ER()

    cols_to_check = target_col + features_X
    mask = np.isfinite(df[cols_to_check]).all(axis=1)
    df = df[mask]
```

```
    cols = list(dict.fromkeys(target_col + features_X  +
                              feats_to_balance + feats_grouping +
                              feats_reference+ feats_postproc))
    df = df[cols]
    #df['campaign'] = "ALL"

    return df
```

```
def import_data_IR():
    if target_col[0] == "i_NH3":
        file_path = "csv_decomposed/innova/df_signal_decomposition_NH3_junesept.csv"
        #file_path = "csv/innova/switched/df_signal_decomposition_10_50_200.csv"
    elif target_col[0] == "i_CO2":
        file_path = "csv_decomposed/innova/df_signal_decomposition_CO2_current.csv"
    else:
        raise ValueError("Target is not supported")

    df = pd.read_csv(file_path, sep=";")
    df["datetime"] = pd.to_datetime(df["datetime"])
    df.sort_values("datetime").reset_index(drop=True)

    if target_col[0] == "i_NH3":
        df["campaign"] = df["datetime"].apply(assign_campaign)
        df = df[df["campaign"] != "other"].copy()
        df["innova_point"] = df.apply(assign_innova_point, axis=1)
        df = df[df["innova_point"] != "no"].copy()
        with open("csv_decomposed/innova/features_decomposition_NH3_list_junesept.json", "r")
        #with open("csv/innova/switched/features_decomposition_list_10_50_200.json", "r")as
            dict_features = json.load(f)
    elif target_col[0] == "i_CO2":
        df["campaign"] = df["datetime"].apply(assign_campaign_iCO2)
        df = df[df["campaign"] != "other"].copy()
        df["innova_point"] = df.apply(assign_innova_point_iCO2, axis=1)
        df = df[df["innova_point"] != "no"].copy()
        with open("csv/innova/features_decomposition_CO2_list.json", "r") as f:
            dict_features = json.load(f)
    else:
        raise ValueError("The target col do not match any existing function")


    df["hot"] = df.apply(assign_hot_season,axis=1)
    df["umidity_range"] = df.apply(assign_humidity, axis = 1)
```

```
    #Removing of rabbit 4
    df["column"] = df.apply(assign_colonna, axis = 1)
    df['height'] = df.apply(assign_altezza, axis = 1)



    return df,dict_features
```

```
def assign_innova_point(row):
    camp = row["campaign"]
    rid = row["id_rabbit"]

    # if camp == "2025_Jun":
    #     mapping = {2: "sl11", 4: "sl5", 6: "sl4", 1: "sl3", 5: "no"}
    # elif camp == "2025_Sep":
    #     mapping = {2: "sl11", 4: "sl5", 3: "sl4", 6: "sl3", 5: "no"}
    if camp == "2025_Apr":
        #mapping = {1: "no", 2: "no", 4: "no", 6: "sl5"}
        #mapping = {1: "no", 2: "sl5", 4: "no", 6: "sl5"}
        mapping = {1: "no", 2: "no", 4: "no", 6: "no"}
    elif camp == "2025_Jun":
        mapping = {2: "sl11", 4: "sl5", 1: "sl3", 6: "sl4", 5: "no"}
        #mapping = {2: "sl11", 4: "no", 1: "sl4", 6: "sl3", 5: "no"}
    elif camp == "2025_Sep":
        mapping = {2: "sl11", 4: "sl5", 6: "sl3", 3: "sl4", 5: "no"}
        #mapping = {2: "sl11", 4: "no", 6: "sl4", 3: "sl3", 5: "no"}
    elif camp == "2026_Gen":
        #mapping = {1: "no", 4: "no", 5: "sl11", 6: "no"}
        #mapping = {1: "sl11", 4: "no", 5: "sl11", 6: "sl11"}
        mapping = {1: "no", 4: "no", 5: "no", 6: "no"}
    else:
        mapping = {}

    return mapping.get(rid, "no")
```

Function print :

1.2 Metadata

Nombre de lignes df	7730

Nombre de lignes df clean	7730
Nombre de features	53
% de NaN	0.0
Taille dataset entraînement	5411
Taille dataset test	2319
Métrique utilisée	r2

Column target : i_NH3

Features_X : r_NH3, r_temperature, r_umidity, r_CO2, r_THI, r_NH3_st_8h_mean, r_NH3_st_8h_std, r_NH3_st_8h_rms, r_NH3_st_8h_median, r_NH3_st_8h_min, r_NH3_st_8h_max, r_NH3_st_8h_diff_last, r_NH3_st_8h_slope, r_NH3_st_12h_mean, r_NH3_st_12h_std, r_NH3_st_12h_rms, r_NH3_st_12h_median, r_NH3_st_12h_min, r_NH3_st_12h_max, r_NH3_st_12h_diff_last, r_NH3_st_12h_slope, r_temperature_st_8h_mean, r_temperature_st_8h_std, r_temperature_st_8h_rms, r_temperature_st_8h_median, r_temperature_st_8h_min, r_temperature_st_8h_max, r_temperature_st_8h_diff_last, r_temperature_st_8h_slope, r_temperature_st_12h_mean, r_temperature_st_12h_std, r_temperature_st_12h_rms, r_temperature_st_12h_median, r_temperature_st_12h_min, r_temperature_st_12h_max, r_temperature_st_12h_diff_last, r_temperature_st_12h_slope, r_umidity_st_8h_mean, r_umidity_st_8h_std, r_umidity_st_8h_rms, r_umidity_st_8h_median, r_umidity_st_8h_min, r_umidity_st_8h_max, r_umidity_st_8h_diff_last, r_umidity_st_8h_slope, r_umidity_st_12h_mean, r_umidity_st_12h_std, r_umidity_st_12h_rms, r_umidity_st_12h_median, r_umidity_st_12h_min, r_umidity_st_12h_max, r_umidity_st_12h_diff_last, r_umidity_st_12h_slope

feats_to_balance :

feats_grouping :

feats_postproc : datetime, id_channel, id_rabbit, campaign

feats_ref :

id_cols : campaign

2 Data correlation and clustering

2.1 Clustermap and cluster list

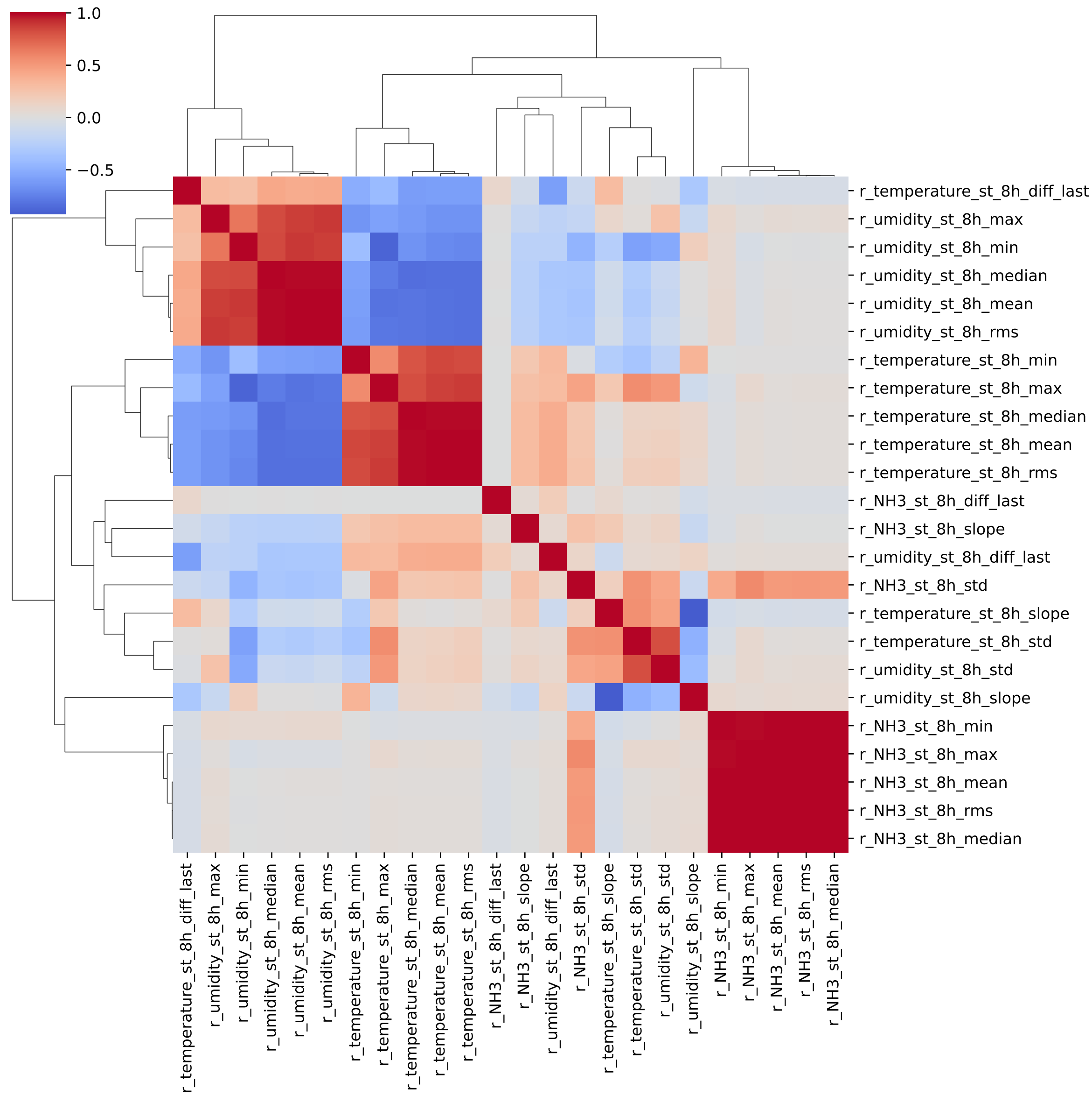
The following section has been done with a threshold choosen of : 0.8

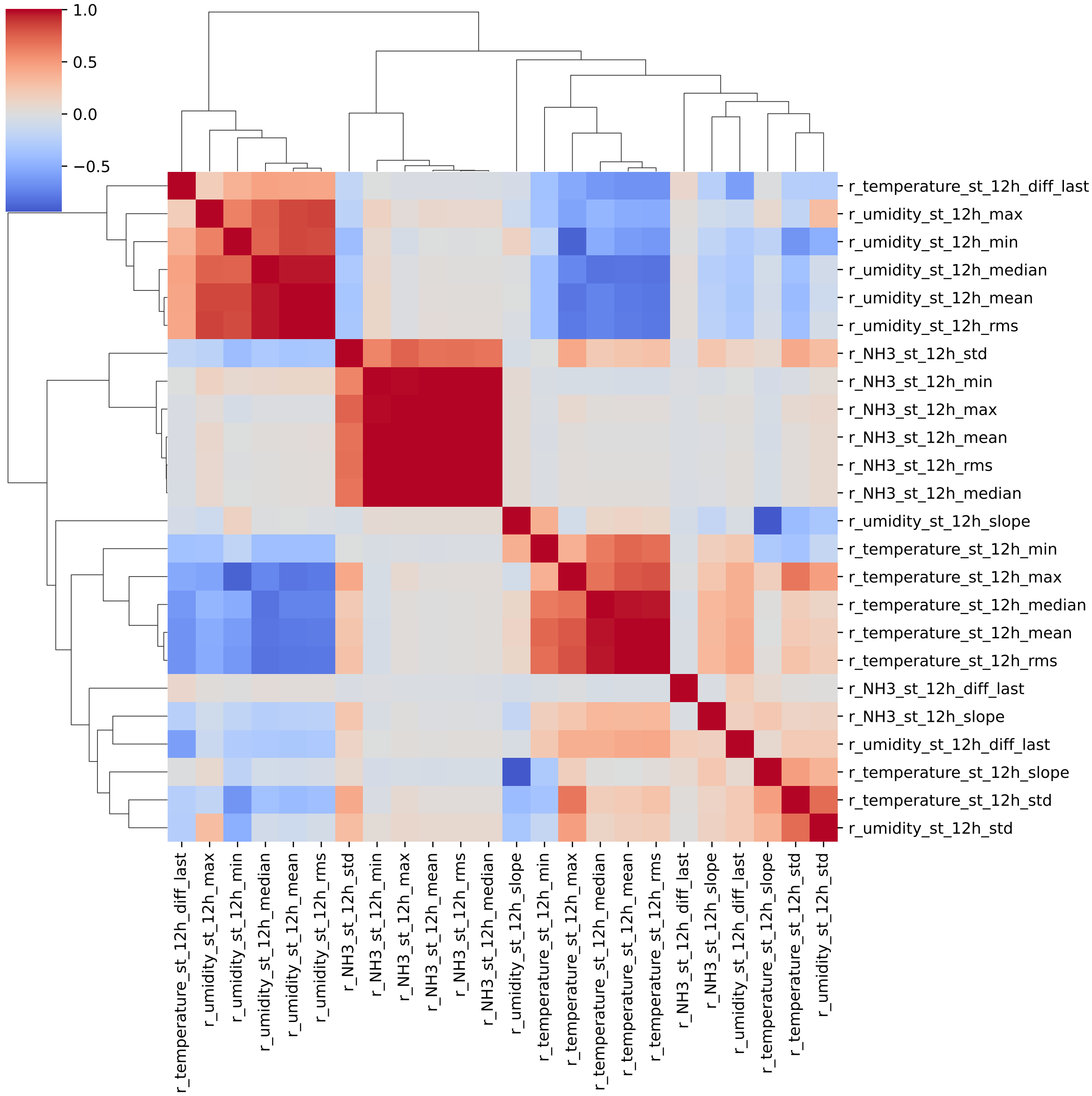
cluster 1: r, *, N, H, 3 ; **cluster 2:** r,* , t, e, m, p, e, r, a, t, u, r, e ; **cluster 3:** r, *, u, m, i, d, i, t, y ; **cluster 4:** r,* , C, O, 2 ; **cluster 5:** r, _, T, H, I ; **cluster 6:** r_NH3_st_8h_median, r_NH3_st_8h_mean, r_NH3_st_8h_max, r_NH3_st_8h_rms, r_NH3_st_8h_min ; **cluster 7:** r_NH3_st_8h_std ; **cluster 8:** r_NH3_st_8h_diff_last ; **cluster 9:** r_NH3_st_8h_slope ; **cluster 10:** r_umidity_st_8h_median, r_temperature_st_8h_min, r_umidity_st_8h_max, r_temperature_st_8h_median, r_temperature_st_8h_rms, r_umidity_st_8h_rms, r_umidity_st_8h_mean, r_temperature_st_8h_mean, r_umidity_st_8h_min, r_temperature_st_8h_max ; **cluster 11:** r_umidity_st_8h_std, r_temperature_st_8h_std ; **cluster 12:** r_temperature_st_8h_diff_last ; **cluster 13:** r_temperature_st_8h_slope, r_umidity_st_8h_slope ; **cluster 14:** r_umidity_st_8h_diff_last ; **cluster 15:** r_NH3_st_12h_median, r_NH3_st_12h_rms, r_NH3_st_12h_min, r_NH3_st_12h_mean, r_NH3_st_12h_max ; **cluster 16:** r_NH3_st_12h_std ; **cluster 17:** r_NH3_st_12h_diff_last ; **cluster 18:** r_NH3_st_12h_slope ; **cluster 19:** r_umidity_st_12h_max, r_temperature_st_12h_mean, r_temperature_st_12h_max, r_temperature_st_12h_median, r_temperature_st_12h_rms, r_umidity_st_12h_min, r_umidity_st_12h_rms, r_umidity_st_12h_median, r_umidity_st_12h_mean ; **cluster 20:** r_temperature_st_12h_std ; **cluster 21:** r_temperature_st_12h_min ; **cluster 22:** r_temperature_st_12h_diff_last ; **cluster 23:** r_umidity_st_12h_slope, r_temperature_st_12h_slope ; **cluster 24:** r_umidity_st_12h_std ; **cluster 25:** r_umidity_st_12h_diff_last ;

2.2 Features selection

Only the first items of the clusters is choosen for the rest of the study

Choosen features_X: r_NH3, r_temperature, r_umidity, r_CO2, r_THI, r_NH3_st_8h_median, r_NH3_st_8h_std, r_NH3_st_8h_diff_last, r_NH3_st_8h_slope, r_umidity_st_8h_median, r_umidity_st_8h_std, r_temperature_st_8h_diff_last, r_temperature_st_8h_slope, r_umidity_st_8h_diff_last, r_NH3_st_12h_median, r_NH3_st_12h_std, r_NH3_st_12h_diff_last, r_NH3_st_12h_slope, r_umidity_st_12h_max, r_temperature_st_12h_std, r_temperature_st_12h_min, r_temperature_st_12h_diff_last, r_umidity_st_12h_slope, r_umidity_st_12h_std, r_umidity_st_12h_diff_last


1.0
0.5
0.0
−0.5
r_temperature_st_8h_diff_last
r_umidity_st_8h_max
r_umidity_st_8h_min
r_umidity_st_8h_median
r_umidity_st_8h_mean
r_umidity_st_8h_rms
r_temperature_st_8h_min
r_temperature_st_8h_max
r_temperature_st_8h_median
r_temperature_st_8h_mean
r_temperature_st_8h_rms
r_NH3_st_8h_diff_last
r_NH3_st_8h_slope
r_umidity_st_8h_diff_last
r_NH3_st_8h_std
r_temperature_st_8h_slope
r_temperature_st_8h_std
r_umidity_st_8h_std
r_umidity_st_8h_slope
r_NH3_st_8h_min
r_NH3_st_8h_max
r_NH3_st_8h_mean
r_NH3_st_8h_rms
r_NH3_st_8h_median



3 Model(s) analysis

3.1 Metadata

Features_X: r_NH3, r_temperature, r_umidity, r_CO2, r_THI, r_NH3_st_8h_median, r_NH3_st_8h_std, r_NH3_st_8h_diff_last, r_NH3_st_8h_slope, r_umidity_st_8h_median, r_umidity_st_8h_std, r_temperature_st_8h_diff_last, r_temperature_st_8h_slope, r__umidity__st__8h__diff__last, r__NH3__st__12h__median, r__NH3__st__12h__std, r__NH3__st__12h__diff__last, r__NH3__st__12h__slope, r__umidity__st__12h__max, r__temperature__st__12h__std, r__temperature__st__12h__min, r__temperature__st__12h__diff__last, r__umidity__st__12h__slope, r__umidity__st__12h__std, r__umidity__st__12h__diff__last

Len(Features__X): 25

Config Gridsearch:{'show': True, 'search__type': 'optuna', 'model__name': None, 'model': None, 'param__grid': None, 'best__param': None, 'cv__func': ShuffleSplit(n__splits=5, random__state=42, test__size=0.3, train__size=None), 'save__models': True, 'sumup-table': True}

3.2 Explanation legend tables

Columns

- rank: ranking regarding cross validation test
- train__cv: mean score train of the CV
- test__cv: mean score test of the CV
- test: score on the testing dataset
- diff: test-test__cv
- r__test: test/test__cv
- r__train: train__cv/test__cv

Colors:

[Orange] r__test < 0.95 (under performance of the test)

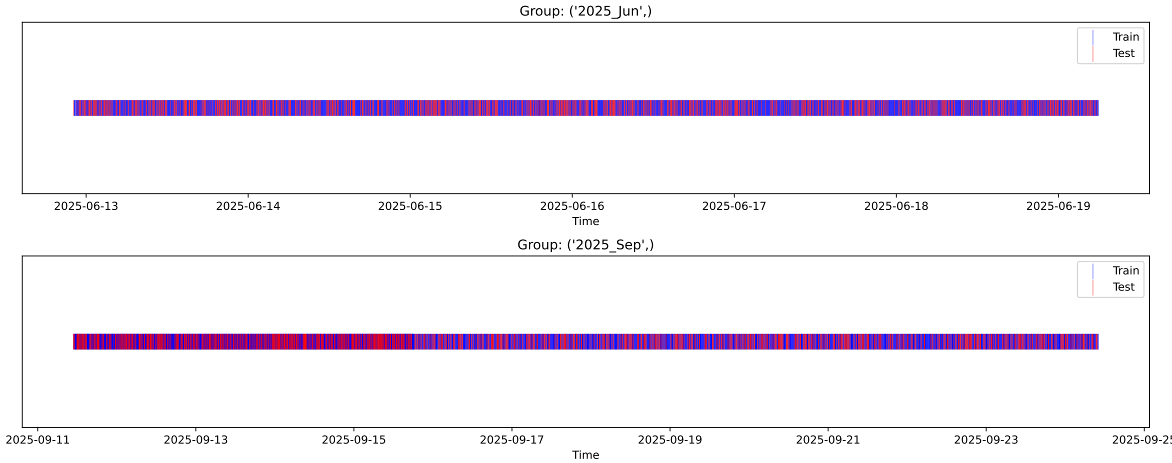
[Green] r__test > 1.05 (over performance of the test)

[Red] r__train not in [0.95, 1.05] (gap between train/test)

3.3 Repartition of train and test

3.3.1 External Split: X__train vs X__test

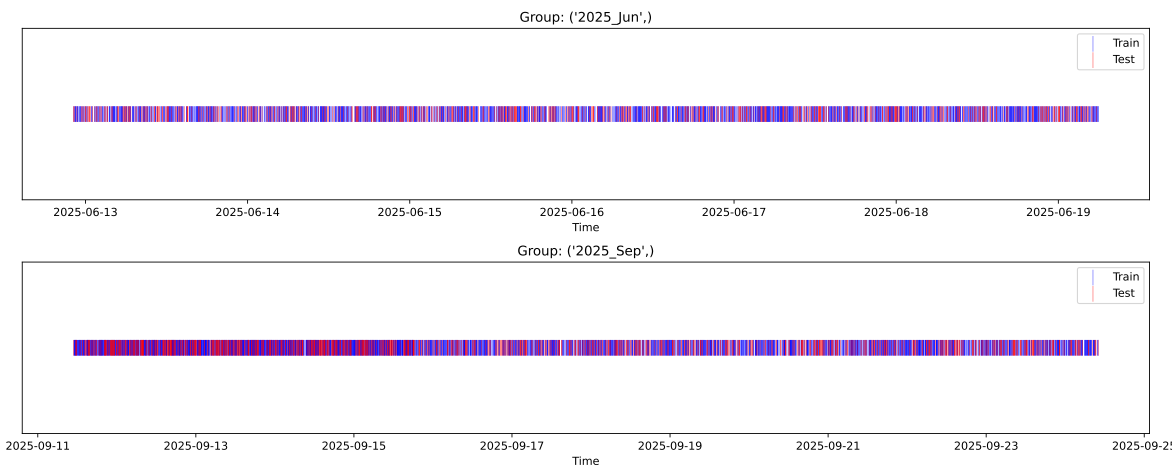
External Split: Train: 5411 samples Test: 2319 samples



3.3.2 Internal CV Folds

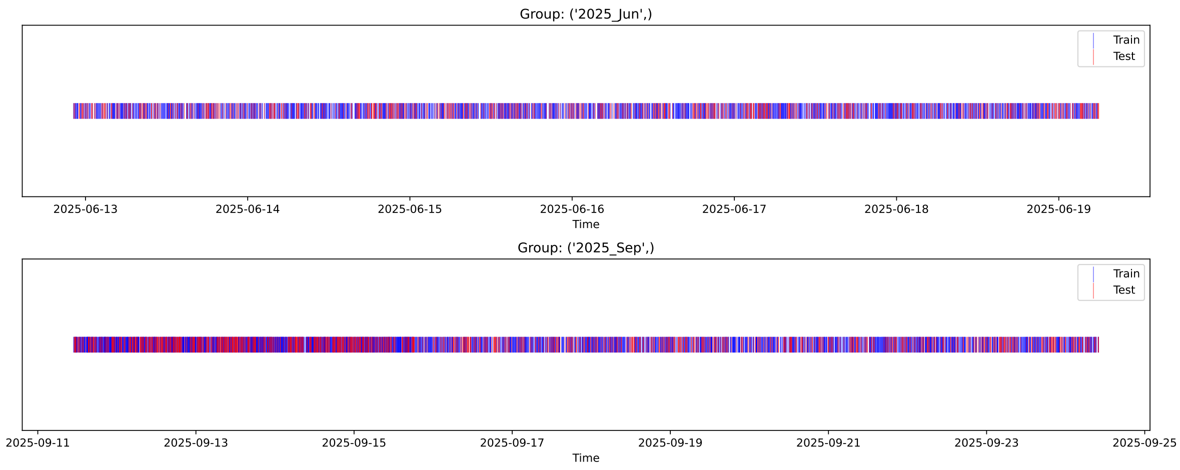
Fold 1/5

Train: 3787 samples Test: 1624 samples Excluded: 0 samples Window exclusion: 0h



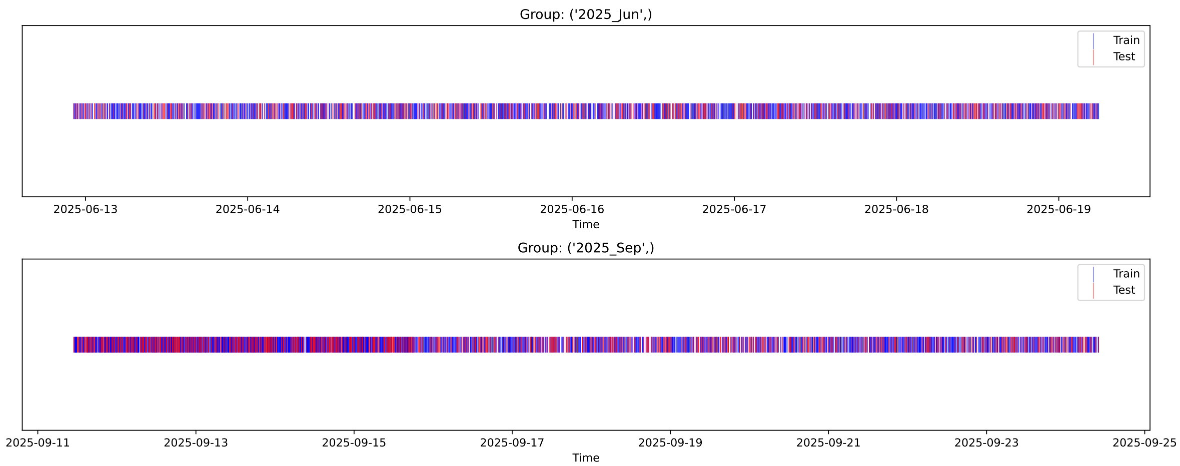
Fold 2/5

Train: 3787 samples Test: 1624 samples Excluded: 0 samples Window exclusion: 0h



Fold 3/5

Train: 3787 samples Test: 1624 samples Excluded: 0 samples Window exclusion: 0h

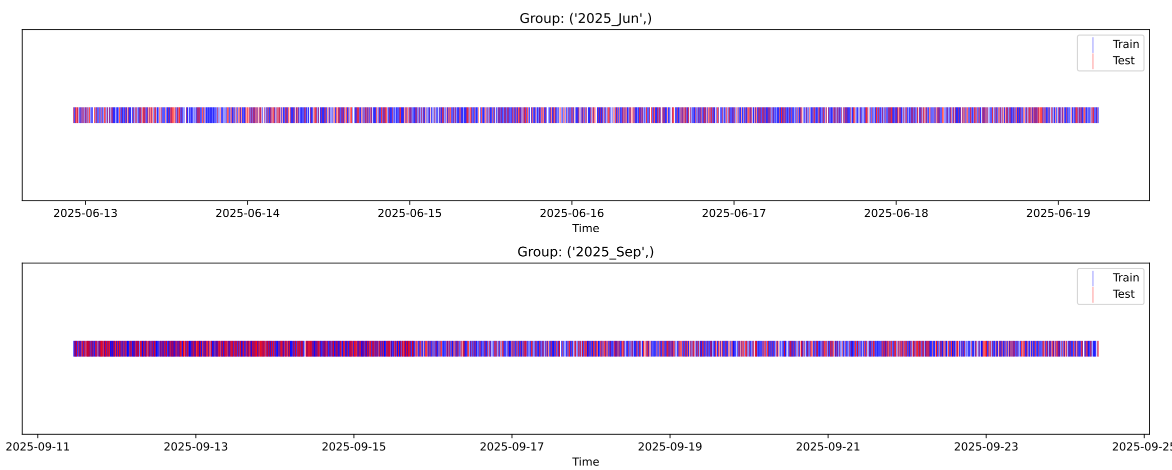


Fold 4/5

Train: 3787 samples Test: 1624 samples Excluded: 0 samples Window exclusion: 0h

Fold 5/5

Train: 3787 samples Test: 1624 samples Excluded: 0 samples Window exclusion: 0h



3.4 Model : Linear

3.4.1 Gridsearch

Distribution y_train vs y_test: y_train: mean=2.342, std=0.800, min=0.959, max=13.845 y_test: mean=2.364, std=0.795, min=1.068, max=7.277

Fold 0 — y_test: mean=2.34, std=0.79, y_train: mean=2.34, std=0.80

Fold 1 — y_test: mean=2.35, std=0.83, y_train: mean=2.34, std=0.79

Fold 2 — y_test: mean=2.35, std=0.79, y_train: mean=2.34, std=0.81

Fold 3 — y_test: mean=2.33, std=0.77, y_train: mean=2.35, std=0.81

Fold 4 — y_test: mean=2.34, std=0.85, y_train: mean=2.34, std=0.78

train scores CV: [0.55635412 0.57999784 0.55407456 0.55265283 0.5674933]

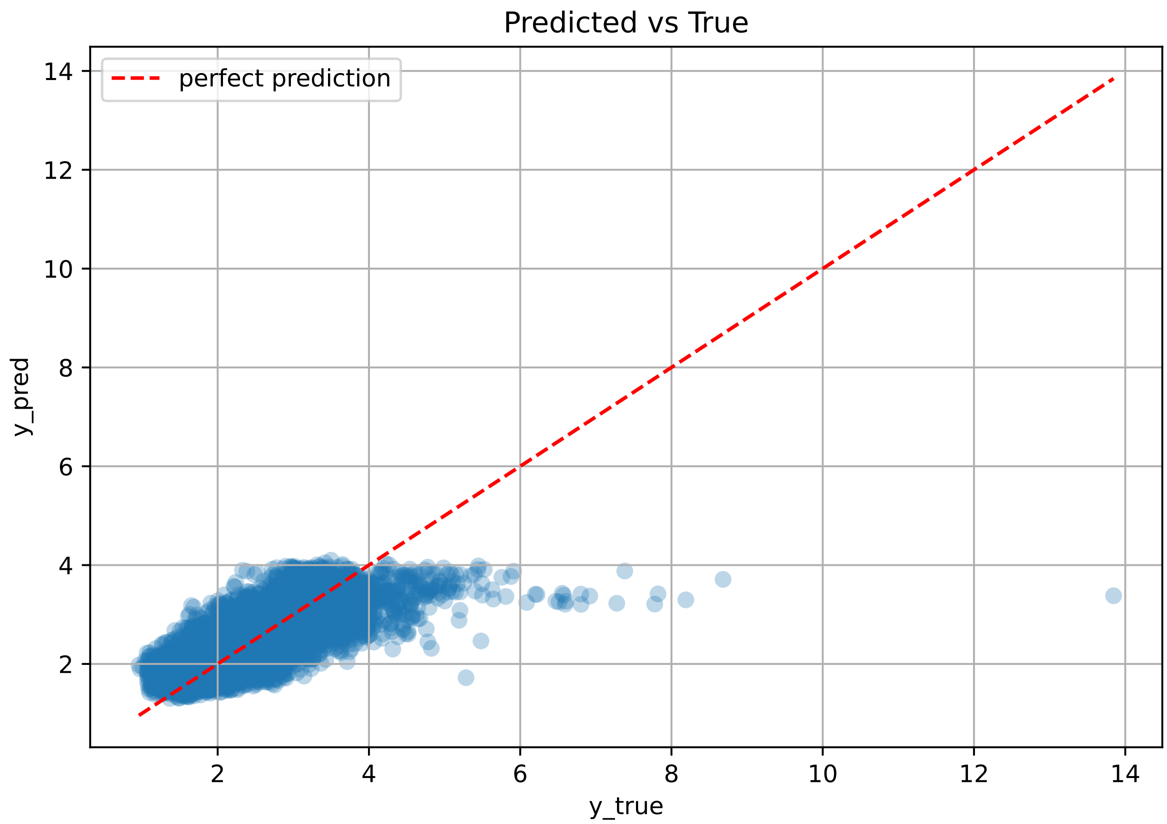
test scores CV: [0.55526403 0.50452289 0.56110386 0.56467286 0.53255807]

rank	mean_train_cv	mean_test_cv	std_test_cv	test_score_cv	test	diff	r_test	r_train
1.0	0.5621	0.5436	0.0225	0.5868	0.5877	0.0441	1.0811	1.034

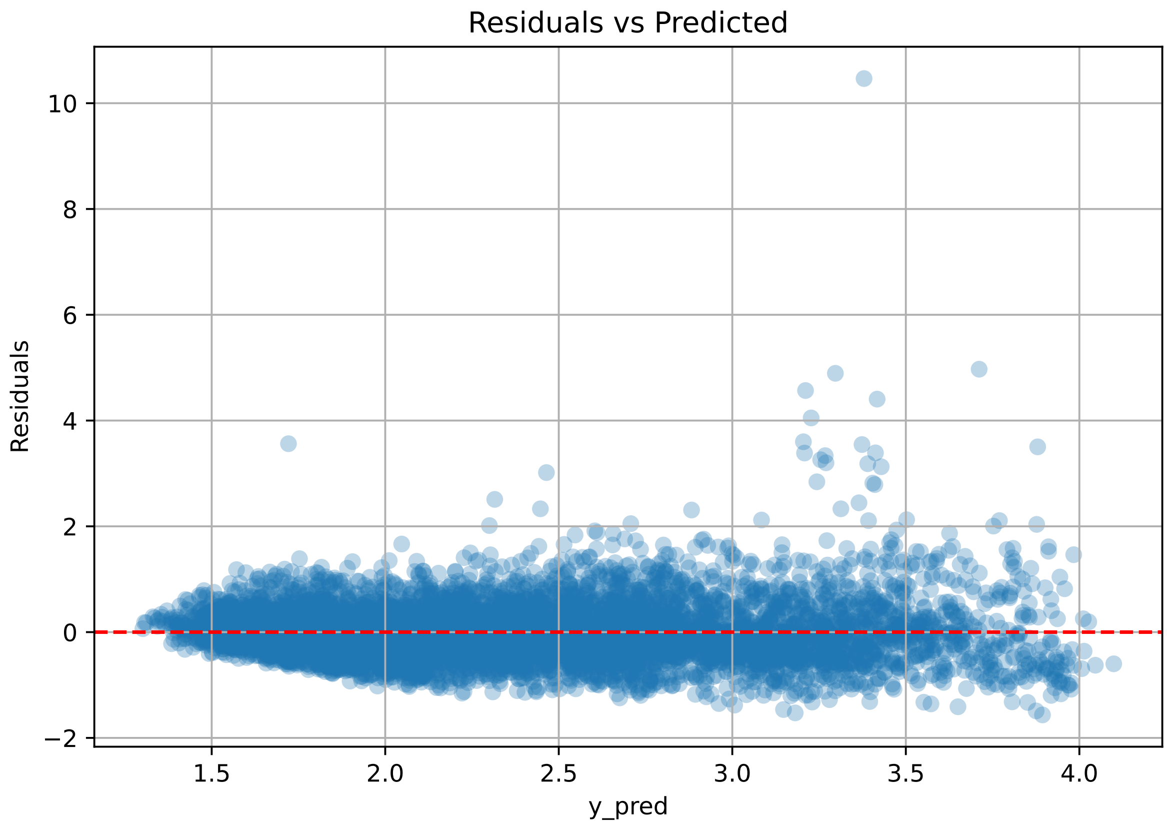
Coefficients:

coef_r_CO2 = -0.0492, coef_r_NH3 = -0.0097, coef_r_NH3_st_12h_diff_last = 0.0062, coef_r_NH3_st_12h_median = 0.2204, coef_r_NH3_st_12h_slope = 0.0458, coef_r_NH3_st_12h_std = 0.0256, coef_r_NH3_st_8h_diff_last = 0.0062, coef_r_NH3_st_8h_median = -0.2819, coef_r_NH3_st_8h_slope = 0.0255, coef_r_NH3_st_8h_std = 0.0179, coef_r_THI = 0.6126, coef_r_temperature = -0.4854, coef_r_temperature_st_12h_diff_last = 0.0242, coef_r_temperature_st_12h_min = -0.1050, coef_r_temperature_st_12h_std = -0.0779, coef_r_temperature_st_8h_diff_last = 0.0242, coef_r_temperature_st_8h_slope = 0.0433, coef_r_umidity = -0.9210, coef_r_umidity_st_12h_diff_last = -0.0067, coef_r_umidity_st_12h_max = 0.2337, coef_r_umidity_st_12h_slope = 0.4904, coef_r_umidity_st_12h_std = -0.2152, coef_r_umidity_st_8h_diff_last = -0.0067, coef_r_umidity_st_8h_median = 0.0318, coef_r_umidity_st_8h_std = -0.0016, intercept = 2.3417,

3.4.2 Scatter


Predicted vs True
perfect prediction
y_pred
y_true
14
12
10
8
6
4
2
2
4
6
8
10
12
14

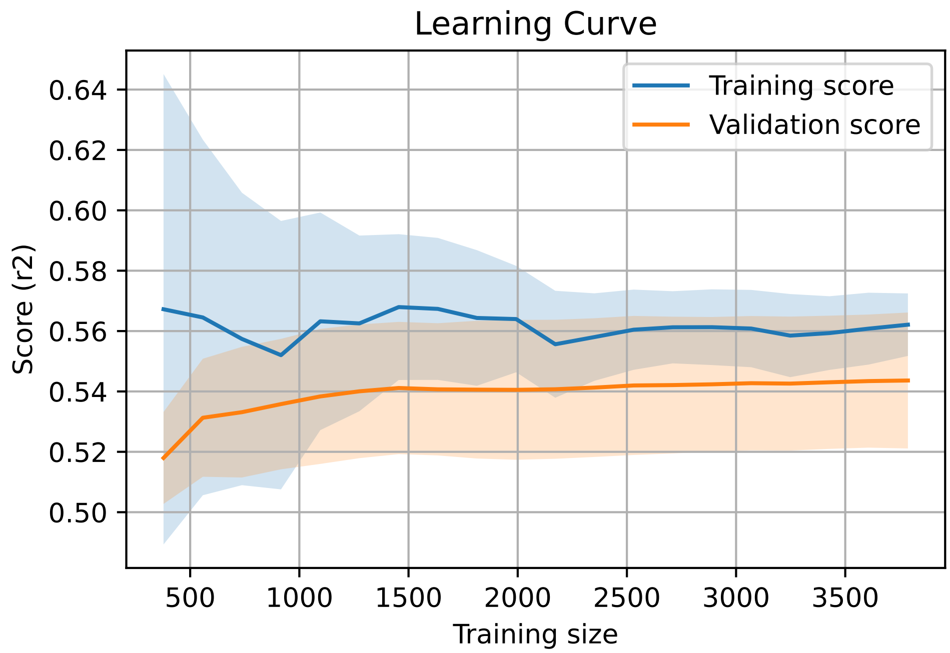
3.4.3 Scatter residuals



3.4.4 Learning curve

$\mathbf{Len(group)} : 7730$, $\mathbf{Len(training)} : 5411$,

$\mathbf{Len(training\_real)} : 3788$, $\mathbf{Len(test\_of\_training)} : 1623$, $\mathbf{Len(test)} : 2319$

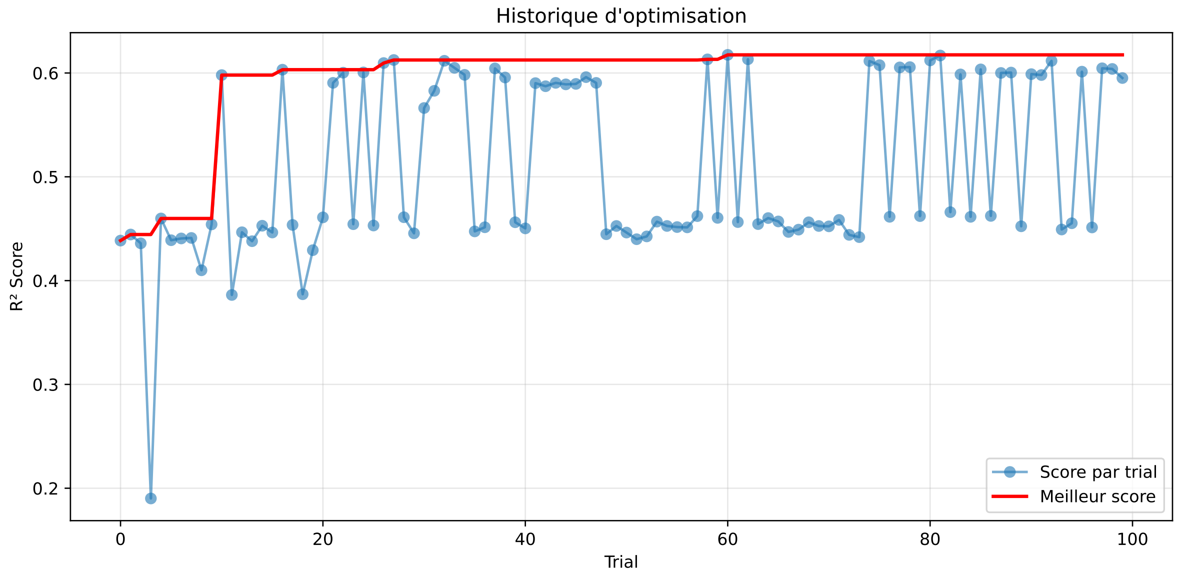


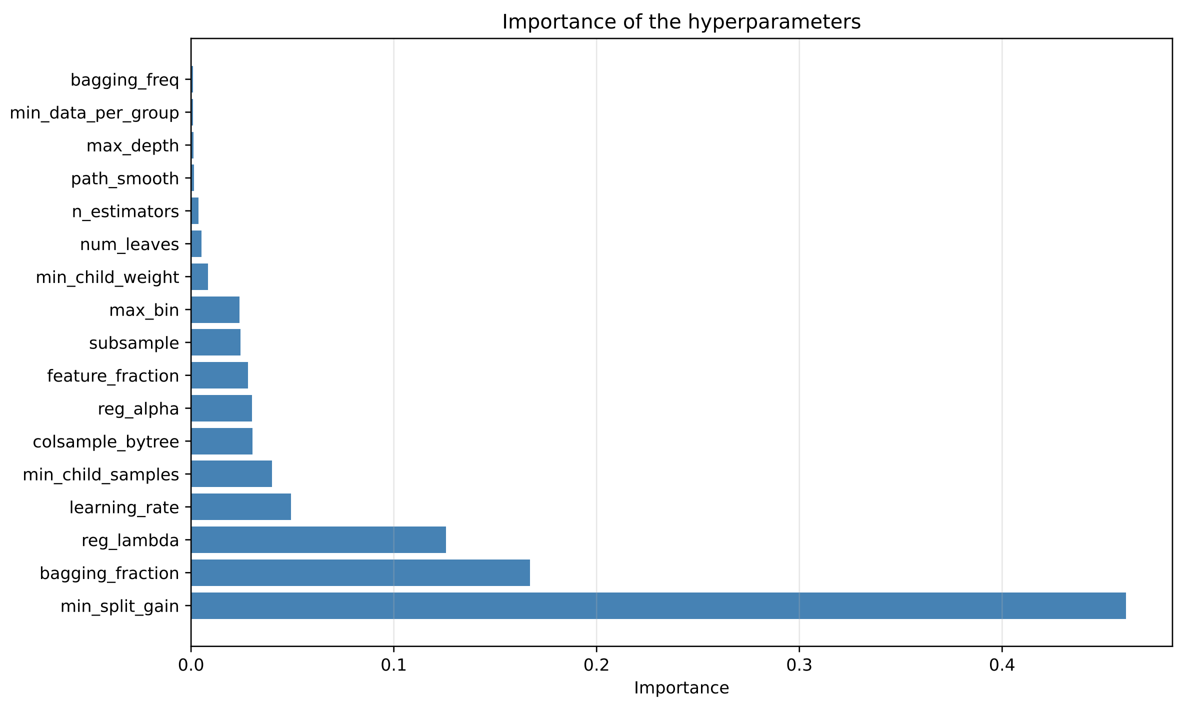
3.5 Model : LGBM

3.5.1 Gridsearch

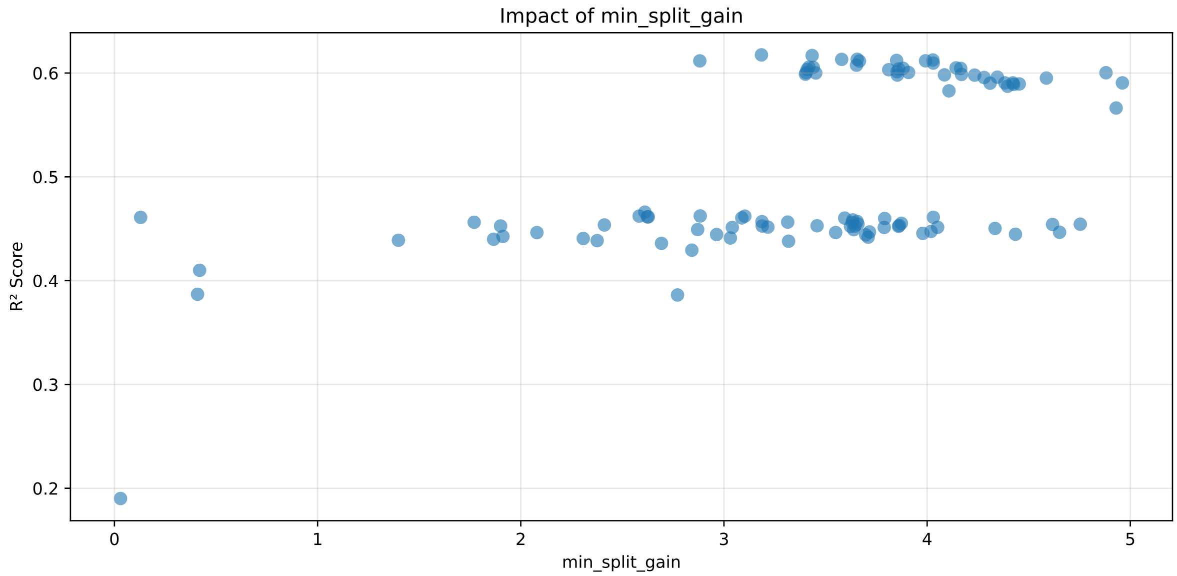
Distribution y_train vs y_test: y_train: mean=2.342, std=0.800, min=0.959, max=13.845 y_test: mean=2.364, std=0.795, min=1.068, max=7.277

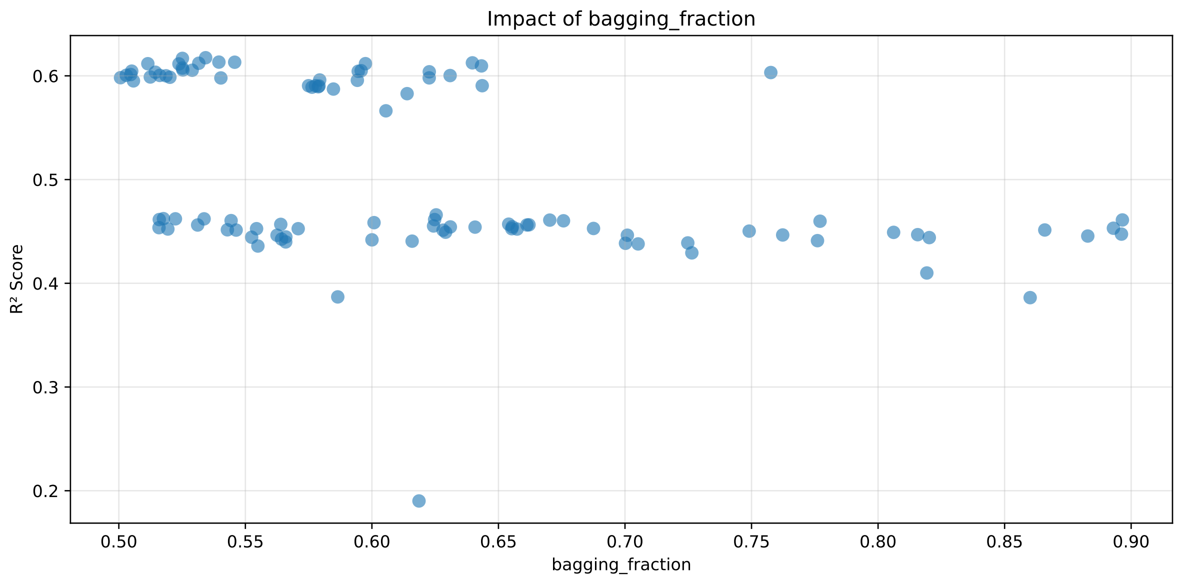
==================== TOP Importance FEATURES ==================== feature importance r_CO2 324 r_umidity 257 r_NH3_st_12h_median 172 r_temperature_st_8h_diff_last 167 r_NH3_st_8h_median 156 r_temperature_st_12h_min 131 r_temperature_st_8h_slope 125 r_umidity_st_12h_max 124 r_temperature 118 r_umidity_st_12h_slope 97 r_umidity_st_8h_median 91 r_NH3_st_8h_std 79 r_NH3_st_8h_slope 77 r_umidity_st_12h_std 74 r_umidity_st_8h_diff_last 62 r_NH3 56 r_THI 56 r_temperature_st_12h_diff_last 51 r_temperature_st_12h_std 46 r_NH3_st_8h_diff_last 45 r_umidity_st_12h_diff_last 40 r_NH3_st_12h_slope 36 r_NH3_st_12h_diff_last 20 r_NH3_st_12h_std 16 r_umidity_st_8h_std 9

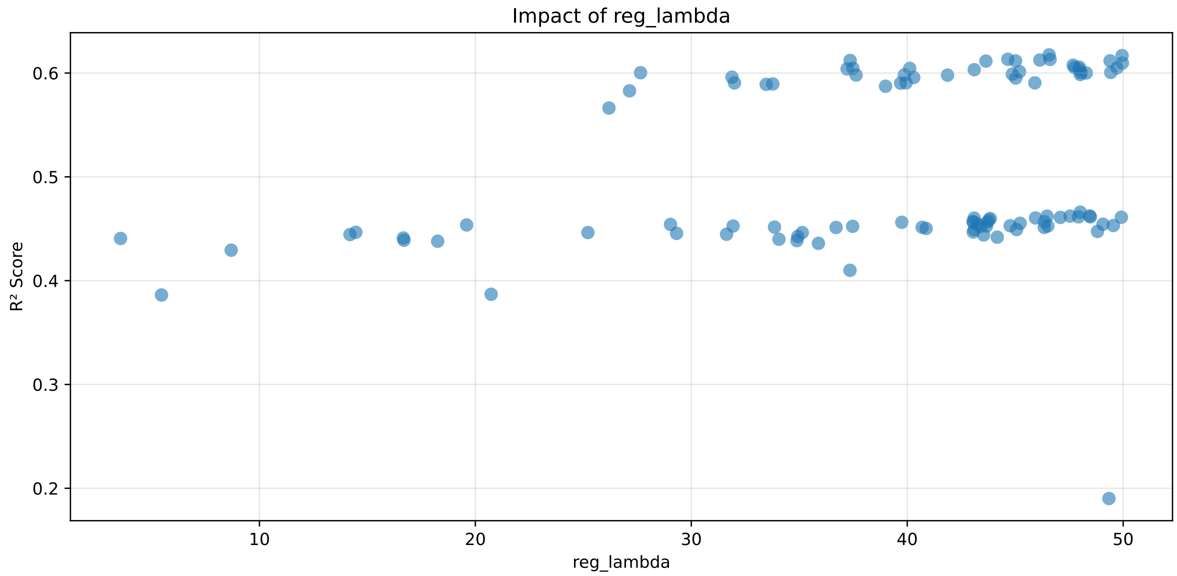




Paramètres avec >5% d'importance : ['min_split_gain', 'bagging_fraction', 'reg_lambda']

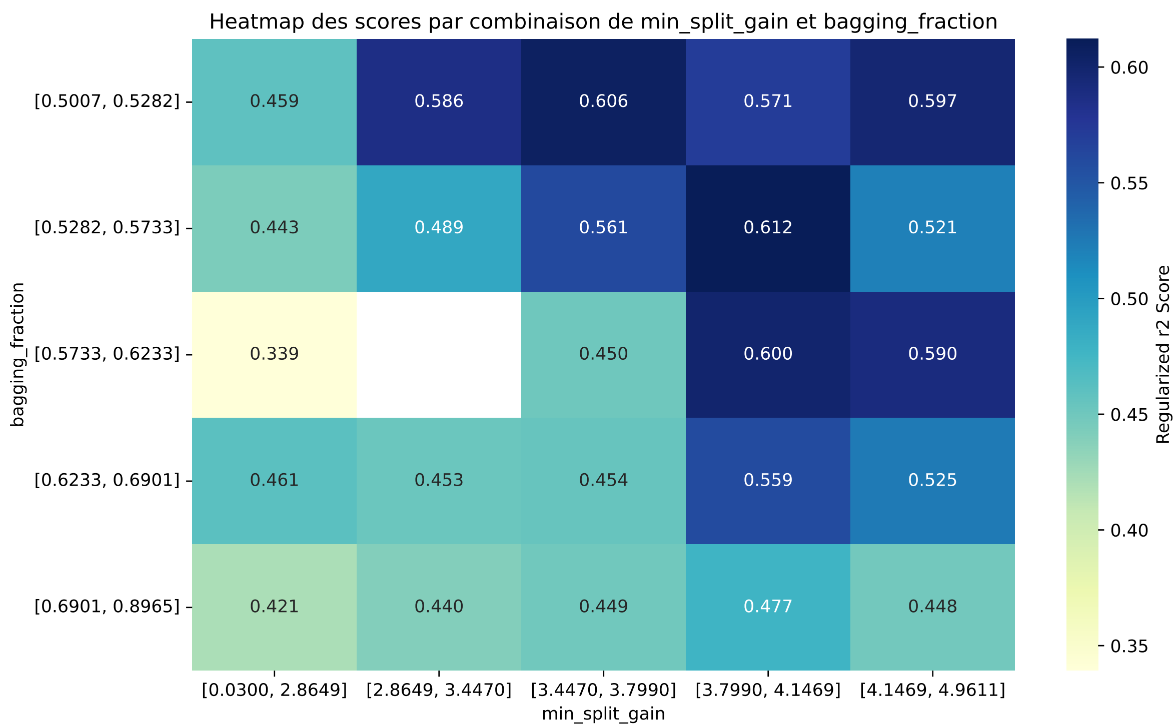

Impact of min_split_gain
R² Score
0.2
0.3
0.4
0.5
0.6
0
1
2
3
4
5
min_split_gain


Impact of bagging_fraction
R² Score
0.2
0.3
0.4
0.5
0.6
0.50
0.55
0.60
0.65
0.70
0.75
0.80
0.85
0.90
bagging_fraction



C:\Users\titou\Desktop\UNICT\1- Gas sensor calibration\Python\machine_learning\workflow_main

The default value of observed=False is deprecated and will change to observed=True in a futu



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.6474	0.6175	0.6595	0.03	0.6175	0.6847	0.0672	1.1089	1.0484
2	0.6471	0.6168	0.6576	0.0302	0.6168	0.6841	0.0672	1.109	1.049
3	0.6429	0.6132	0.6544	0.0263	0.6132	0.6819	0.0686	1.1119	1.0484
4	0.6435	0.6131	0.6551	0.0279	0.6131	0.6809	0.0678	1.1105	1.0495
5	0.642	0.6125	0.6539	0.0285	0.6125	0.6818	0.0692	1.113	1.0481
6	0.6423	0.6121	0.6535	0.0295	0.6121	0.6833	0.0712	1.1163	1.0492
7	0.641	0.6118	0.6528	0.0279	0.6118	0.682	0.0702	1.1148	1.0478
8	0.6417	0.6117	0.6523	0.026	0.6117	0.6777	0.066	1.1078	1.0491
9	0.6416	0.6115	0.6531	0.0289	0.6115	0.6756	0.0641	1.1048	1.0492
10	0.6378	0.6096	0.65	0.0279	0.6096	0.6811	0.0715	1.1173	1.0462
---	---	---	---	---	---	---	---	---	---
100	0.8821	0.7668	0.8102	0.0324	0.1901	0.8301	0.0633	1.0826	1.1504

Hyperparameters:

Rank 1: {'reg___n_estimators': 500, 'reg___max_depth': 6, 'reg___learning_rate': 0.013838622880103898, 'reg___num_leaves': 19, 'reg___subsample': 0.7916172688989415, 'reg___colsample_bytree': 0.8108737150873865, 'reg___min_child_samples': 13, 'reg___reg_alpha': 1.471687507227033, 'reg___reg_lambda': 46.572684627532595, 'reg___min_split_gain': 3.1845540636815013, 'reg___path_smooth': 4.596257871028001, 'reg___min_child_weight': 0.1615934899137523, 'reg___feature_fraction': 0.6168457119267005, 'reg___bagging_fraction': 0.5342945408919771, 'reg___bagging_freq': 4, 'reg___max_bin': 111, 'reg___min_data_per_group': 142}

Rank 2: {'reg___n_estimators': 600, 'reg___max_depth': 5, 'reg___learning_rate': 0.01260638625069077, 'reg___num_leaves': 16, 'reg___subsample': 0.7423430018693655, 'reg___colsample_bytree': 0.8497432303134734, 'reg___min_child_samples': 24, 'reg___reg_alpha': 0.23962314417493552, 'reg___reg_lambda': 49.95127318132368, 'reg___min_split_gain': 3.434071122238829, 'reg___path_smooth': 2.137707126094318, 'reg___min_child_weight': 0.06226029928452509, 'reg___feature_fraction': 0.6193794963637468, 'reg___bagging_fraction': 0.5251273920161875, 'reg___bagging_freq': 3, 'reg___max_bin': 162, 'reg___min_data_per_group': 128}

Rank 3: {'reg___n_estimators': 600, 'reg___max_depth': 6, 'reg___learning_rate': 0.014471932243469952, 'reg___num_leaves': 20, 'reg___subsample': 0.8686752221024022, 'reg___colsample_bytree': 0.7967187653796545, 'reg___min_child_samples': 13, 'reg___reg_alpha': 0.7092882319264255, 'reg___reg_lambda': 44.65953436733339, 'reg___min_split_gain': 3.65586819547518, 'reg___path_smooth': 4.590845279954879, 'reg___min_child_weight': 1.4159812803470027, 'reg___feature_fraction': 0.6156607465893011, 'reg___bagging_fraction': 0.5395340903038738, 'reg___bagging_freq': 4, 'reg___max_bin': 98, 'reg___min_data_per_group': 142}

Rank 4: {'reg__n_estimators': 500, 'reg__max_depth': 6, 'reg__learning_rate': 0.014670299046521996, 'reg__num_leaves': 20, 'reg__subsample': 0.7986541841021365, 'reg__colsample_bytree': 0.8159485086566937, 'reg__min_child_samples': 13, 'reg__reg_alpha': 1.2456014864219322, 'reg__reg_lambda': 46.61564454693803, 'reg__min_split_gain': 3.5794351559223694, 'reg__path_smooth': 4.704414405341765, 'reg__min_child_weight': 0.5954301033074771, 'reg__feature_fraction': 0.707161414296002, 'reg__bagging_fraction': 0.5457786718436188, 'reg__bagging_freq': 4, 'reg__max_bin': 125, 'reg__min_data_per_group': 141}

Rank 5: {'reg__n_estimators': 800, 'reg__max_depth': 7, 'reg__learning_rate': 0.0216898424357084, 'reg__num_leaves': 46, 'reg__subsample': 0.8982147632356332, 'reg__colsample_bytree': 0.7322419403198958, 'reg__min_child_samples': 49, 'reg__reg_alpha': 0.06557159996582262, 'reg__reg_lambda': 46.141624312658685, 'reg__min_split_gain': 4.028230861132452, 'reg__path_smooth': 3.2981468018216993, 'reg__min_child_weight': 2.12236455664041, 'reg__feature_fraction': 0.6445580626735684, 'reg__bagging_fraction': 0.6397099118108015, 'reg__bagging_freq': 5, 'reg__max_bin': 74, 'reg__min_data_per_group': 194}

Rank 6: {'reg__n_estimators': 600, 'reg__max_depth': 5, 'reg__learning_rate': 0.012551700200986848, 'reg__num_leaves': 15, 'reg__subsample': 0.7374839335013097, 'reg__colsample_bytree': 0.844282732274814, 'reg__min_child_samples': 21, 'reg__reg_alpha': 0.05469999736604231, 'reg__reg_lambda': 37.356034979570964, 'reg__min_split_gain': 3.8497671323644975, 'reg__path_smooth': 4.982508831634398, 'reg__min_child_weight': 4.546402733910402, 'reg__feature_fraction': 0.6176373051958249, 'reg__bagging_fraction': 0.5315792425437592, 'reg__bagging_freq': 3, 'reg__max_bin': 142, 'reg__min_data_per_group': 111}

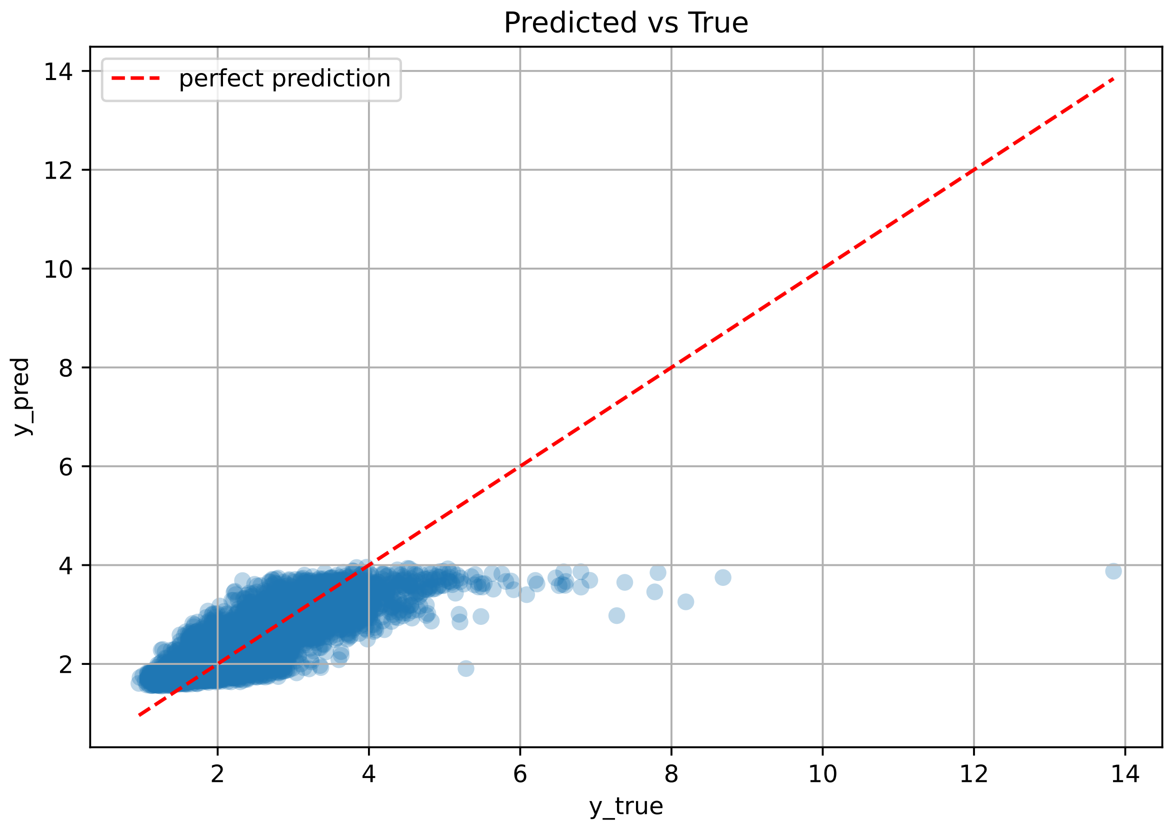
Rank 7: {'reg__n_estimators': 700, 'reg__max_depth': 7, 'reg__learning_rate': 0.010387543559730308, 'reg__num_leaves': 44, 'reg__subsample': 0.889536477892824, 'reg__colsample_bytree': 0.7339787196487539, 'reg__min_child_samples': 11, 'reg__reg_alpha': 0.3406856453947551, 'reg__reg_lambda': 49.395263287479196, 'reg__min_split_gain': 3.9921339787281145, 'reg__path_smooth': 3.774703744028751, 'reg__min_child_weight': 2.760034946669333, 'reg__feature_fraction': 0.6462589356383085, 'reg__bagging_fraction': 0.5974506190284985, 'reg__bagging_freq': 5, 'reg__max_bin': 68, 'reg__min_data_per_group': 156}

Rank 8: {'reg__n_estimators': 300, 'reg__max_depth': 5, 'reg__learning_rate': 0.012287392257045416, 'reg__num_leaves': 25, 'reg__subsample': 0.7484765388686934, 'reg__colsample_bytree': 0.8953992710035461, 'reg__min_child_samples': 11, 'reg__reg_alpha': 1.153092842762065, 'reg__reg_lambda': 45.01634605987324, 'reg__min_split_gain': 2.881171672750476, 'reg__path_smooth': 2.214342159387316, 'reg__min_child_weight': 8.054009987707625, 'reg__feature_fraction': 0.6504360705401757, 'reg__bagging_fraction': 0.5114588690433305, 'reg__bagging_freq': 5, 'reg__max_bin': 157, 'reg__min_data_per_group': 120}

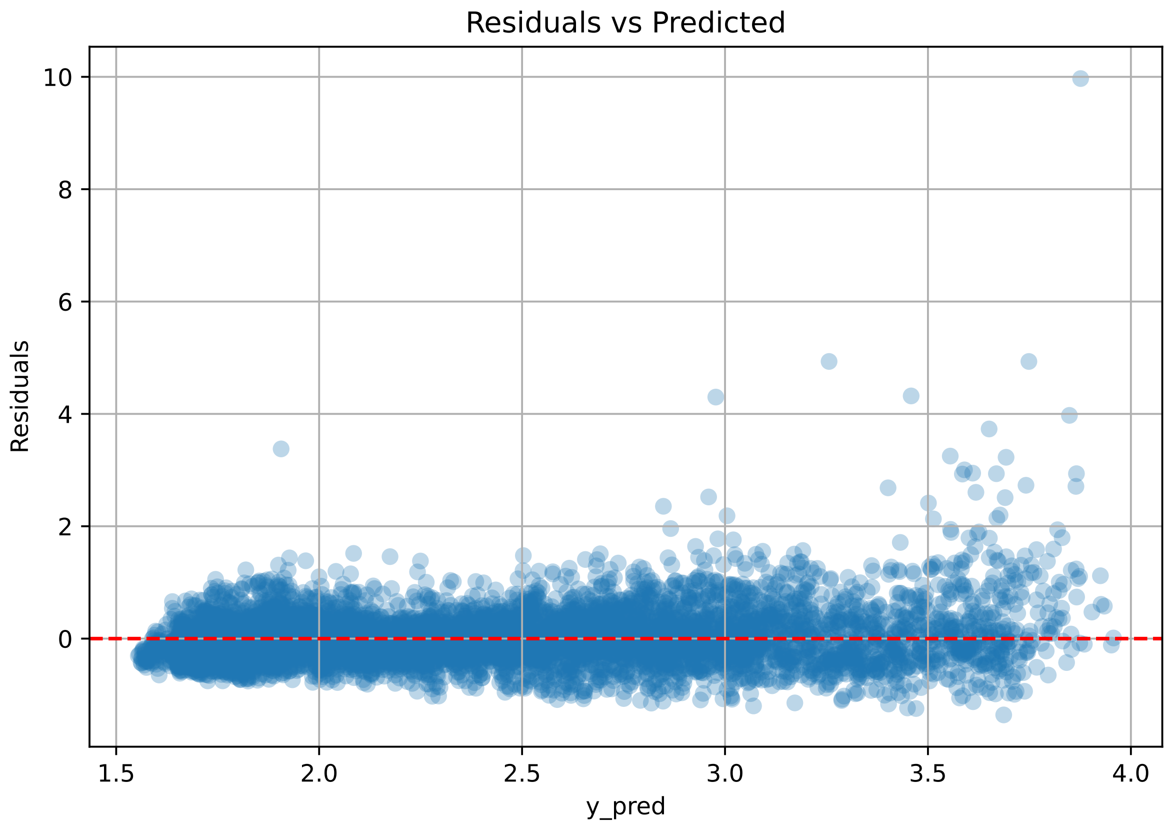
Rank 9: {'reg__n_estimators': 400, 'reg__max_depth': 5, 'reg__learning_rate': 0.01272912272997617, 'reg__num_leaves': 16, 'reg__subsample': 0.7482530681528333, 'reg__colsample_bytree': 0.8149129300703752, 'reg__min_child_samples': 17, 'reg__reg_alpha': 0.3661921197471353, 'reg__reg_lambda': 43.64355099343897, 'reg__min_split_gain': 3.666822402033409, 'reg__path_smooth': 4.394916254966148, 'reg__min_child_weight': 0.6012112194986653, 'reg__feature_fraction': 0.6629236666274638, 'reg__bagging_fraction': 0.523753788231448, 'reg__bagging_freq': 5, 'reg__max_bin': 132, 'reg__min_data_per_group': 128}

Rank 10: {'reg__n_estimators': 700, 'reg__max_depth': 7, 'reg__learning_rate': 0.0231033784004885, 'reg__num_leaves': 24, 'reg__subsample': 0.8987003327874773, 'reg__colsample_bytree': 0.7390480816912725, 'reg__min_child_samples': 47, 'reg__reg_alpha': 0.32454752927043384, 'reg__reg_lambda': 49.96488380813423, 'reg__min_split_gain': 4.029926060016425, 'reg__path_smooth': 3.318076706326533, 'reg__min_child_weight': 1.5451830207533086, 'reg__feature_fraction': 0.6441207494649155, 'reg__bagging_fraction': 0.643291491273942, 'reg__bagging_freq': 5, 'reg__max_bin': 65, 'reg__min_data_per_group': 152}

3.5.2 Scatter


Predicted vs True
perfect prediction
y_pred
y_true
14
12
10
8
6
4
2
2
4
6
8
10
12
14

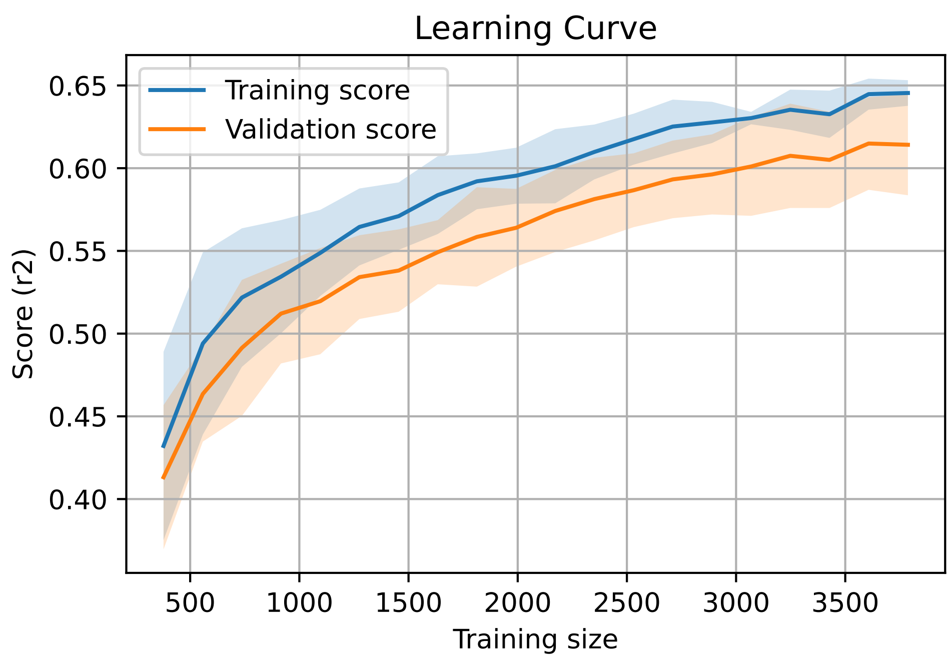
3.5.3 Scatter residuals



3.5.4 Learning curve

Len(group) : 7730, **Len(training)** : 5411,

Len(training_real) : 3788, **Len(test_of_training)** : 1623, **Len(test)** : 2319

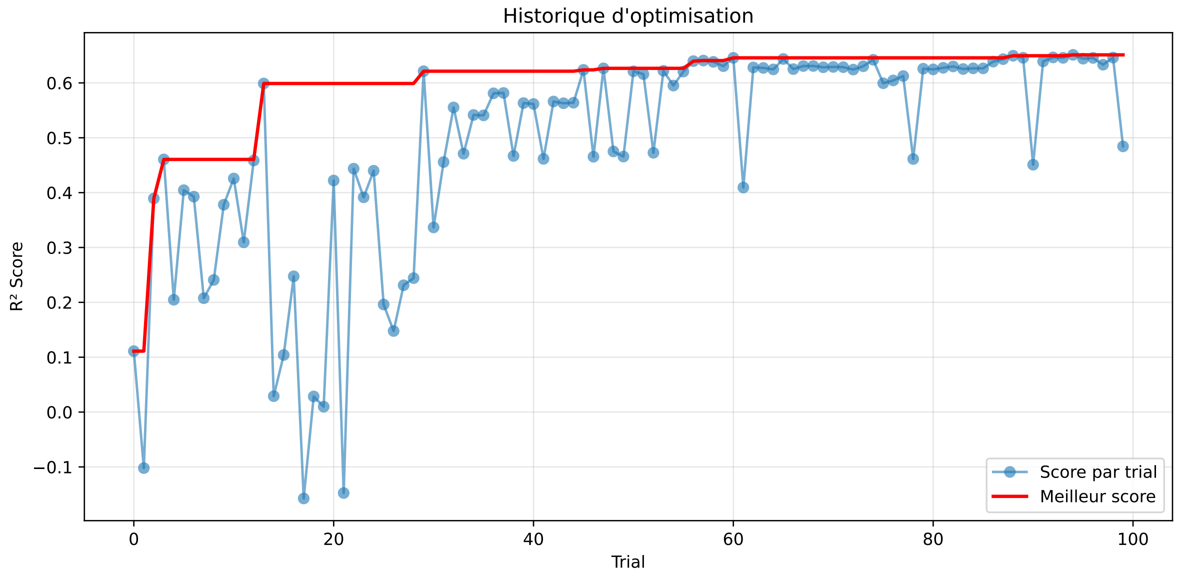


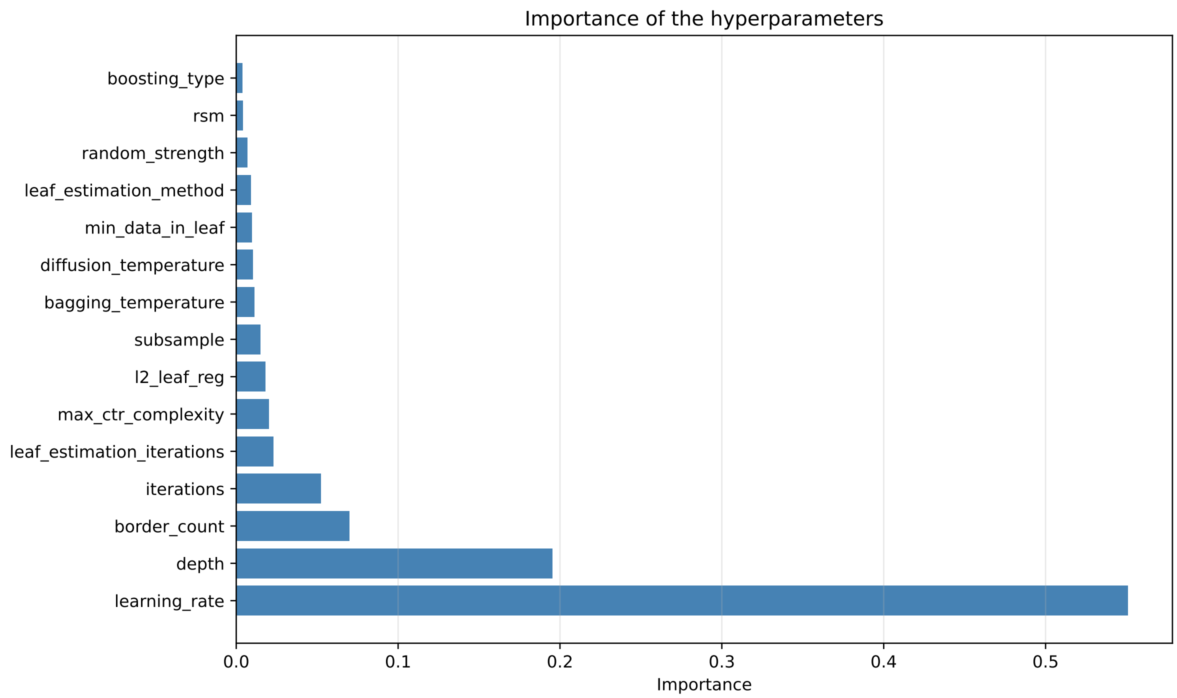
3.6 Model : CatBoost

3.6.1 Gridsearch

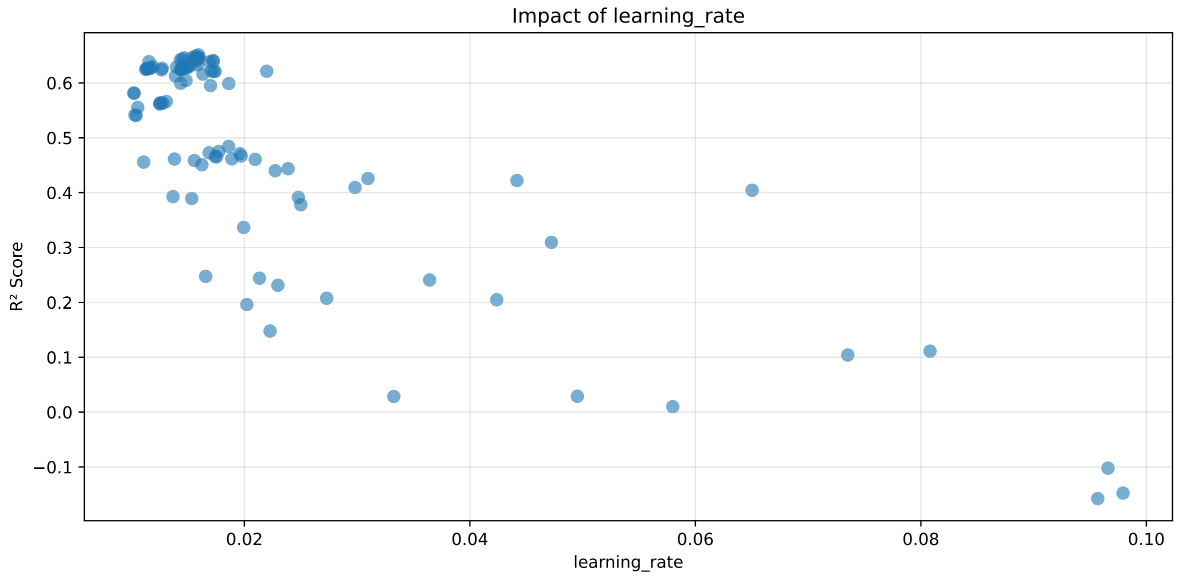
Distribution y_train vs y_test: y_train: mean=2.342, std=0.800, min=0.959, max=13.845 y_test: mean=2.364, std=0.795, min=1.068, max=7.277

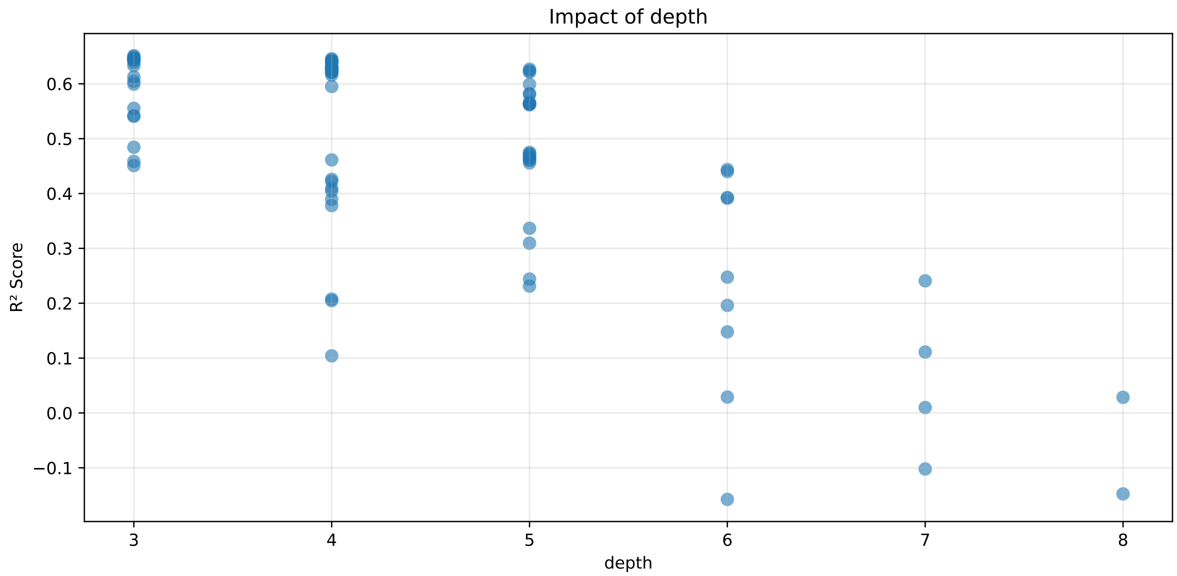
==================== TOP Importance FEATURES ==================== feature importance r_umidity 23.656130 r_temperature_st_8h_slope 10.711988 r_temperature_st_8h_diff_last 9.581528 r_temperature 6.838676 r_temperature_st_12h_diff_last 6.502301 r_CO2 5.959706 r_NH3_st_12h_median 3.727676 r_umidity_st_12h_max 3.658620 r_umidity_st_12h_std 2.953880 r_NH3 2.949698 r_NH3_st_8h_median 2.769853 r_THI 2.542144 r_umidity_st_12h_slope 2.482044 r_umidity_st_8h_median 2.444723 r_umidity_st_8h_diff_last 2.228618 r_temperature_st_12h_min 1.879330 r_umidity_st_12h_diff_last 1.823303 r_NH3_st_8h_std 1.506732 r_temperature_st_12h_std 1.342563 r_NH3_st_8h_slope 1.227136 r_NH3_st_12h_slope 0.879266 r_NH3_st_12h_std 0.771181 r_umidity_st_8h_std 0.664808 r_NH3_st_8h_diff_last 0.468729 r_NH3_st_12h_diff_last 0.429367

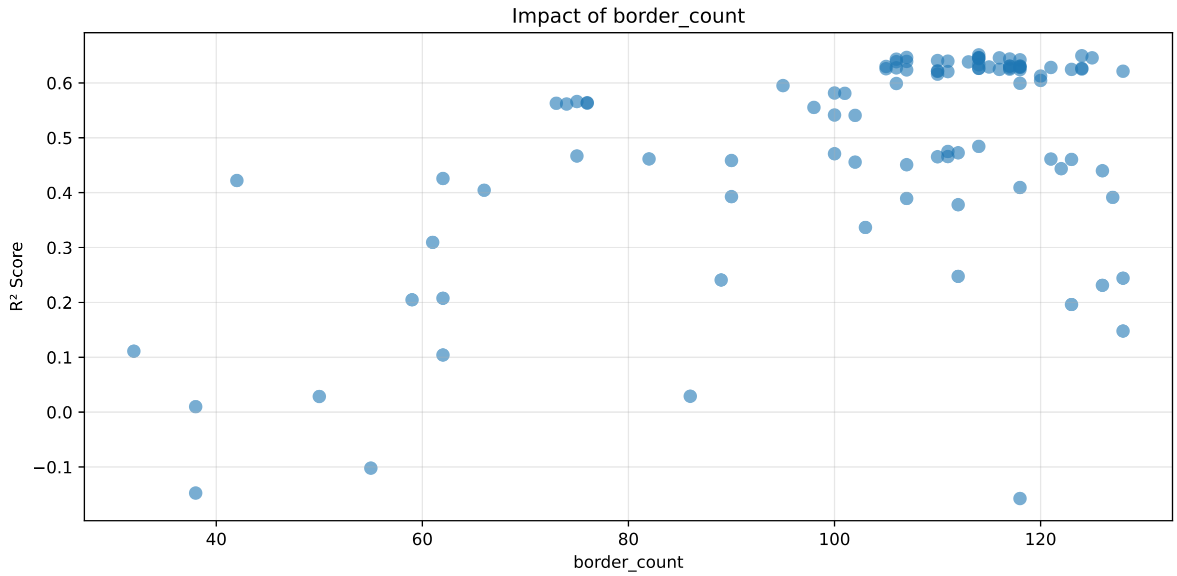


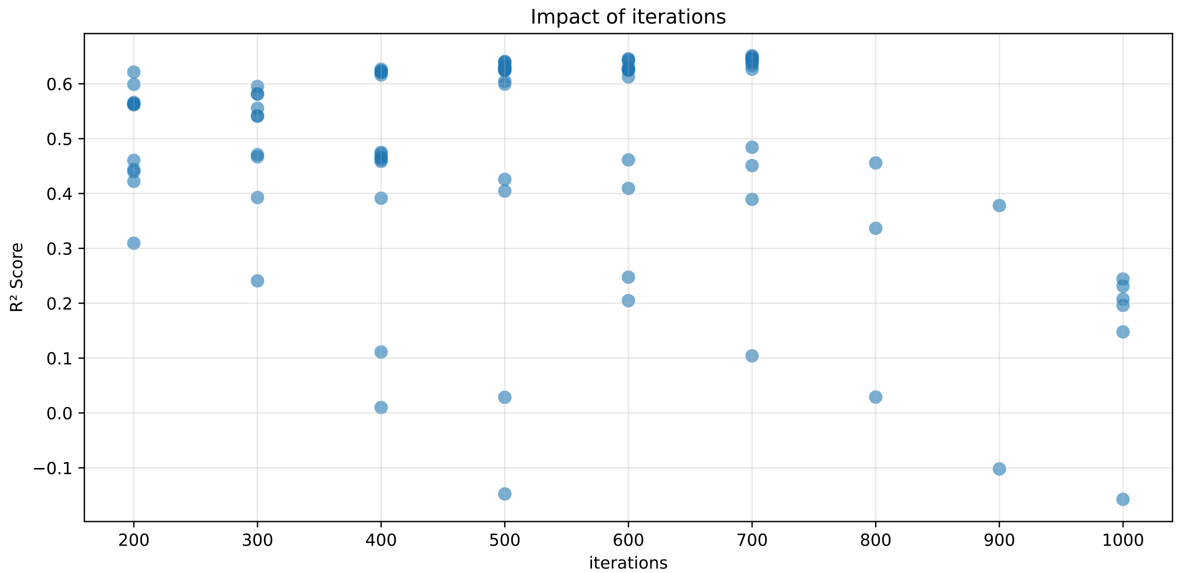


Paramètres avec >5% d'importance : ['learning_rate', 'depth', 'border_count', 'iterations']


Impact of learning_rate
R² Score
learning_rate
0.6
0.5
0.4
0.3
0.2
0.1
0.0
−0.1
0.02
0.04
0.06
0.08
0.10

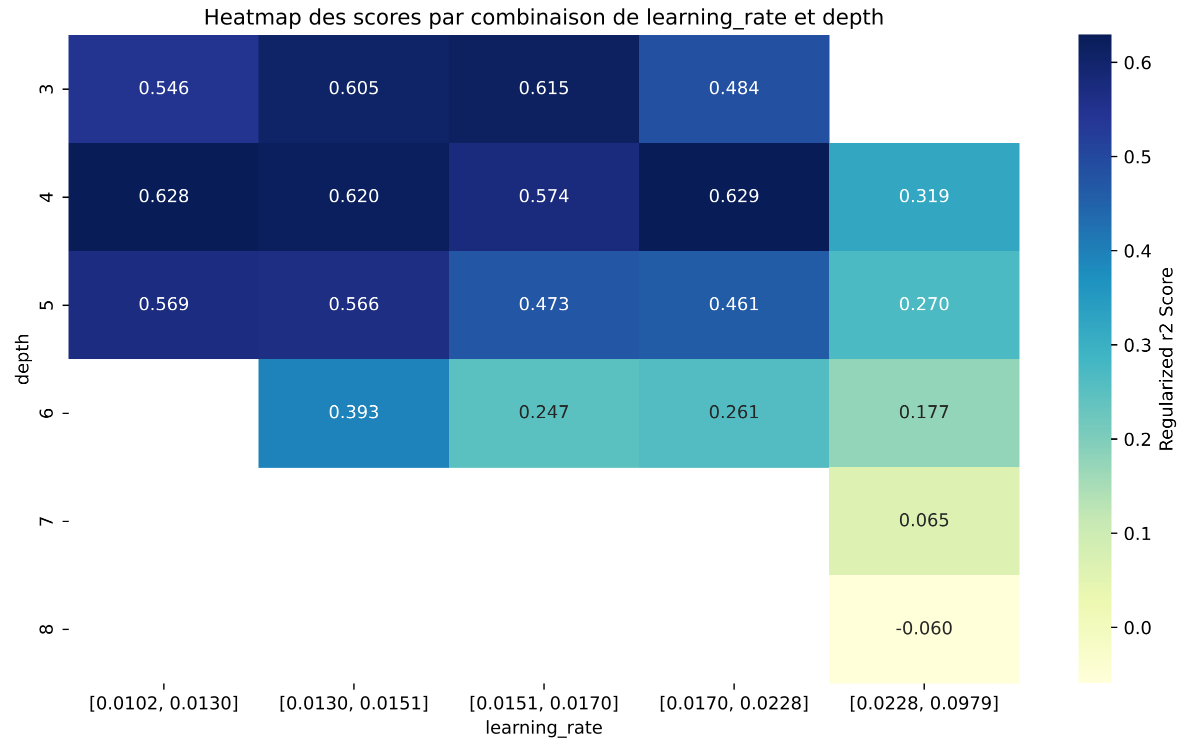

Impact of depth
R² Score
depth
0.6
0.5
0.4
0.3
0.2
0.1
0.0
−0.1
3
4
5
6
7
8






```
C:\Users\titou\Desktop\UNICT\1- Gas sensor calibration\Python\machine_learning\workflow_main

The default value of observed=False is deprecated and will change to observed=True in a futu
```



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.682	0.6511	0.6952	0.0236	0.6511	0.6996	0.0485	1.0744	1.0474
2	0.6811	0.6495	0.6931	0.0245	0.6495	0.6963	0.0468	1.072	1.0487
3	0.6759	0.6465	0.6902	0.0239	0.6465	0.6962	0.0497	1.0769	1.0456
4	0.6745	0.6461	0.6886	0.0233	0.6461	0.6921	0.046	1.0713	1.044
5	0.6766	0.6458	0.6903	0.0242	0.6458	0.6942	0.0484	1.075	1.0478
6	0.6762	0.6457	0.6902	0.024	0.6457	0.6966	0.0509	1.0788	1.0472
7	0.6741	0.6457	0.6886	0.024	0.6457	0.6936	0.0479	1.0742	1.0441
8	0.6755	0.6455	0.6895	0.0245	0.6455	0.6942	0.0487	1.0755	1.0465
9	0.6736	0.6442	0.6879	0.0238	0.6442	0.6907	0.0466	1.0723	1.0457
10	0.6738	0.6436	0.6886	0.024	0.6436	0.6973	0.0537	1.0834	1.0469
---	---	---	---	---	---	---	---	---	---
100	0.983	0.7929	0.8389	0.0308	-0.1575	0.8477	0.0548	1.0691	1.2397

Hyperparameters:

Rank 1: {'reg___iterations': 700, 'reg___depth': 3, 'reg___learning_rate': 0.015945024286809155, 'reg___l2_leaf_reg': 18.70608887342023, 'reg___bagging_temperature': 4.340179108950572, 'reg___random_strength': 3.6882005864830214, 'reg___subsample': 0.5750986331752183, 'reg___min_data_in_leaf': 32, 'reg___leaf_estimation_iterations': 7, 'reg___rsm': 0.6540505714007525, 'reg___border_count': 114, 'reg___leaf_estimation_method': 'Gradient',

’reg__boosting_type’: ‘Ordered’, ’reg__max_ctr_complexity’: 1, ’reg__diffusion_temperature’: 8963.66041265247}

Rank 2: {’reg__iterations’: 700, ’reg__depth’: 3, ’reg__learning_rate’: 0.015723123409608315, ’reg__l2_leaf_reg’: 27.678635819169084, ’reg__bagging_temperature’: 4.325212223800385, ’reg__random_strength’: 3.6477904481201335, ’reg__subsample’: 0.6731550980968671, ’reg__min_data_in_leaf’: 14, ’reg__leaf_estimation_iterations’: 6, ’reg__rsm’: 0.698022224400379, ’reg__border_count’: 124, ’reg__leaf_estimation_method’: ‘Gradient’, ’reg__boosting_type’: ‘Ordered’, ’reg__max_ctr_complexity’: 1, ’reg__diffusion_temperature’: 8870.667526497542}

Rank 3: {’reg__iterations’: 700, ’reg__depth’: 3, ’reg__learning_rate’: 0.015408880989303113, ’reg__l2_leaf_reg’: 29.83132208502438, ’reg__bagging_temperature’: 4.3258865620182085, ’reg__random_strength’: 3.639975045734789, ’reg__subsample’: 0.6031325424171627, ’reg__min_data_in_leaf’: 14, ’reg__leaf_estimation_iterations’: 7, ’reg__rsm’: 0.6581506936451624, ’reg__border_count’: 107, ’reg__leaf_estimation_method’: ‘Gradient’, ’reg__boosting_type’: ‘Ordered’, ’reg__max_ctr_complexity’: 1, ’reg__diffusion_temperature’: 8984.99770724462}

Rank 4: {’reg__iterations’: 700, ’reg__depth’: 3, ’reg__learning_rate’: 0.015715067149792372, ’reg__l2_leaf_reg’: 14.643077298146352, ’reg__bagging_temperature’: 0.44558624663450264, ’reg__random_strength’: 4.768479152504356, ’reg__subsample’: 0.6685258187122063, ’reg__min_data_in_leaf’: 14, ’reg__leaf_estimation_iterations’: 7, ’reg__rsm’: 0.6789789101036552, ’reg__border_count’: 114, ’reg__leaf_estimation_method’: ‘Gradient’, ’reg__boosting_type’: ‘Ordered’, ’reg__max_ctr_complexity’: 1, ’reg__diffusion_temperature’: 8925.49042987497}

Rank 5: {’reg__iterations’: 700, ’reg__depth’: 3, ’reg__learning_rate’: 0.015723202288274507, ’reg__l2_leaf_reg’: 19.148730968677302, ’reg__bagging_temperature’: 0.49848219742118927, ’reg__random_strength’: 3.6496548828162183, ’reg__subsample’: 0.5718002605059889, ’reg__min_data_in_leaf’: 14, ’reg__leaf_estimation_iterations’: 6, ’reg__rsm’: 0.6944465652721789, ’reg__border_count’: 125, ’reg__leaf_estimation_method’: ‘Gradient’, ’reg__boosting_type’: ‘Ordered’, ’reg__max_ctr_complexity’: 1, ’reg__diffusion_temperature’: 960.5618865217807}

Rank 6: {’reg__iterations’: 600, ’reg__depth’: 4, ’reg__learning_rate’: 0.014682464907704475, ’reg__l2_leaf_reg’: 17.00603155714087, ’reg__bagging_temperature’: 4.912584537684613, ’reg__random_strength’: 4.546840430083462, ’reg__subsample’: 0.585657375497121, ’reg__min_data_in_leaf’: 12, ’reg__leaf_estimation_iterations’: 7, ’reg__rsm’: 0.6202439937652168, ’reg__border_count’: 116, ’reg__leaf_estimation_method’: ‘Newton’, ’reg__boosting_type’: ‘Ordered’, ’reg__max_ctr_complexity’: 1, ’reg__diffusion_temperature’: 9219.053375008993}

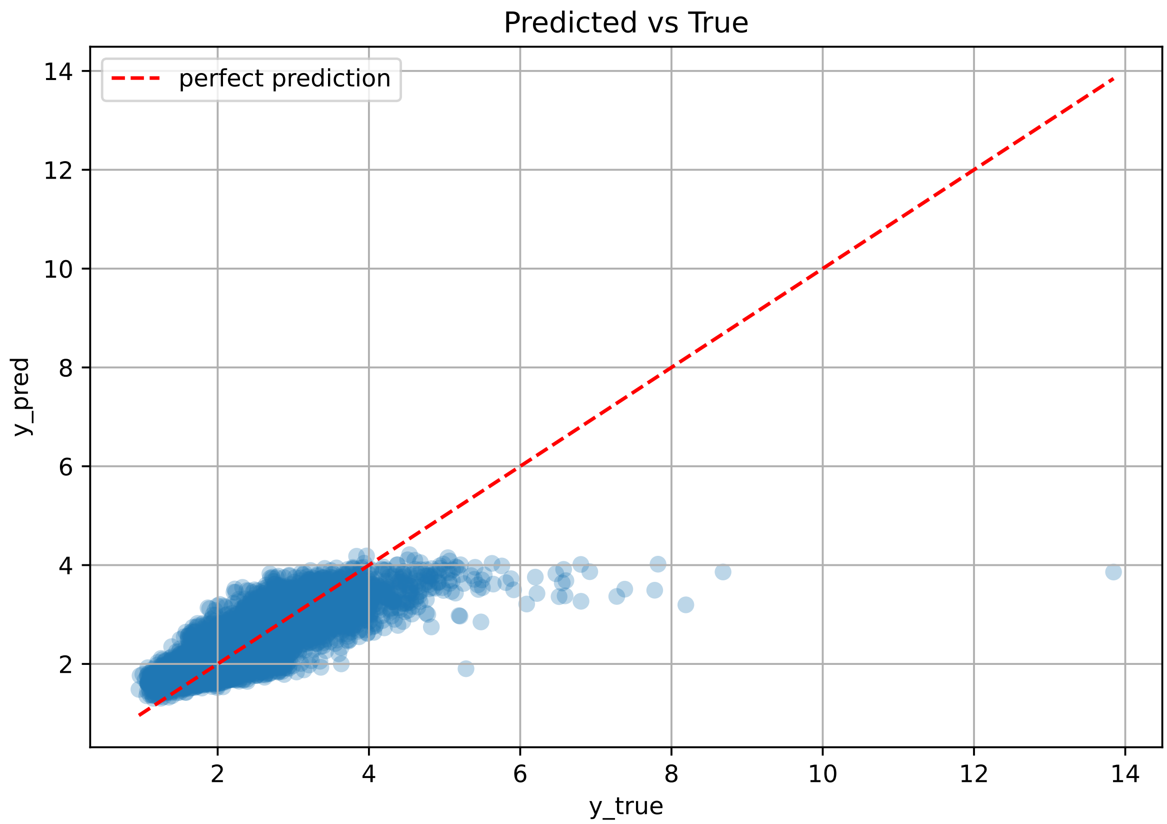
Rank 7: {’reg__iterations’: 700, ’reg__depth’: 3, ’reg__learning_rate’: 0.015991236161372848, ’reg__l2_leaf_reg’: 19.038353132403863, ’reg__bagging_temperature’: 4.814572018331458, ’reg__random_strength’: 3.642854795298393, ’reg__subsample’: 0.5738098340466181, 'reg___min_data_in_leaf': 14, 'reg___leaf_estimation_iterations': 6, 'reg___rsm': 0.6508141842747072, 'reg___border_count': 114, 'reg___leaf_estimation_method': 'Gradient', 'reg___boosting_type': 'Ordered', 'reg___max_ctr_complexity': 1, 'reg___diffusion_temperature': 419.1980893779564}

Rank 8: {'reg___iterations': 700, 'reg___depth': 3, 'reg___learning_rate': 0.015894800825188245, 'reg___l2_leaf_reg': 19.49352229176866, 'reg___bagging_temperature': 4.803343422472756, 'reg___random_strength': 4.766786578295653, 'reg___subsample': 0.6643822566156213, 'reg___min_data_in_leaf': 14, 'reg___leaf_estimation_iterations': 6, 'reg___rsm': 0.6964588833126989, 'reg___border_count': 114, 'reg___leaf_estimation_method': 'Gradient', 'reg___boosting_type': 'Ordered', 'reg___max_ctr_complexity': 1, 'reg___diffusion_temperature': 8841.86327489688}

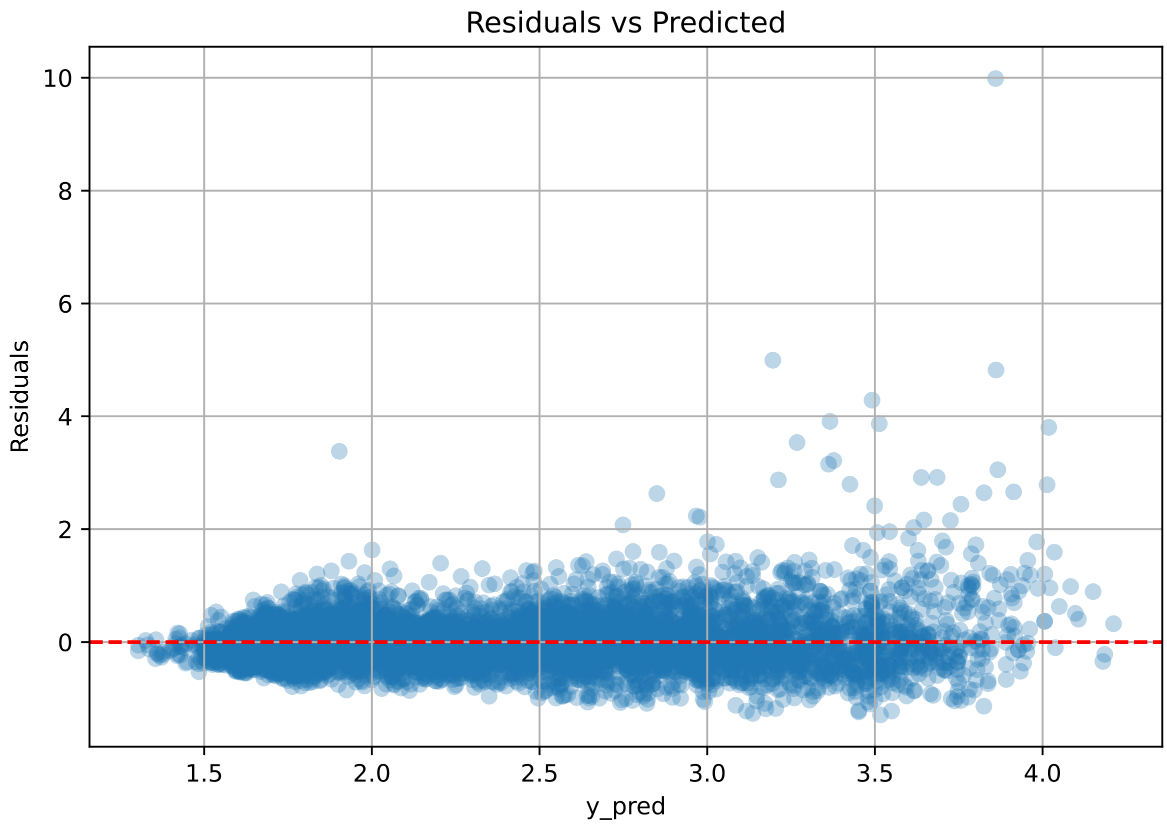
Rank 9: {'reg___iterations': 700, 'reg___depth': 3, 'reg___learning_rate': 0.015637490821488895, 'reg___l2_leaf_reg': 18.774731609905203, 'reg___bagging_temperature': 4.2823543829836765, 'reg___random_strength': 4.766607532867727, 'reg___subsample': 0.658253138867601, 'reg___min_data_in_leaf': 14, 'reg___leaf_estimation_iterations': 7, 'reg___rsm': 0.7239407881306584, 'reg___border_count': 114, 'reg___leaf_estimation_method': 'Gradient', 'reg___boosting_type': 'Ordered', 'reg___max_ctr_complexity': 1, 'reg___diffusion_temperature': 8974.625971923571}

Rank 10: {'reg___iterations': 600, 'reg___depth': 4, 'reg___learning_rate': 0.014485043530835967, 'reg___l2_leaf_reg': 24.093258467579734, 'reg___bagging_temperature': 4.945691099740031, 'reg___random_strength': 4.258710102817461, 'reg___subsample': 0.5826484727628904, 'reg___min_data_in_leaf': 13, 'reg___leaf_estimation_iterations': 7, 'reg___rsm': 0.6127862005382343, 'reg___border_count': 117, 'reg___leaf_estimation_method': 'Gradient', 'reg___boosting_type': 'Ordered', 'reg___max_ctr_complexity': 1, 'reg___diffusion_temperature': 9271.867818033712}

3.6.2 Scatter


Predicted vs True
perfect prediction
y_pred
y_true
2
4
6
8
10
12
14
2
4
6
8
10
12
14

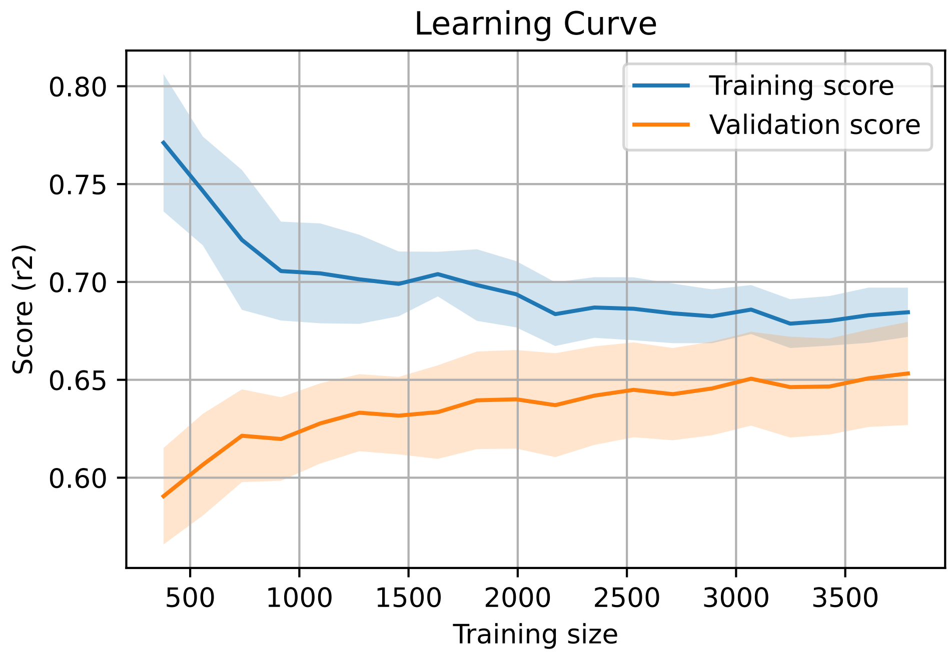
3.6.3 Scatter residuals



3.6.4 Learning curve

$\mathbf{Len(group)}$: 7730, $\mathbf{Len(training)}$: 5411,

$\mathbf{Len(training\_real)}$: 3788, $\mathbf{Len(test\_of\_training)}$: 1623, $\mathbf{Len(test)}$: 2319



3.7 Sumup-table

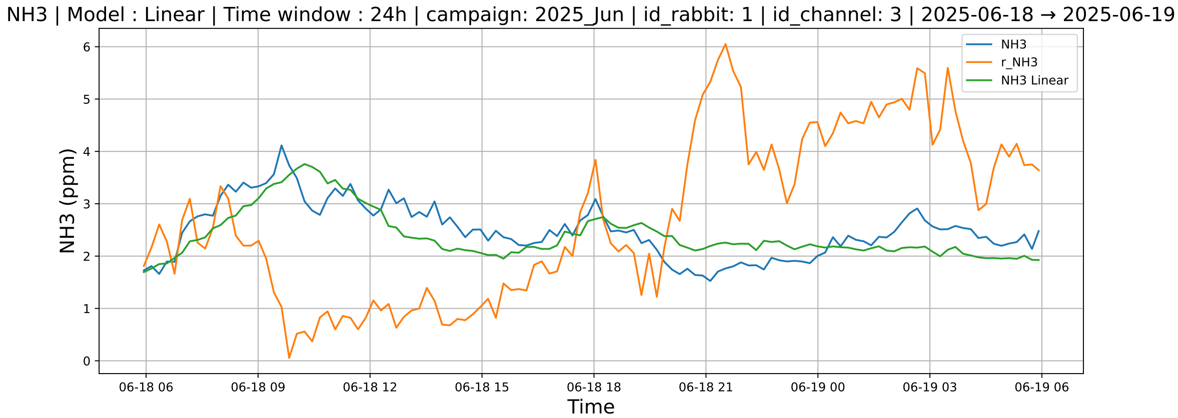
model	mean_train_cv	mean_test_cv	std_test_cv	test_cv	test	diff	r_test	r_train	pen_score
Linear	0.562	0.544	0.022	0.587	0.588	0.044	1.081	1.034	nan
LGBM	0.647	0.618	0.03	0.66	0.685	0.067	1.109	1.048	0.6175
CatBoost	0.682	0.651	0.024	0.695	0.7	0.048	1.074	1.047	0.6511

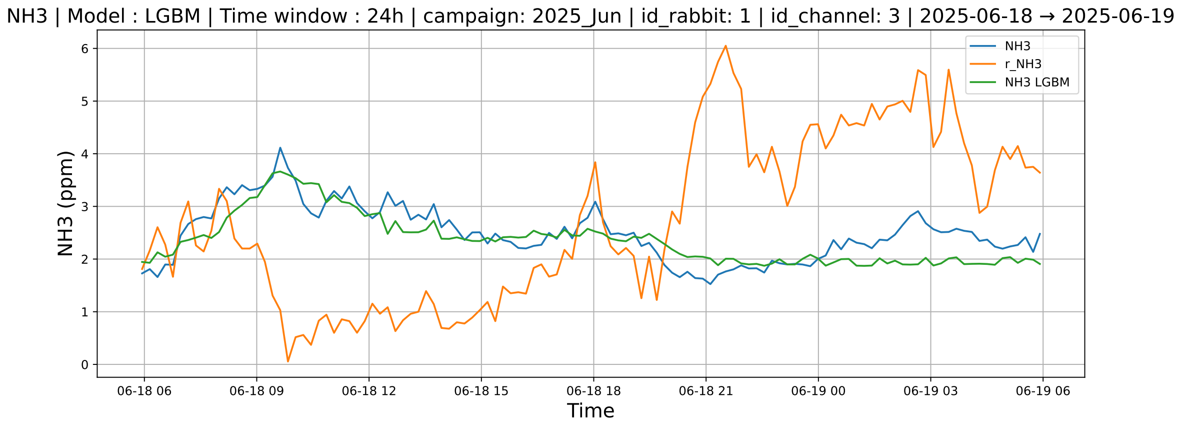
4 Plot

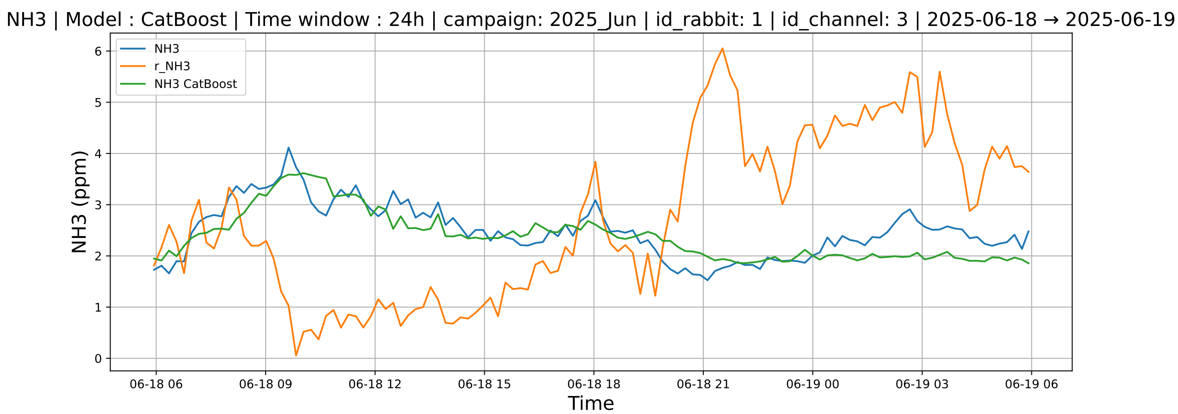
4.1 Standalone plots

4.1.1 Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3

4.1.1.1 Time window : 24

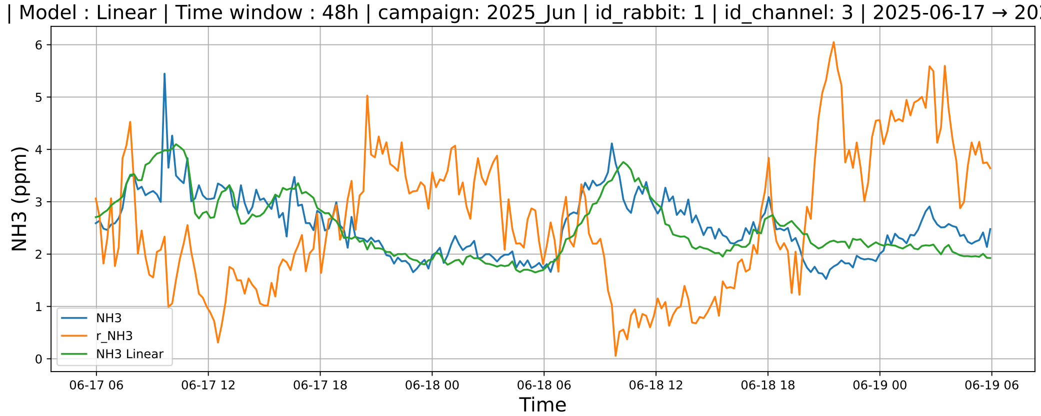




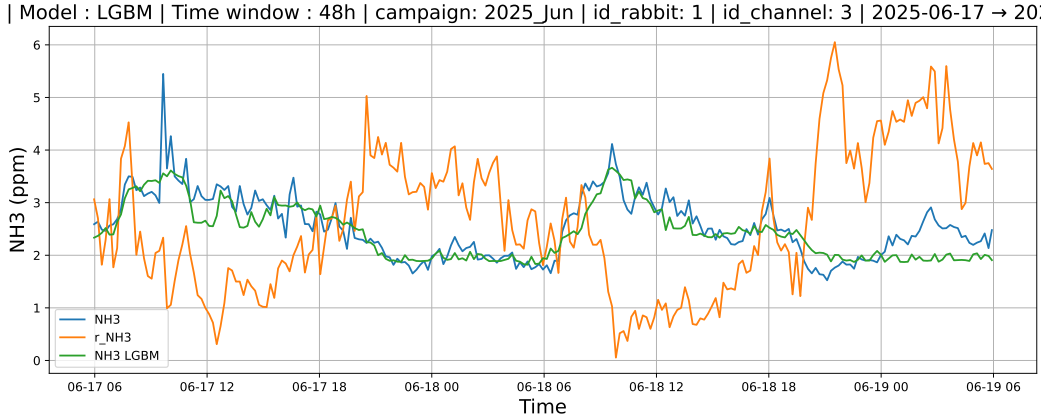


4.1.1.2 Time window : 48

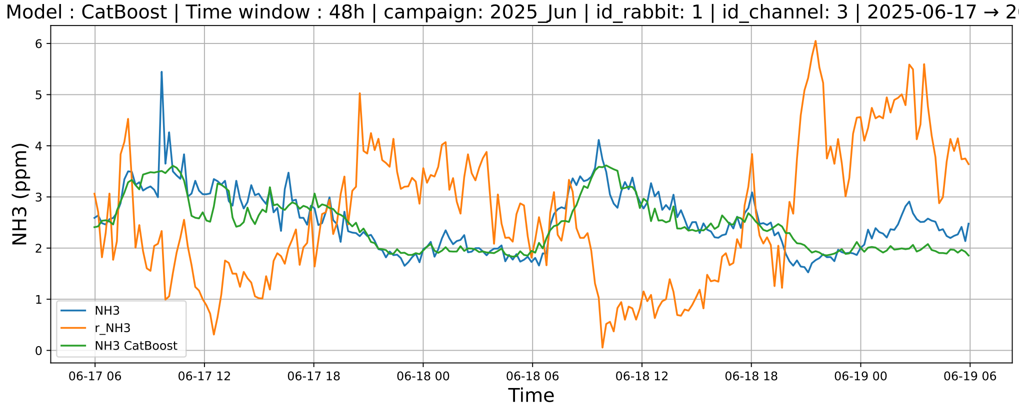
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19

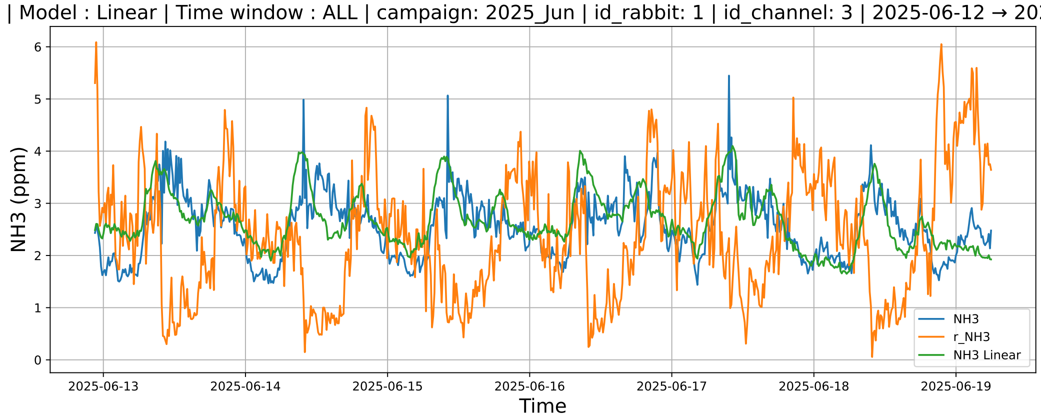


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19

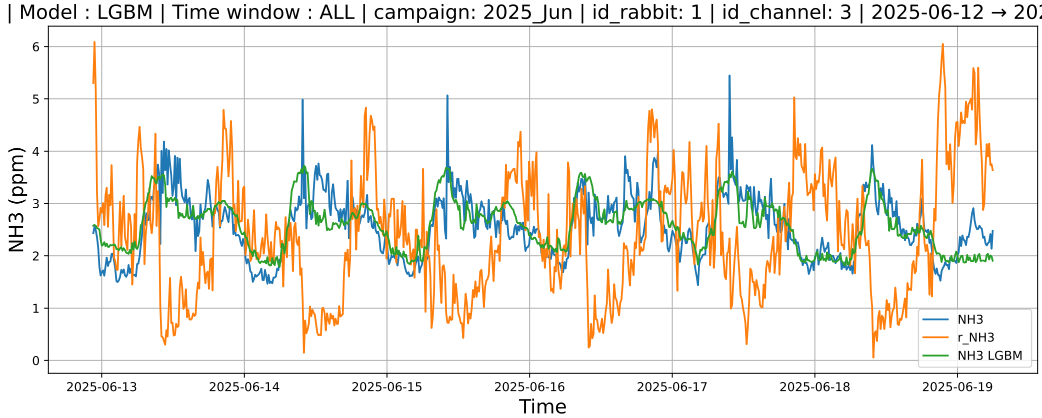


4.1.1.3 Time window : ALL

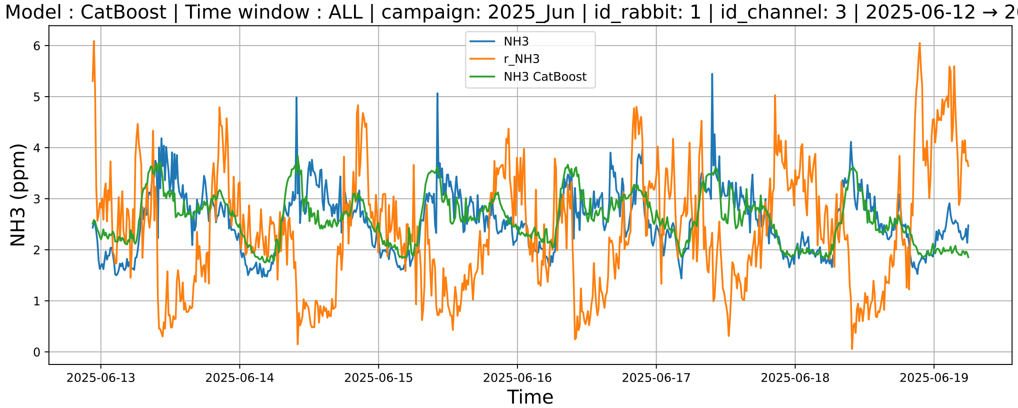
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19

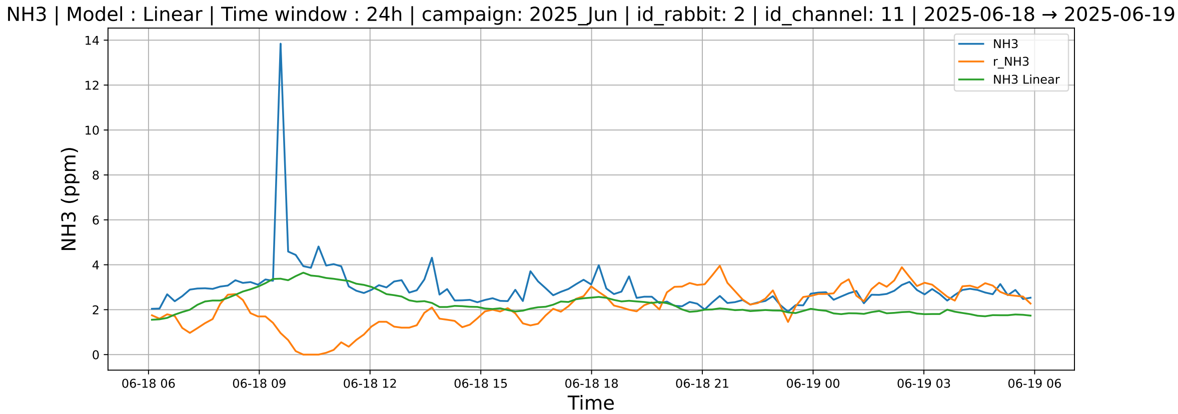


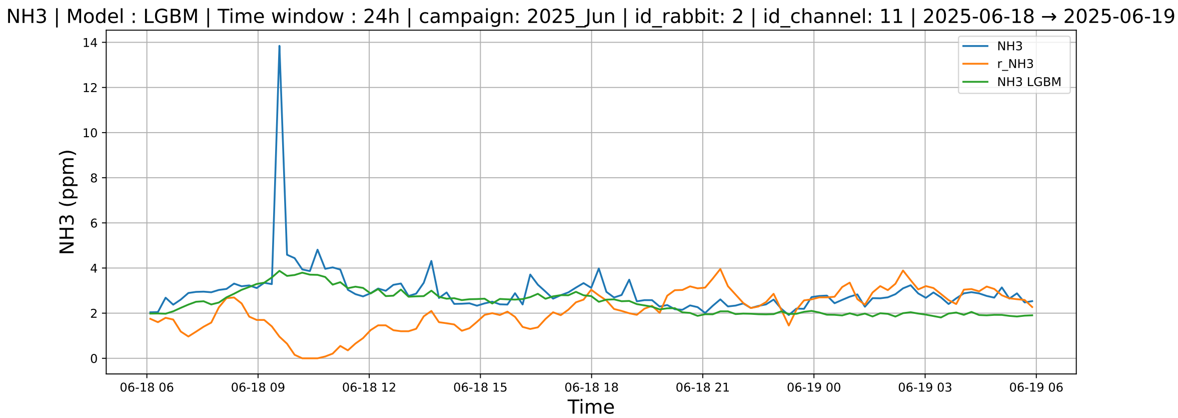
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19

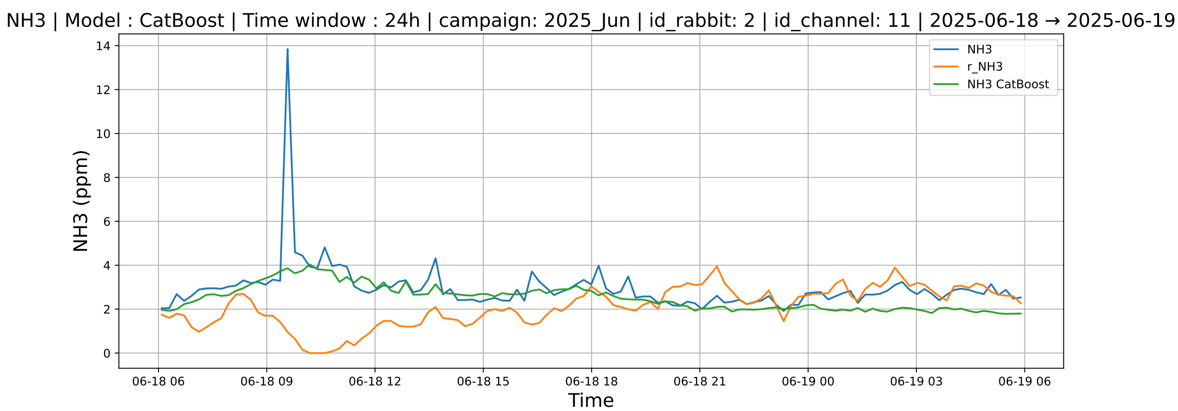


4.1.2 Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11

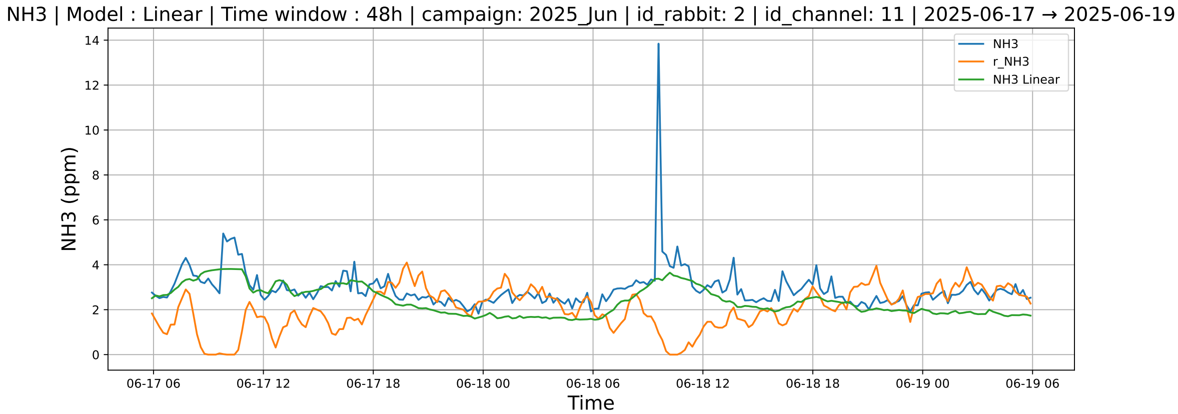
4.1.2.1 Time window : 24

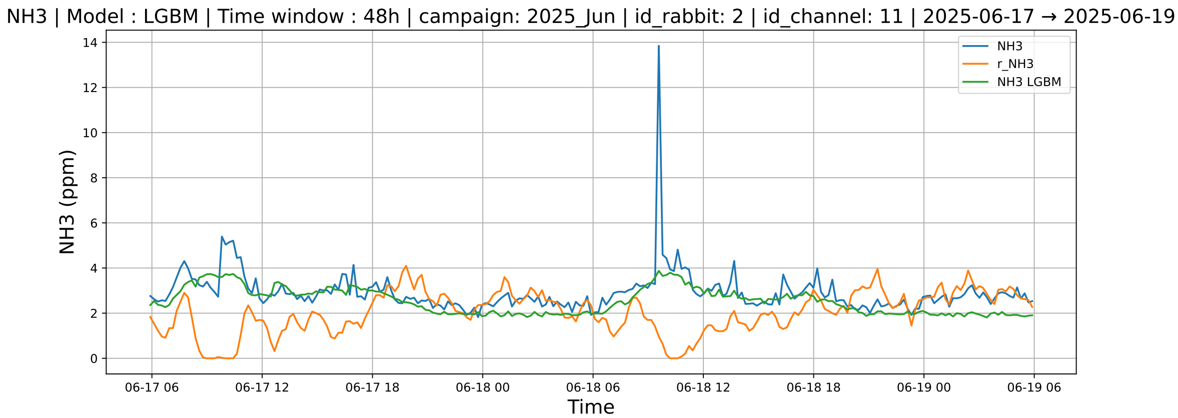


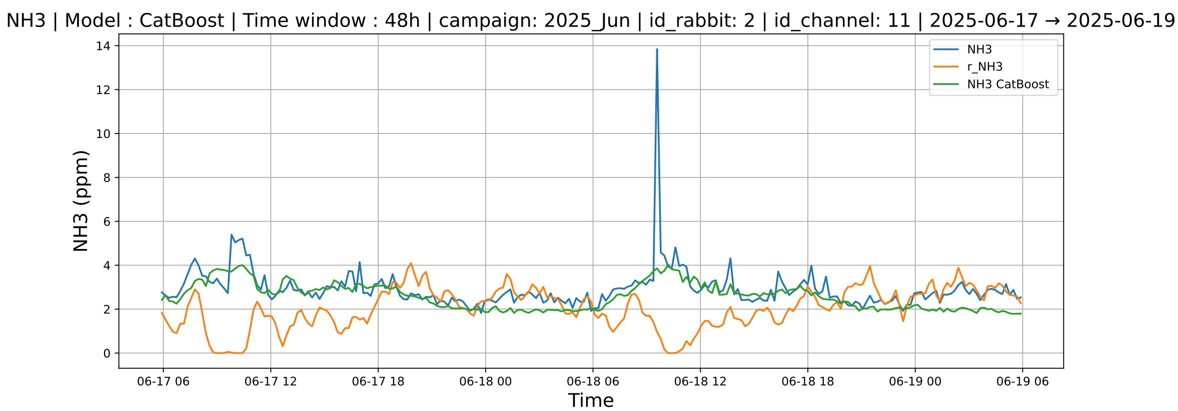




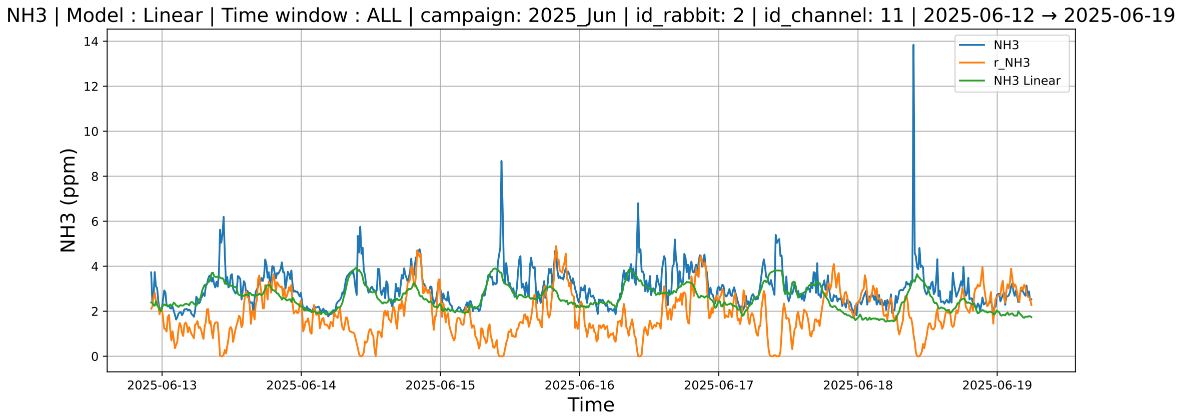
4.1.2.2 Time window : 48

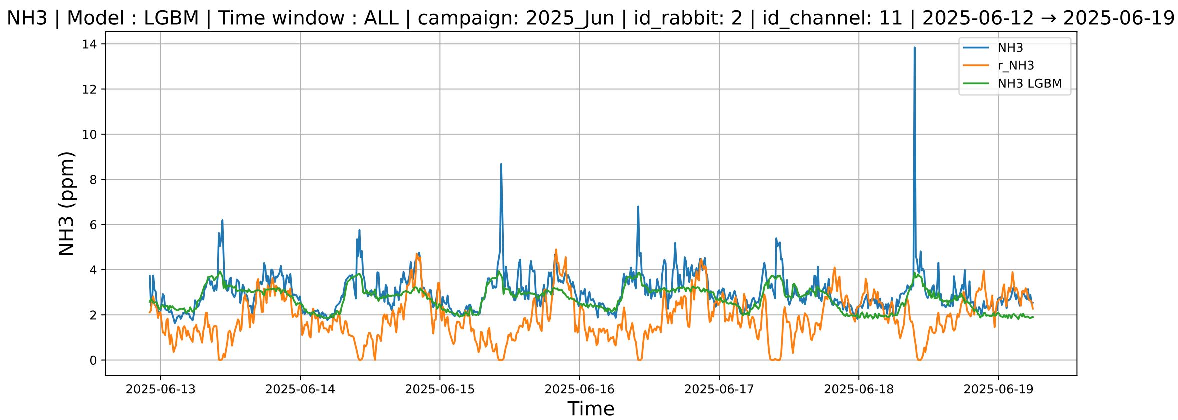


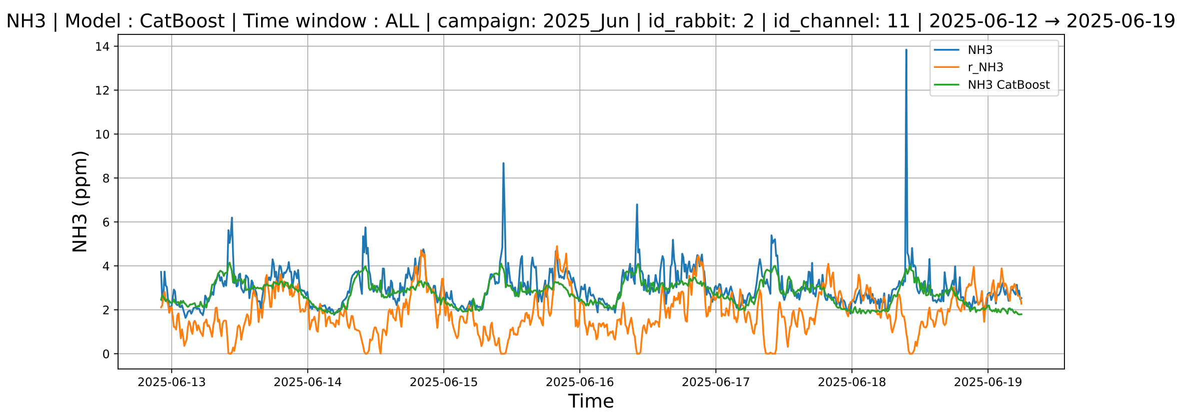




4.1.2.3 Time window : ALL

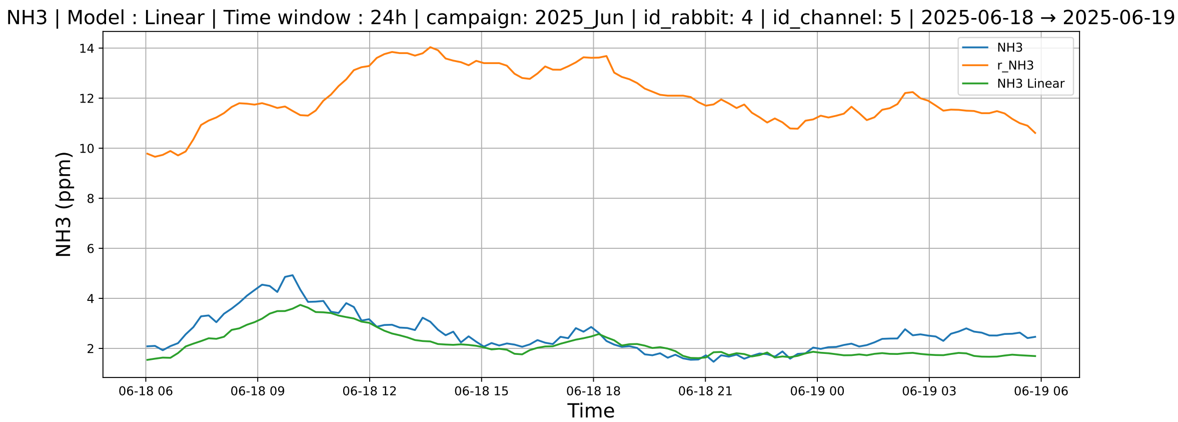


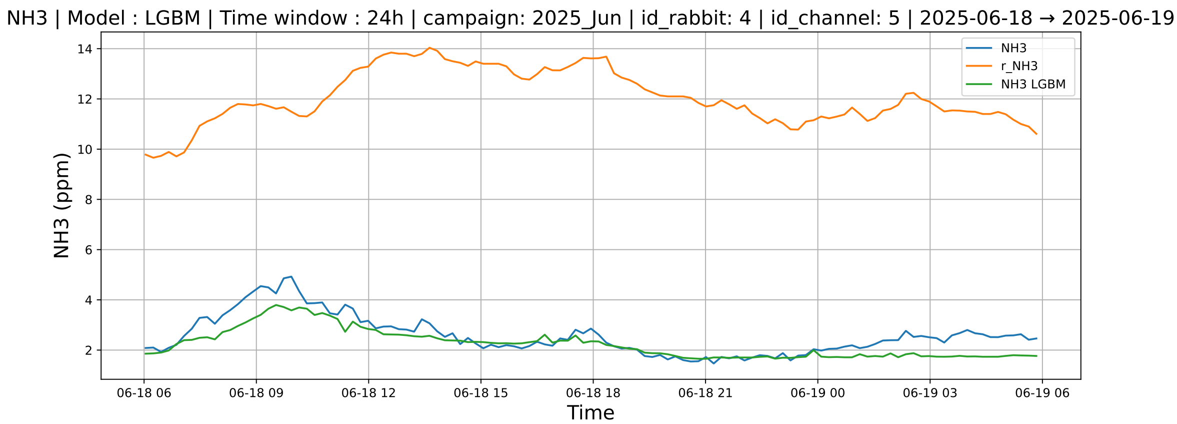


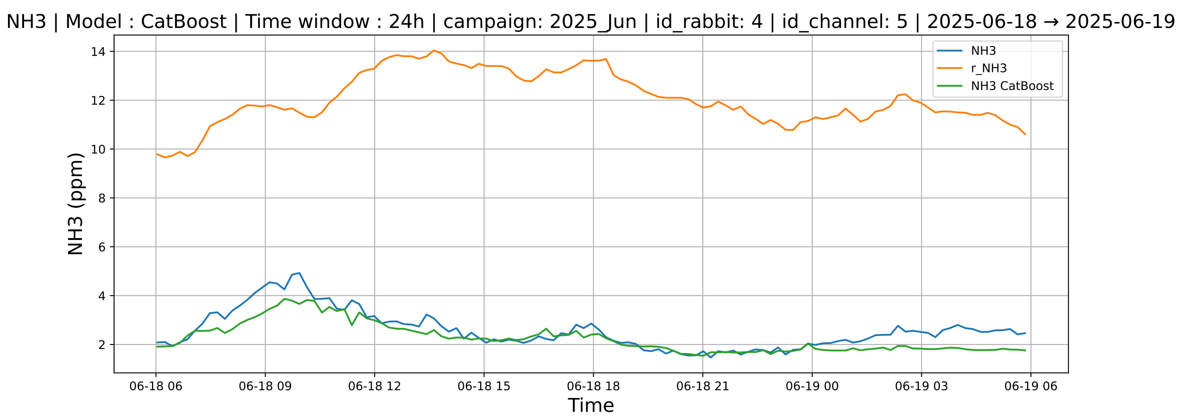


4.1.3 Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5

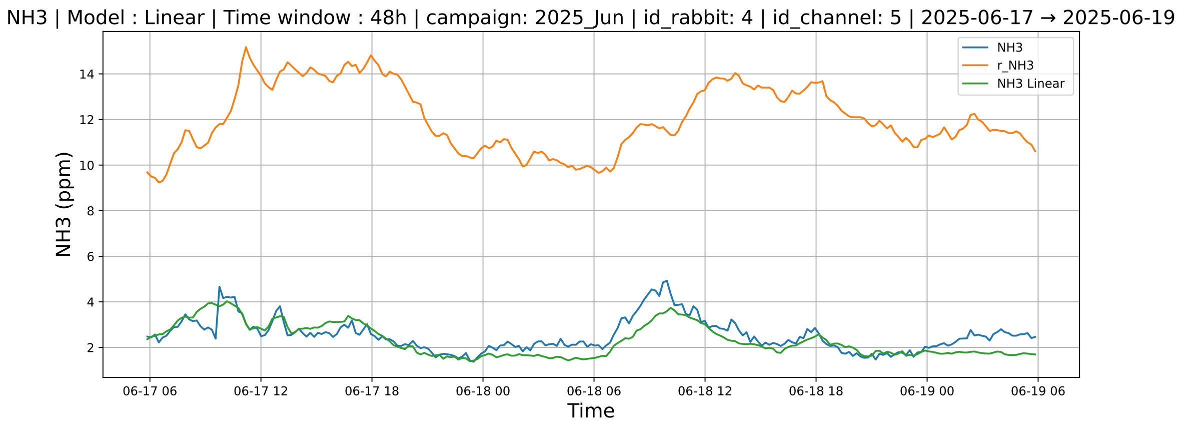
4.1.3.1 Time window : 24

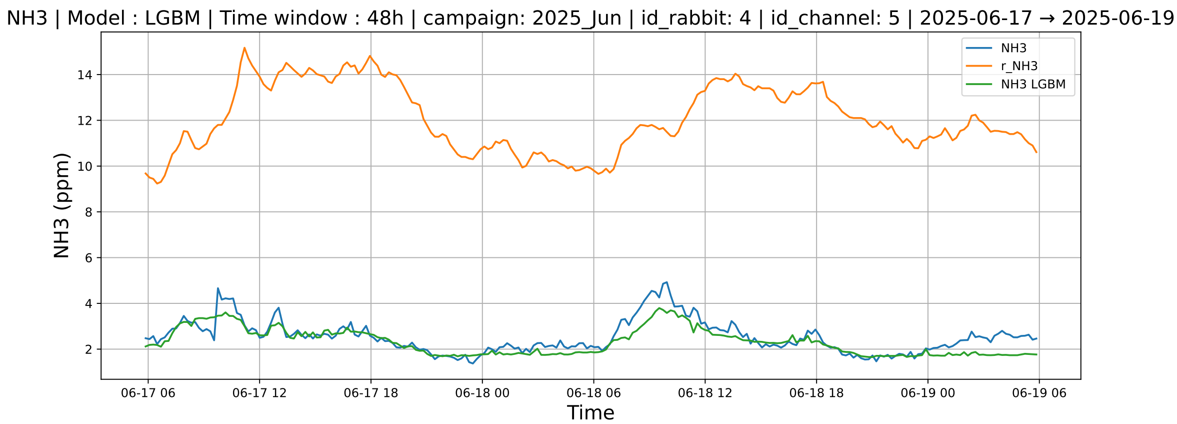


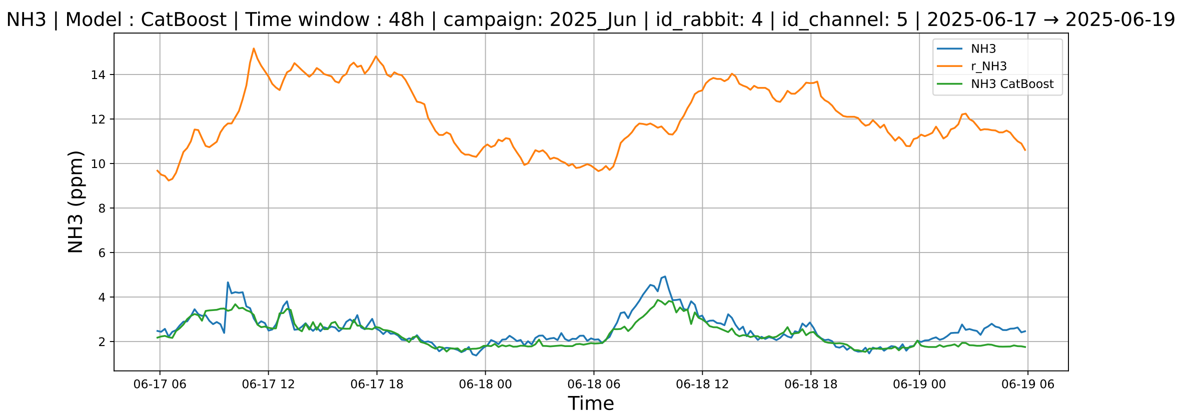




4.1.3.2 Time window : 48

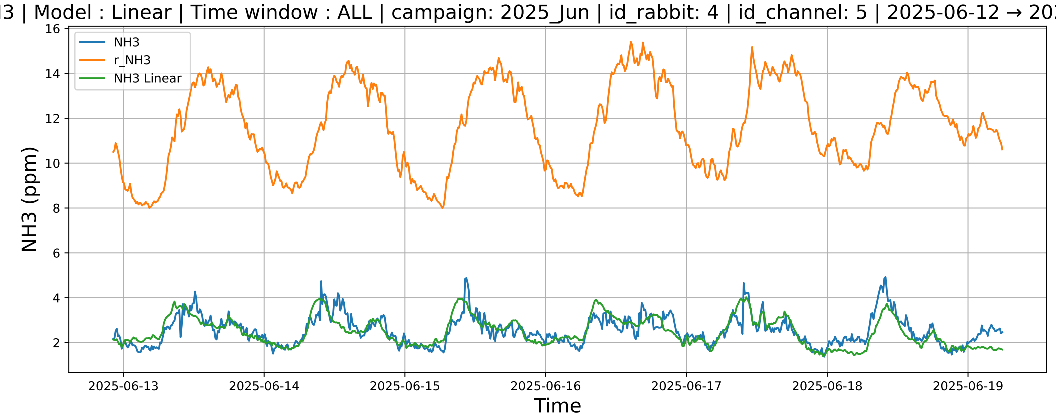




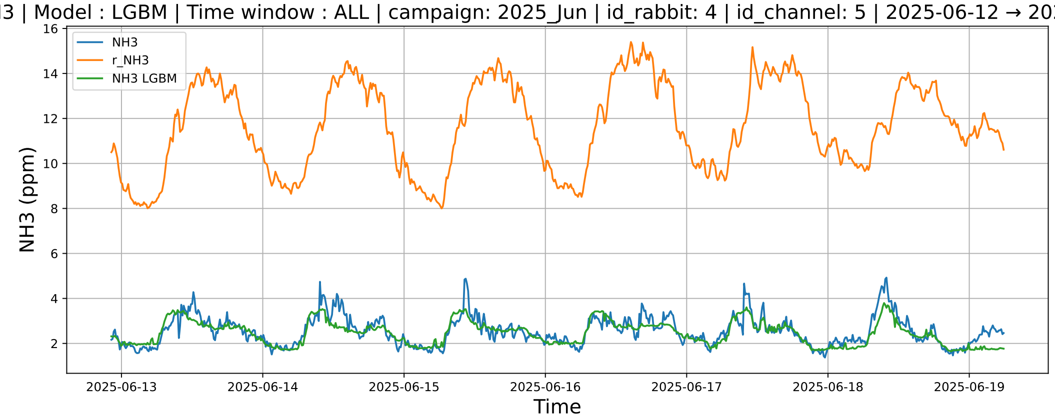


4.1.3.3 Time window : ALL

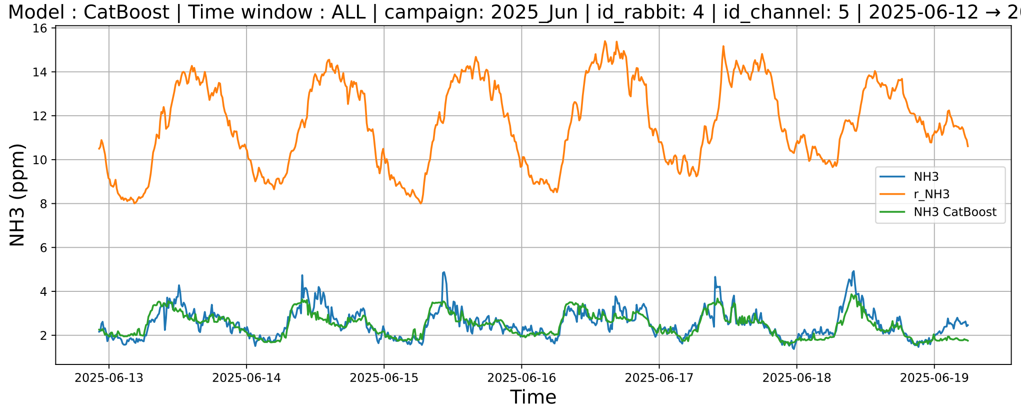
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19

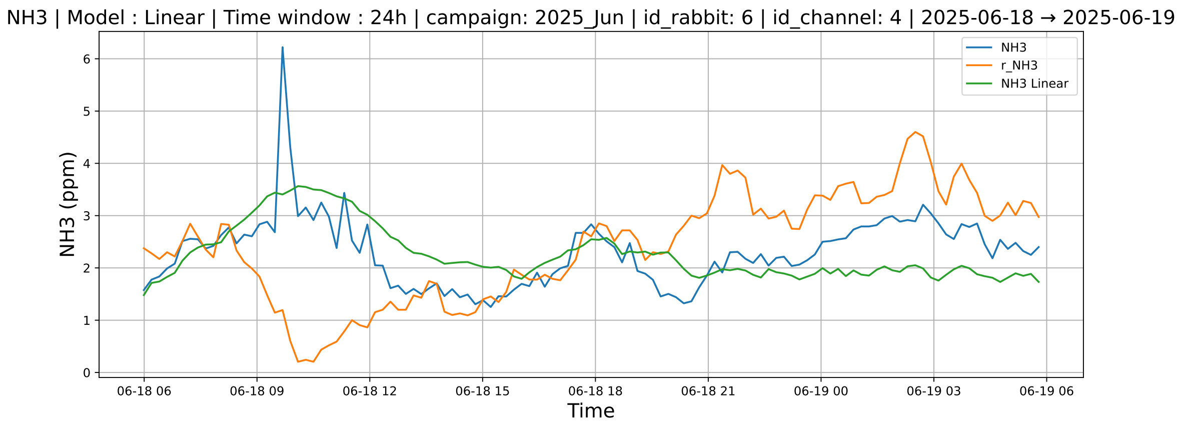


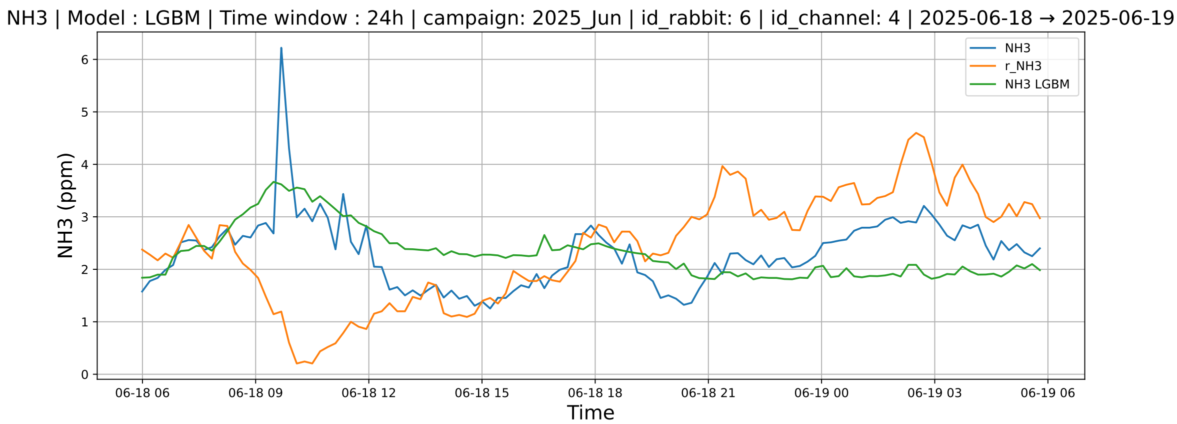
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19

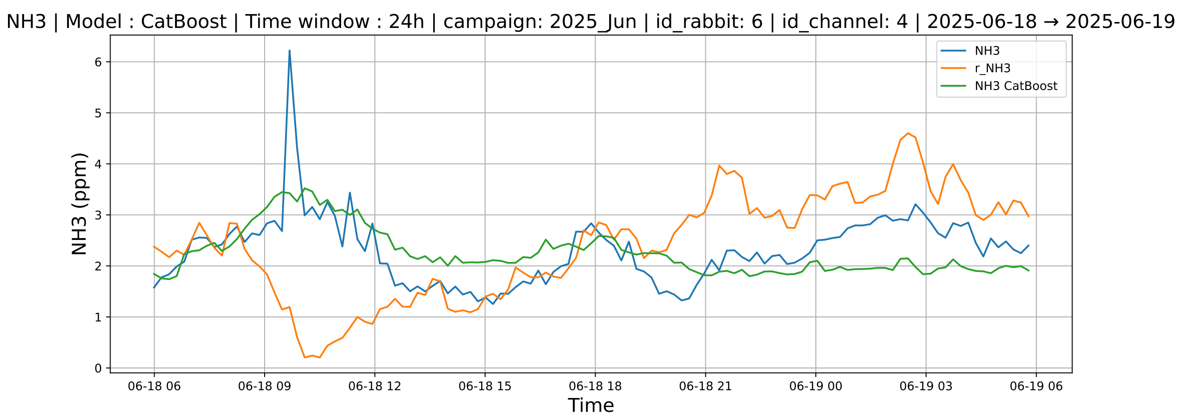


4.1.4 Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4

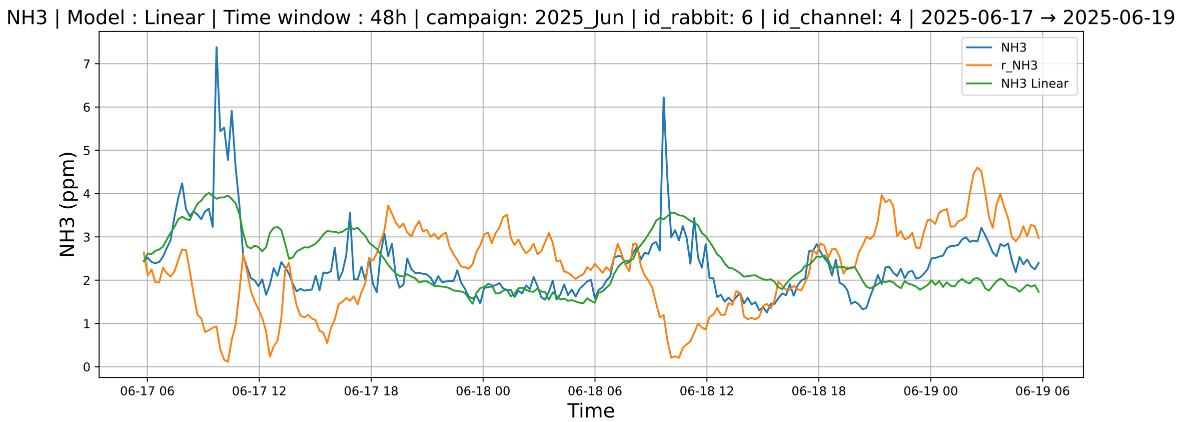
4.1.4.1 Time window : 24

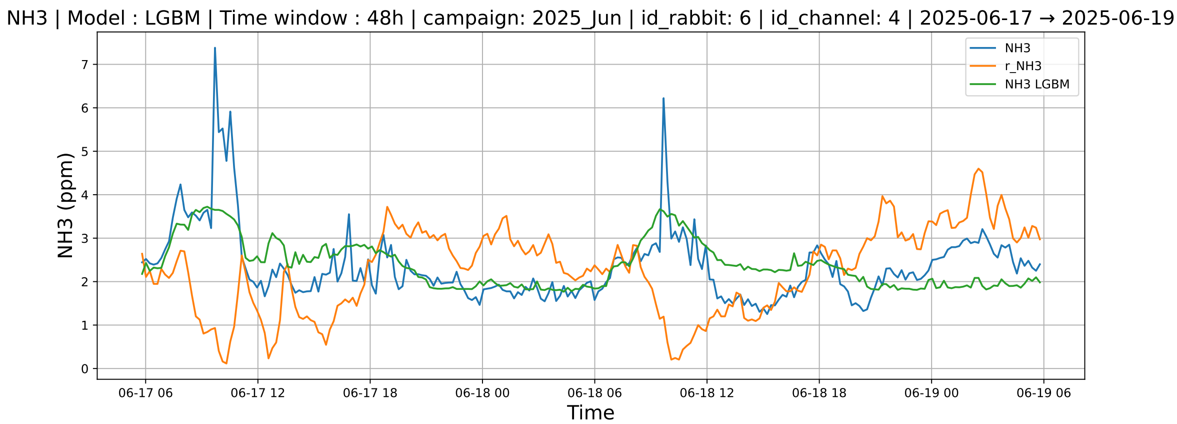


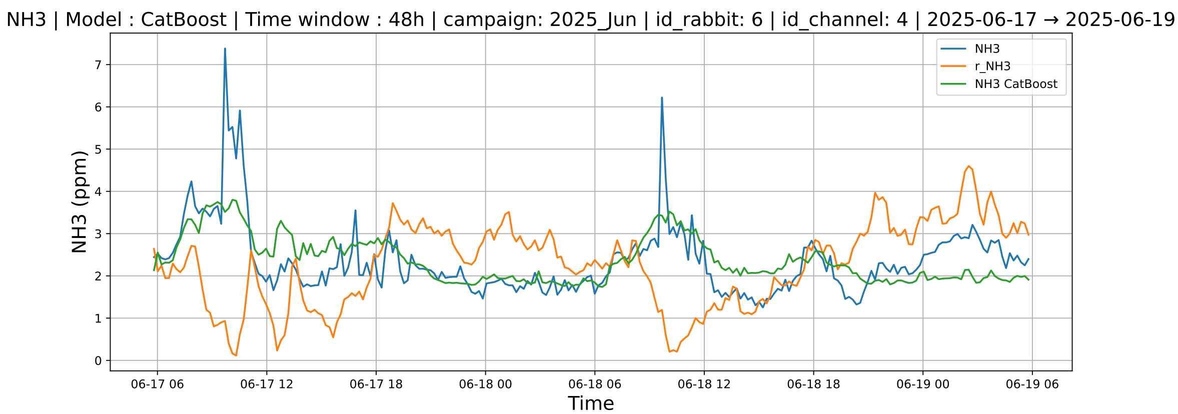




4.1.4.2 Time window : 48

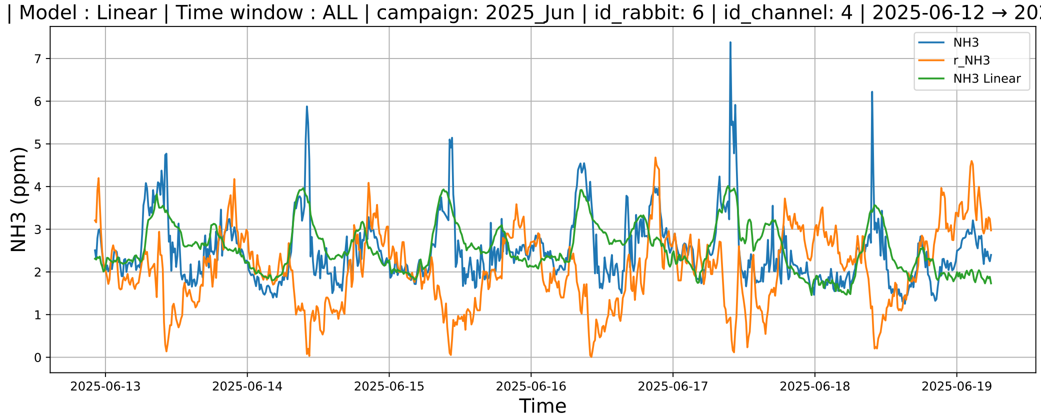




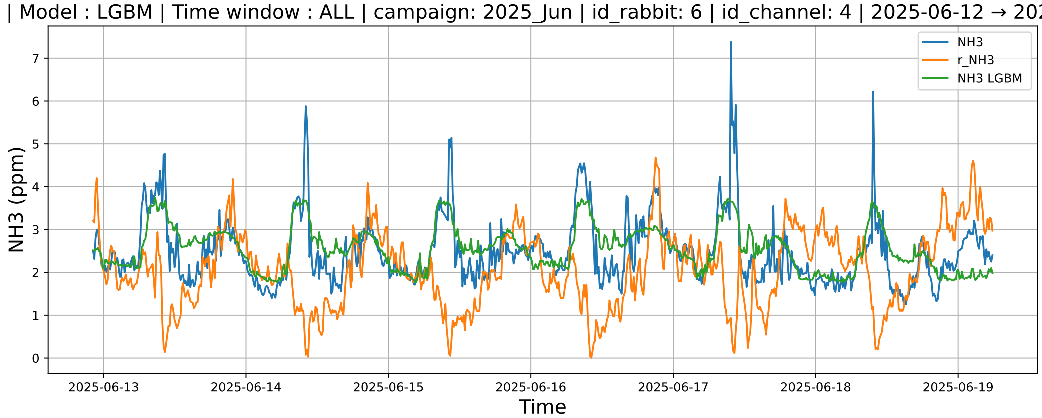


4.1.4.3 Time window : ALL

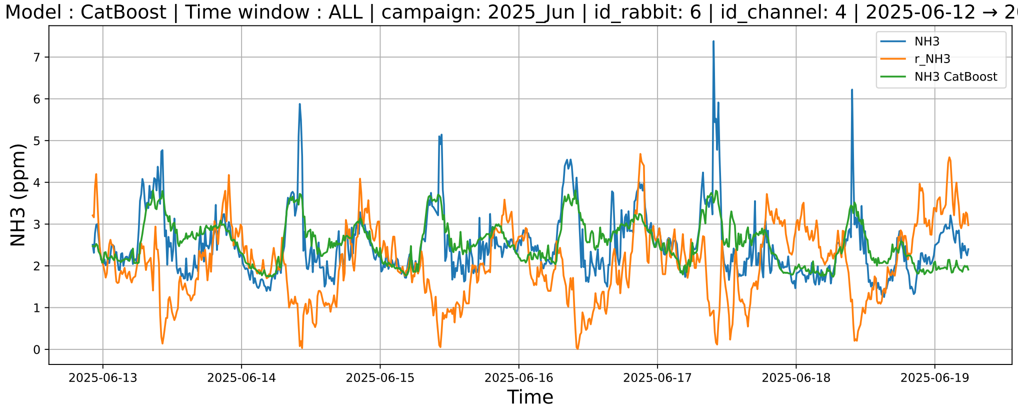
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



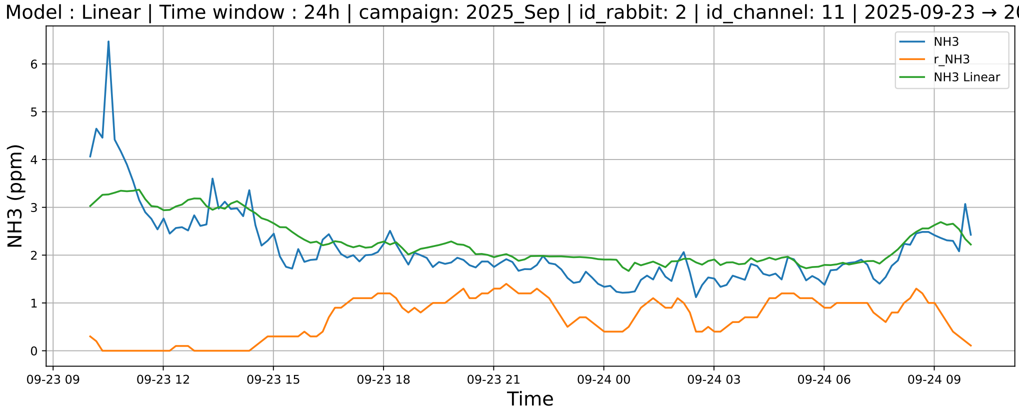
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



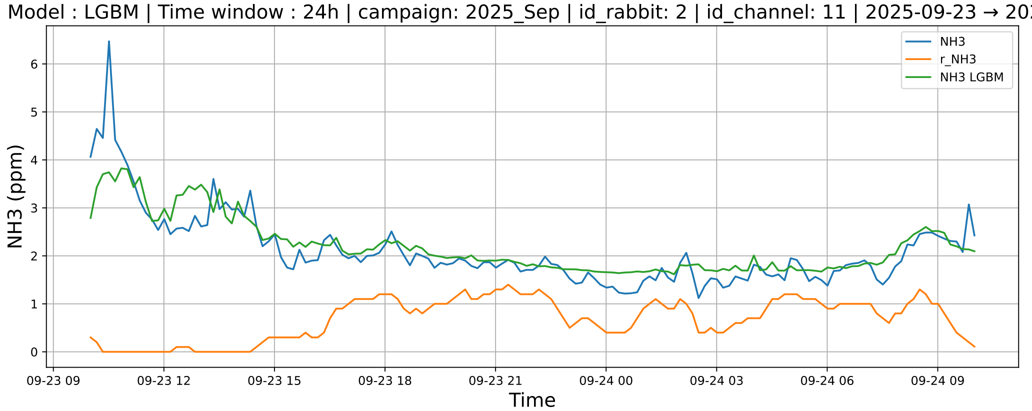
4.1.5 Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11

4.1.5.1 Time window : 24

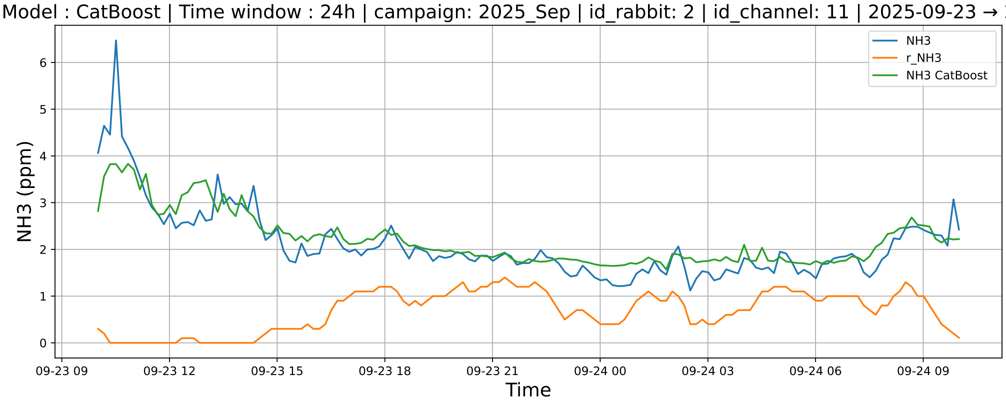
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24

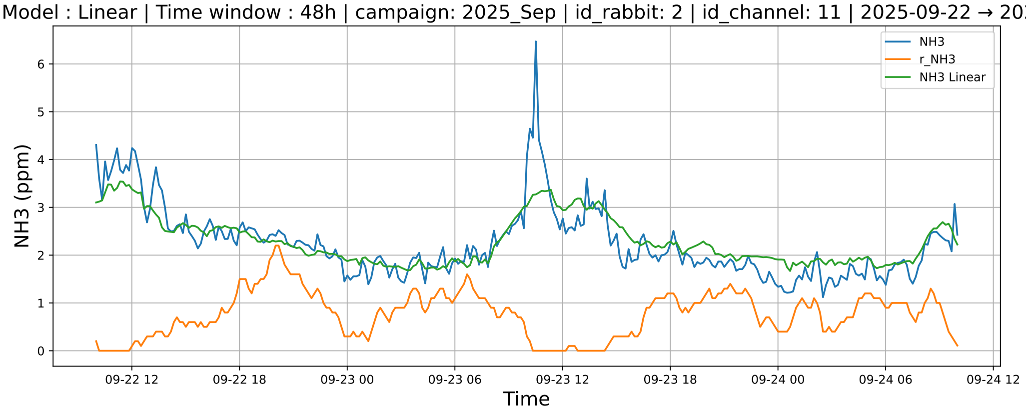


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24

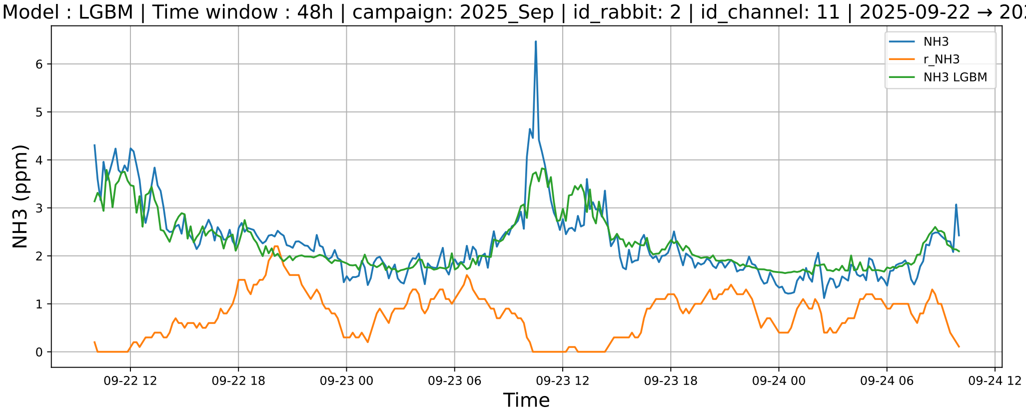


4.1.5.2 Time window : 48

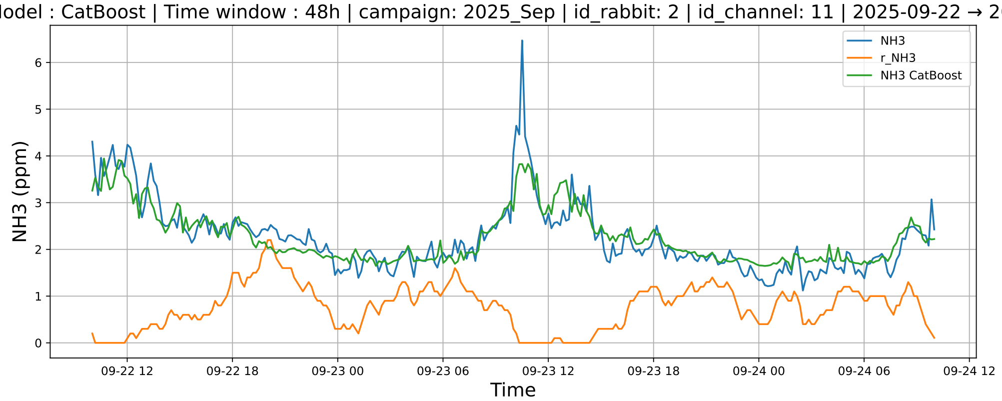
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24

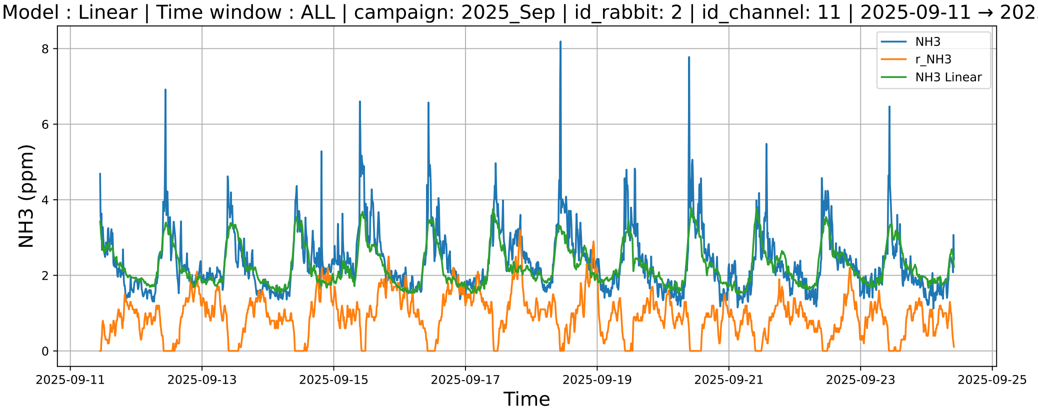


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24

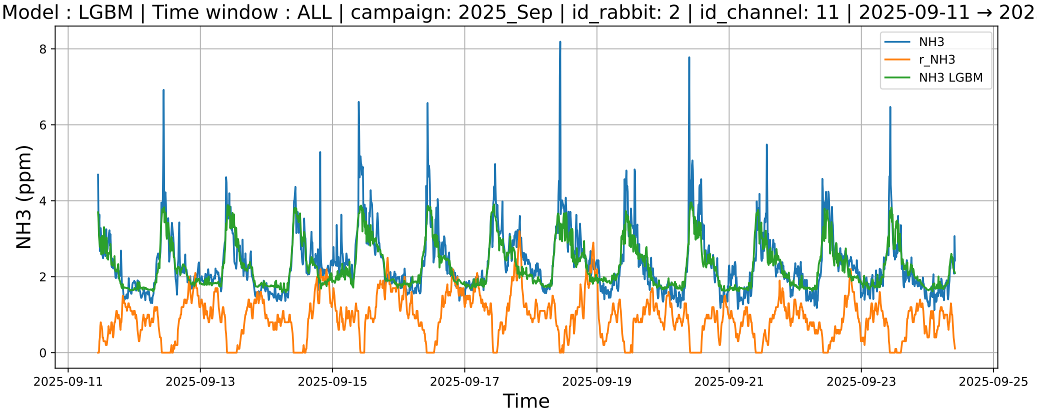


4.1.5.3 Time window : ALL

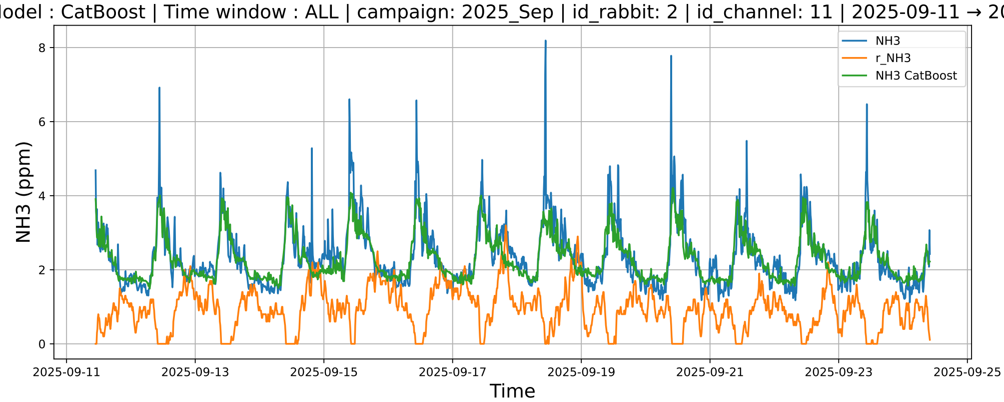
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



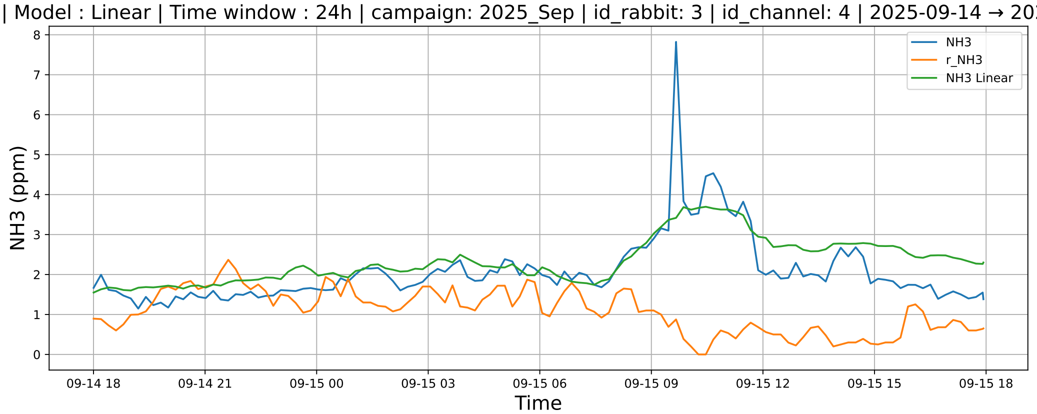
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



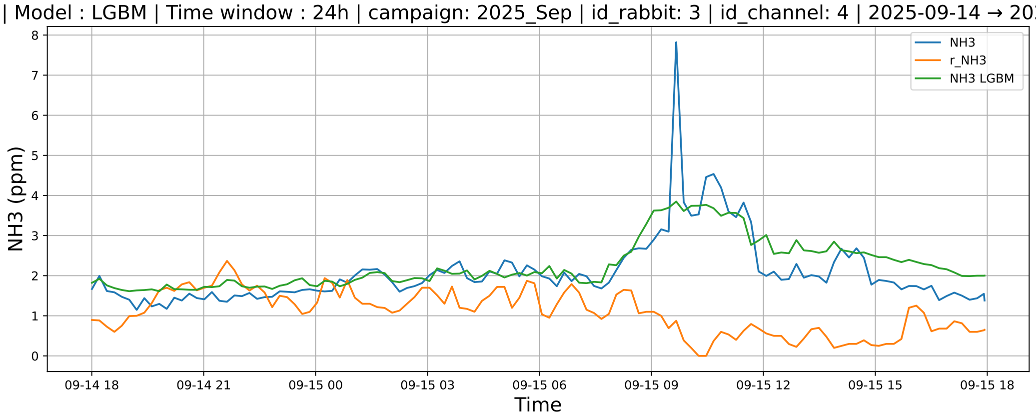
4.1.6 Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4

4.1.6.1 Time window : 24

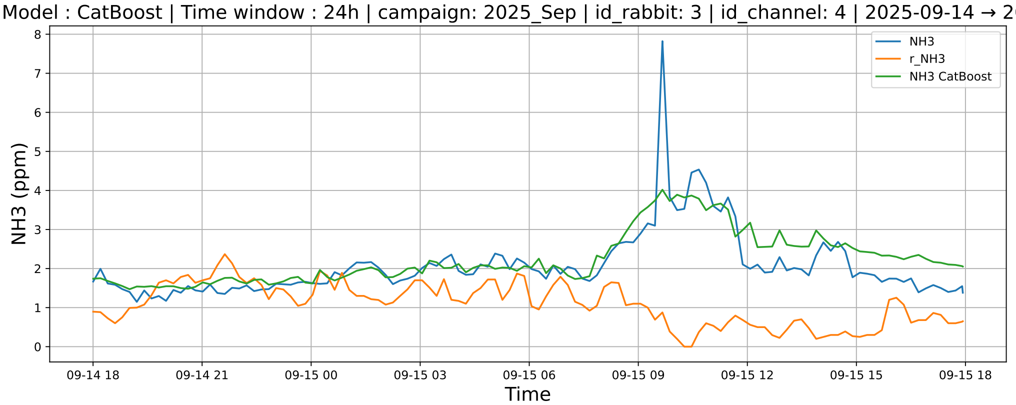
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15



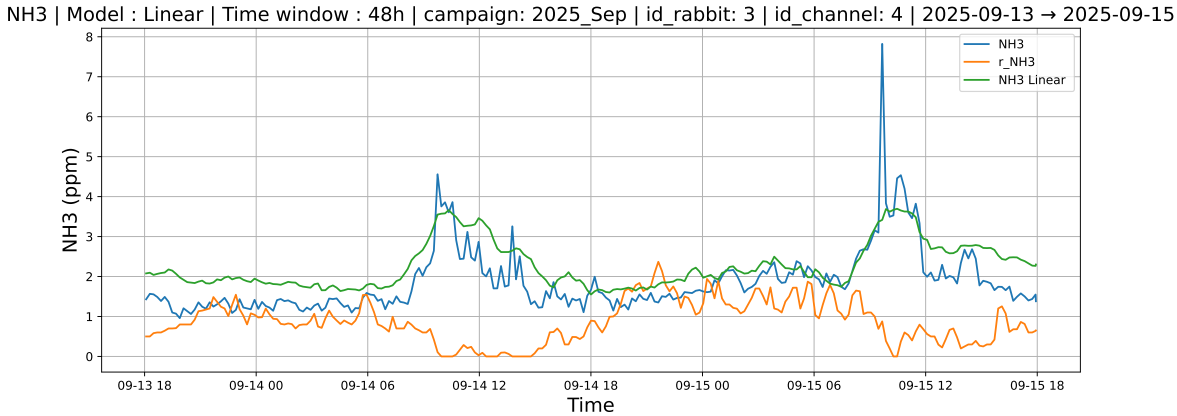
NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15

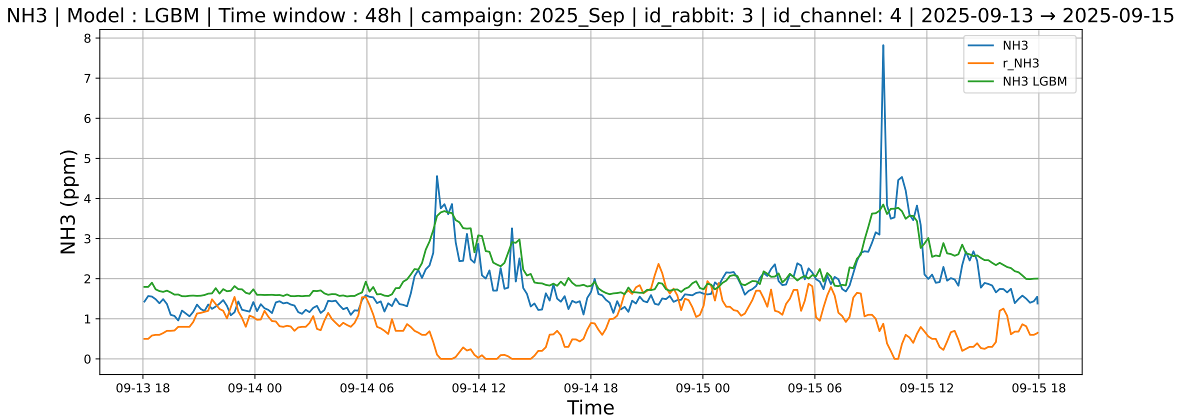


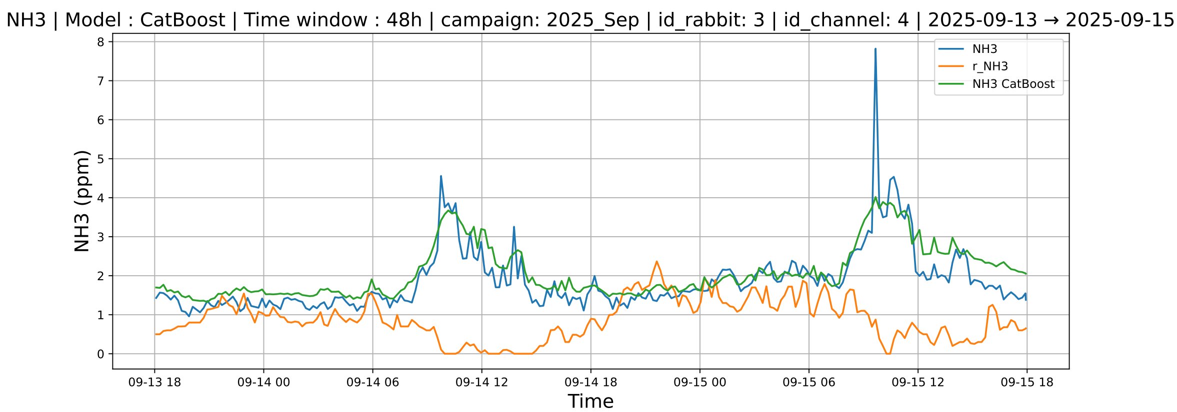
NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15



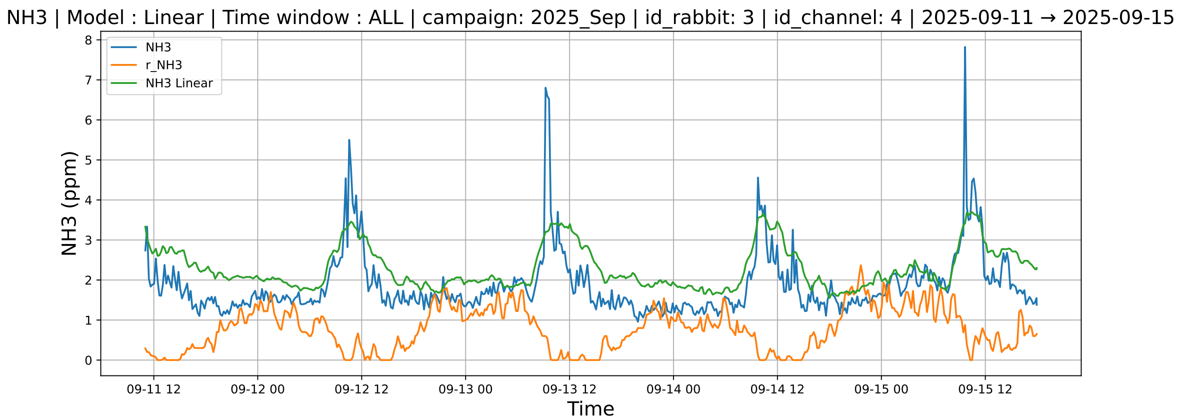
4.1.6.2 Time window : 48

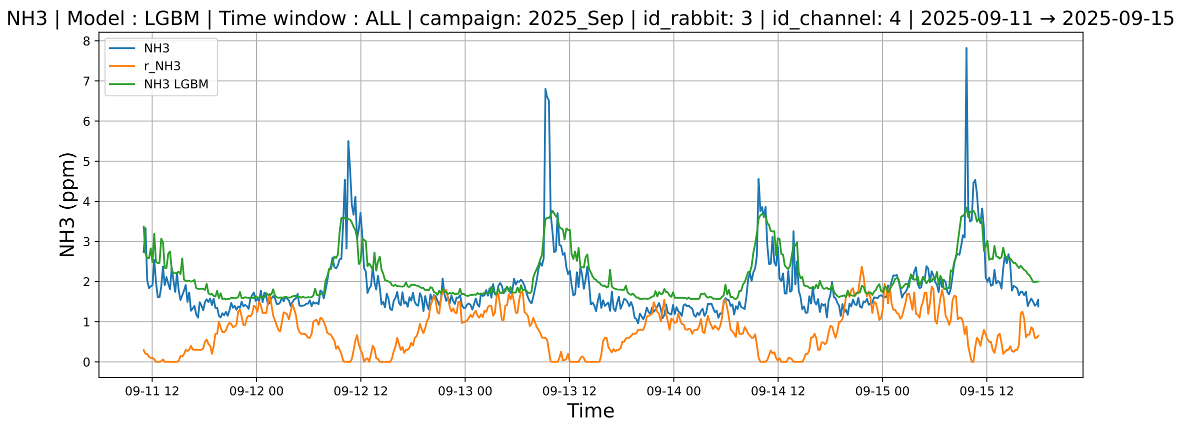


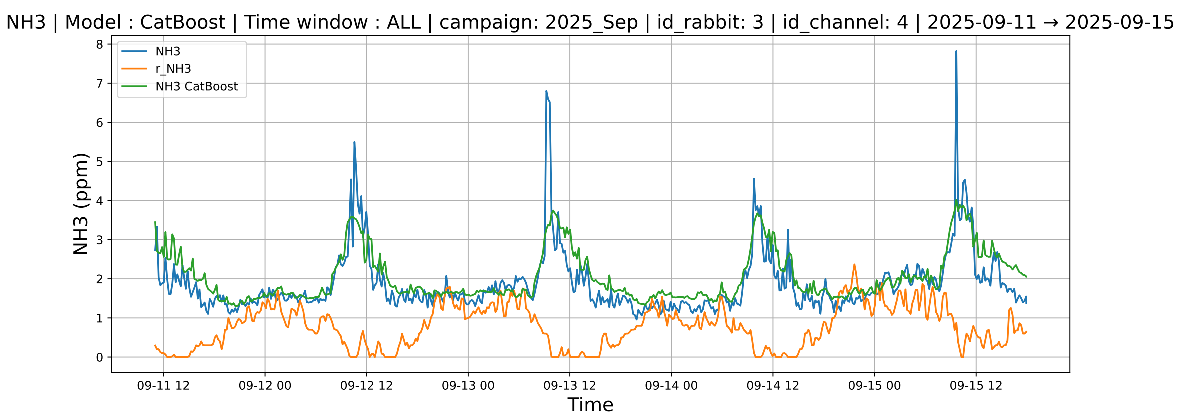




4.1.6.3 Time window : ALL

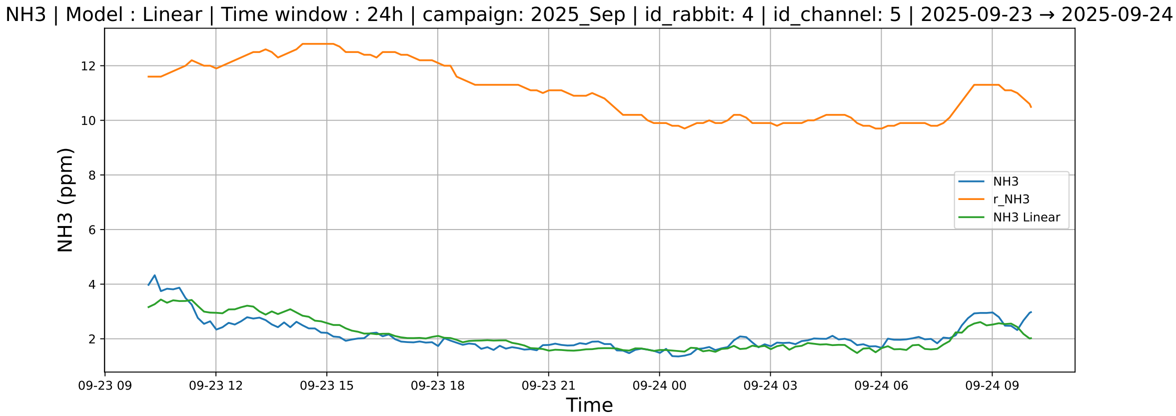


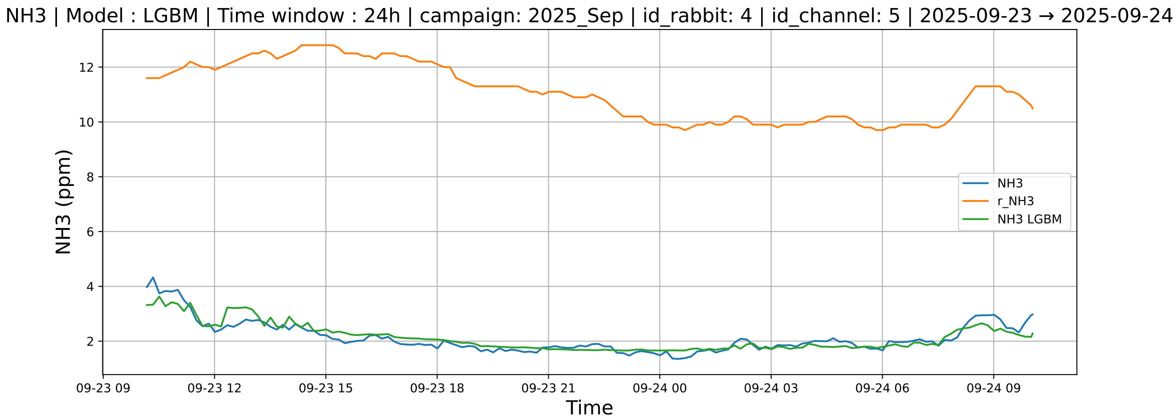


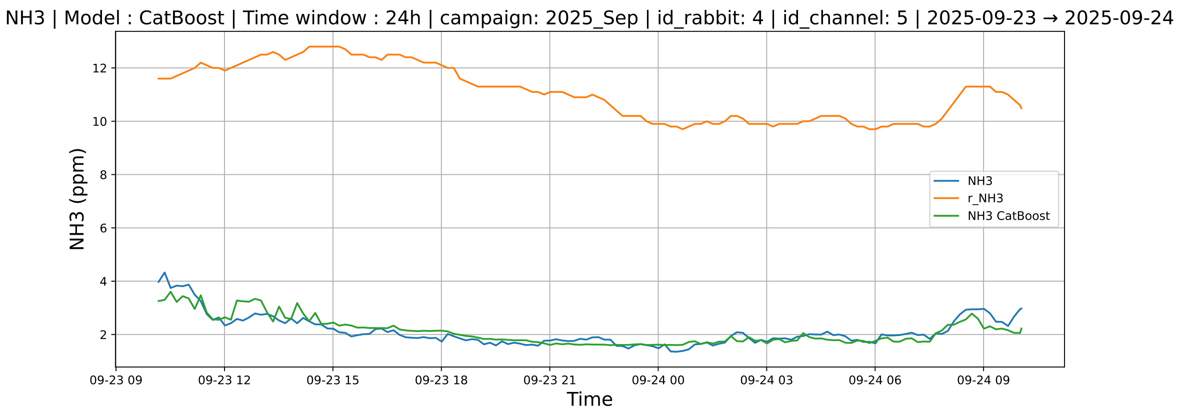


4.1.7 Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5

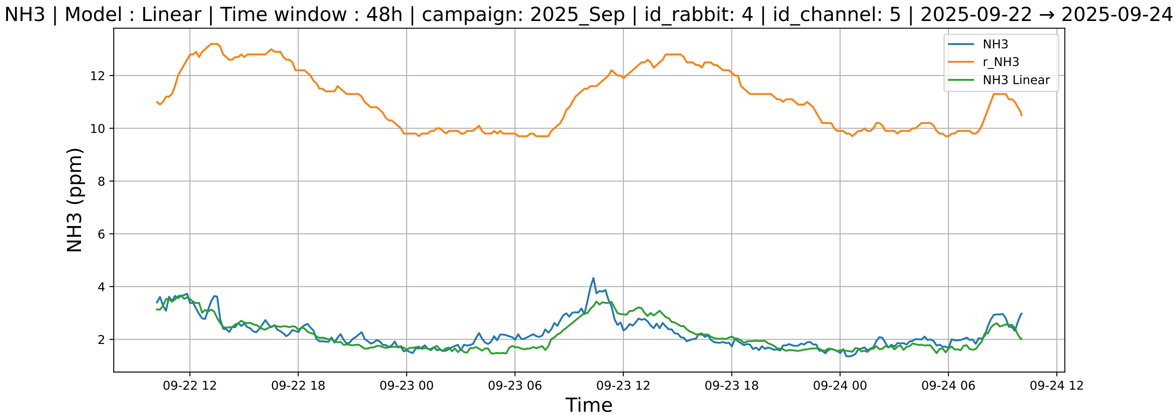
4.1.7.1 Time window : 24

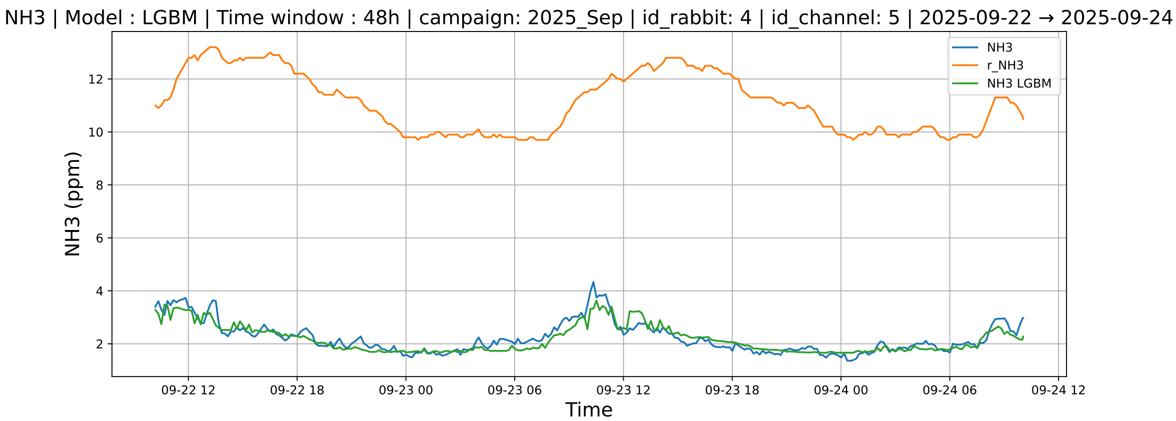


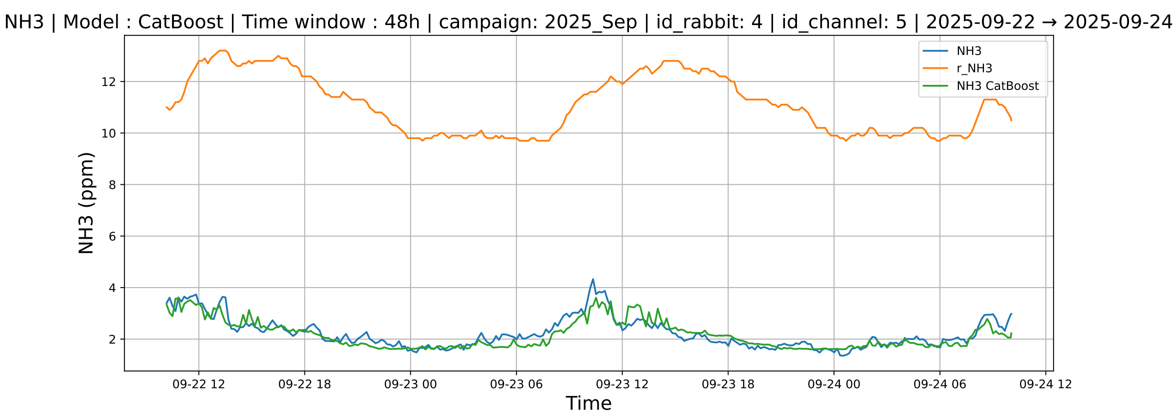




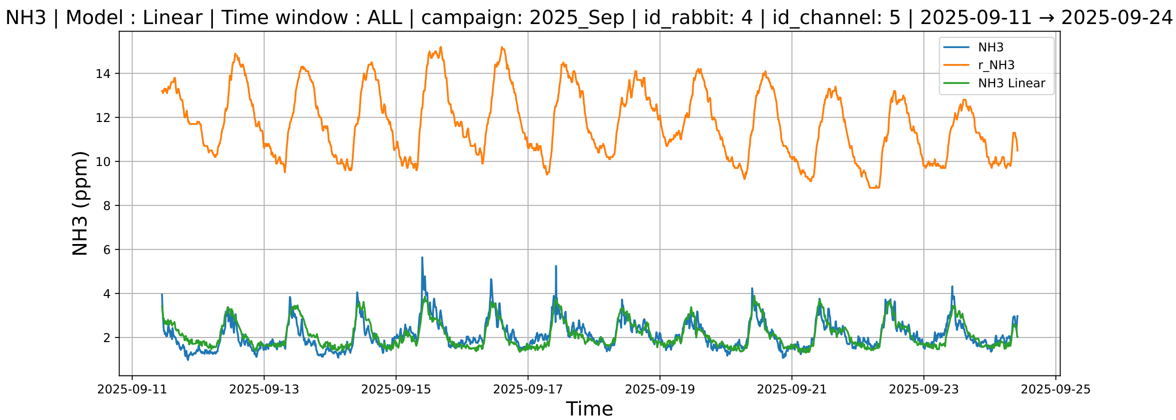
4.1.7.2 Time window : 48

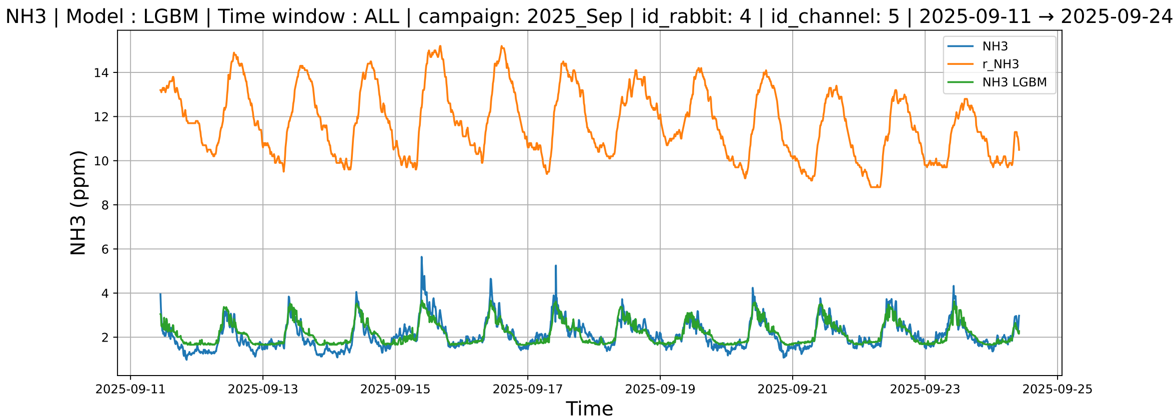


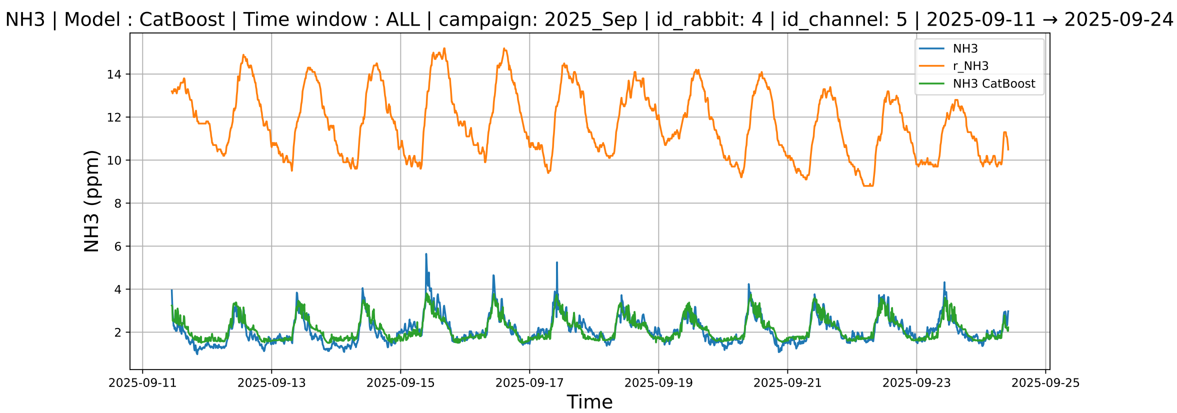




4.1.7.3 Time window : ALL



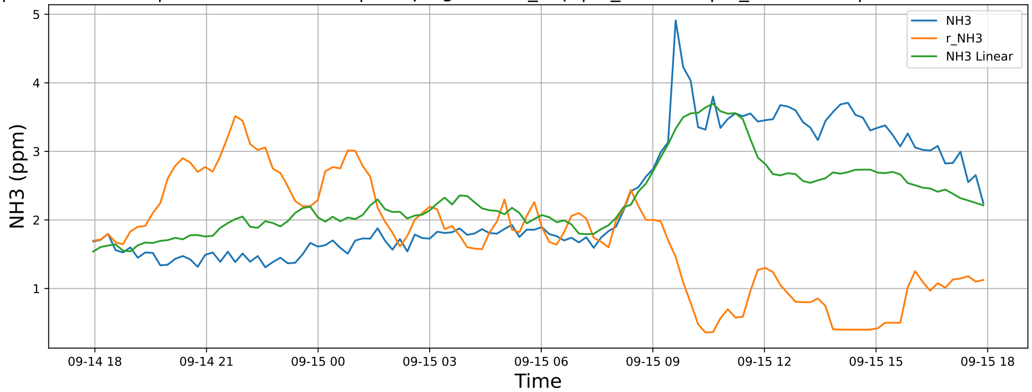




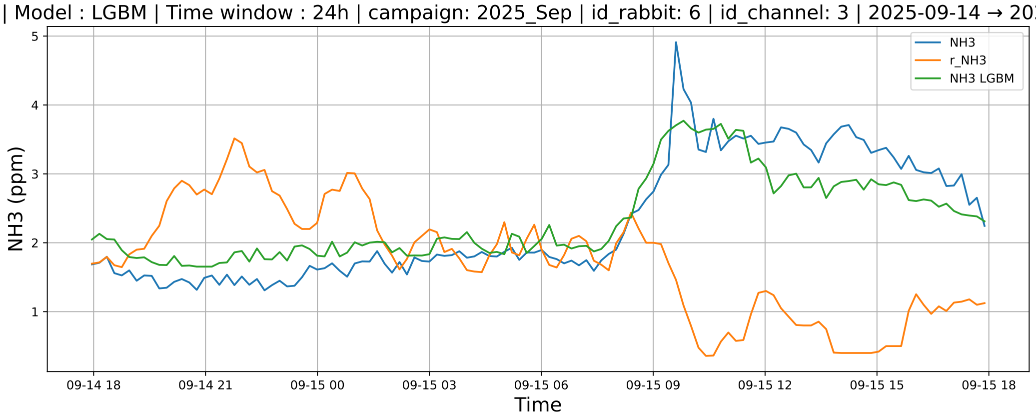
4.1.8 Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3

4.1.8.1 Time window : 24

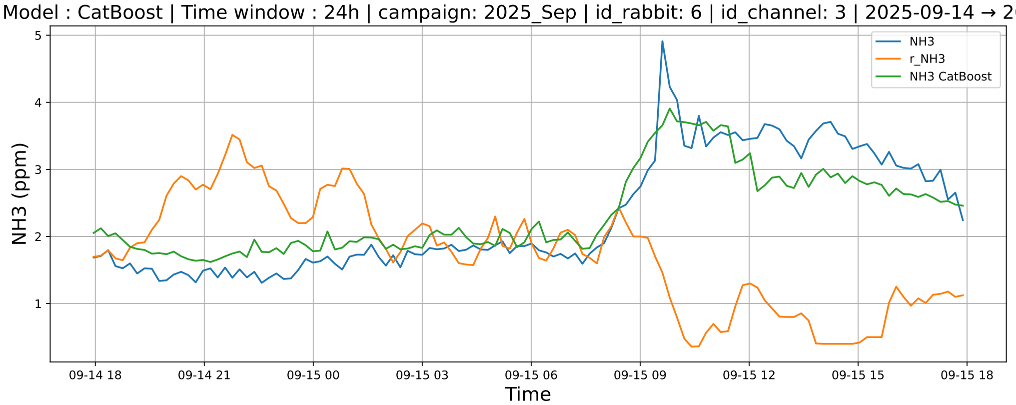
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15

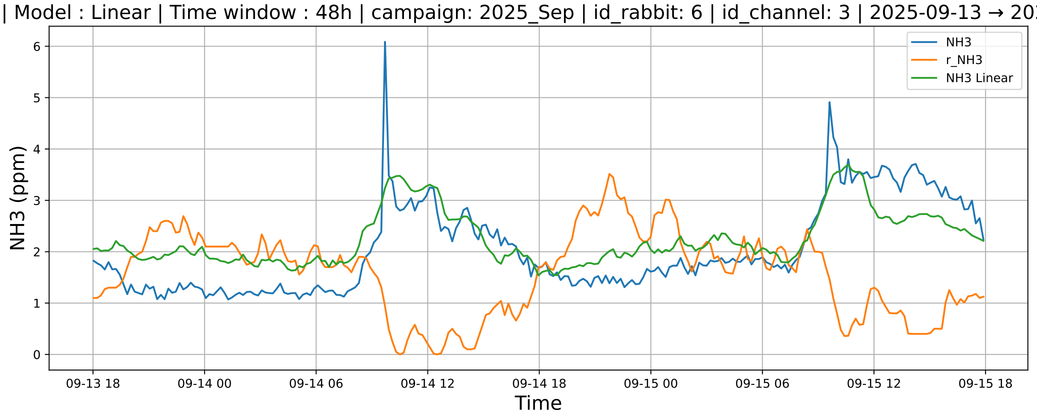


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15

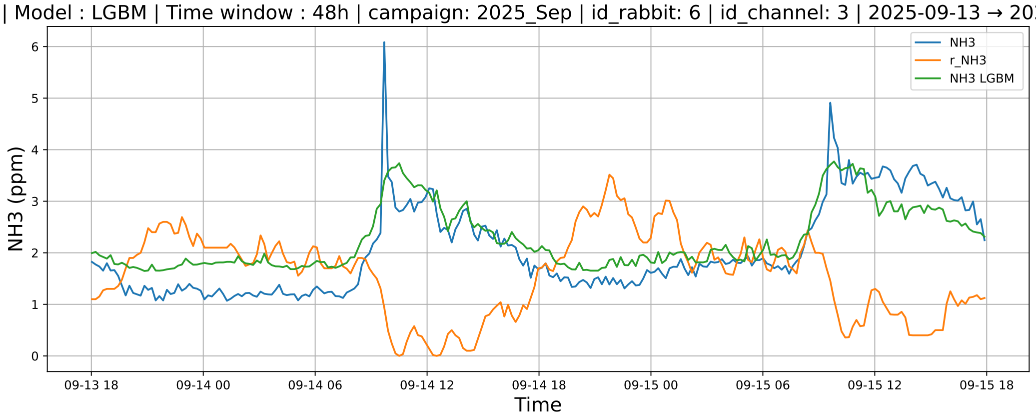


4.1.8.2 Time window : 48

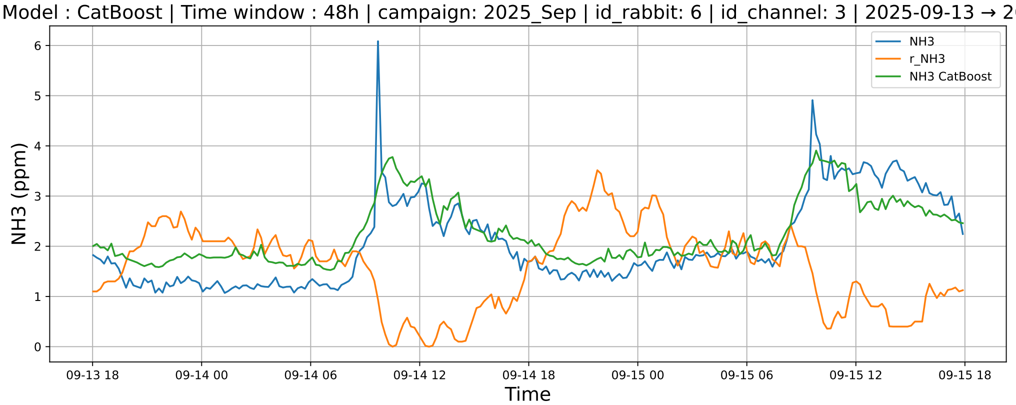
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15



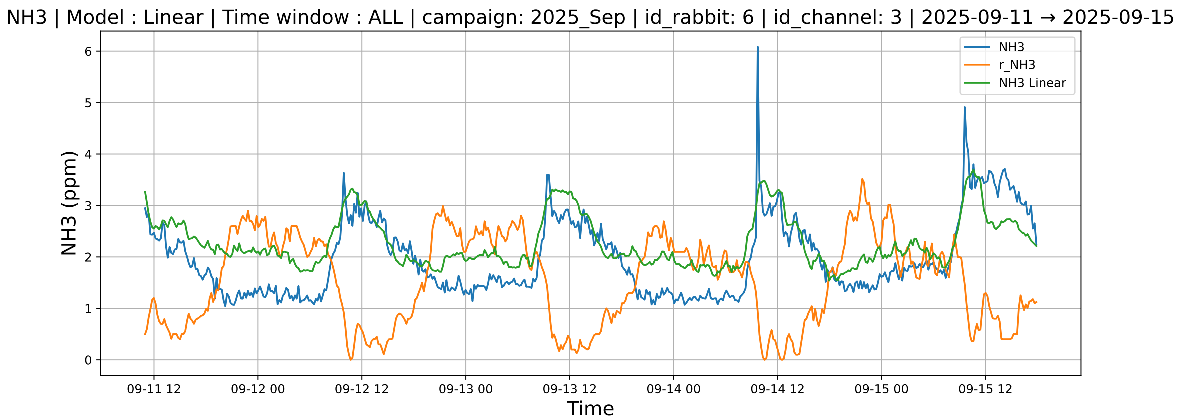
NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15

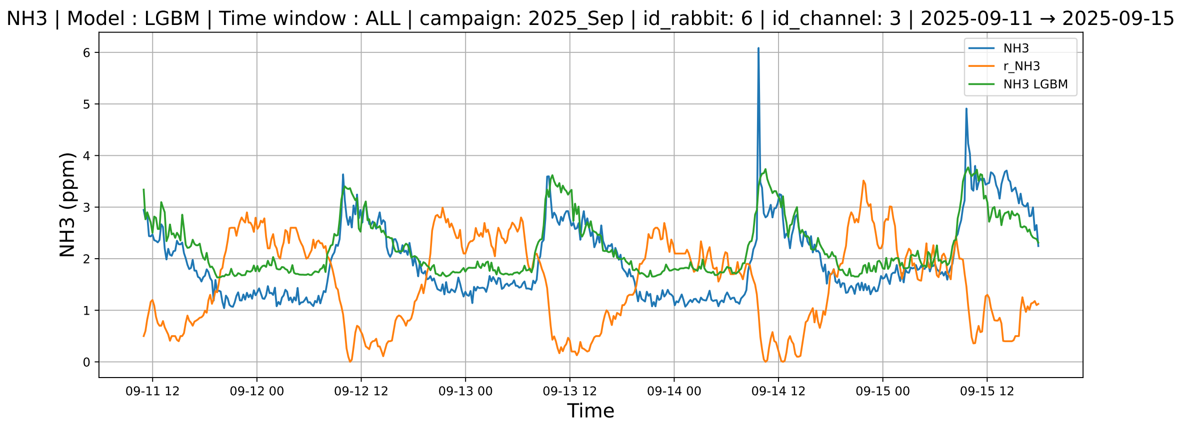


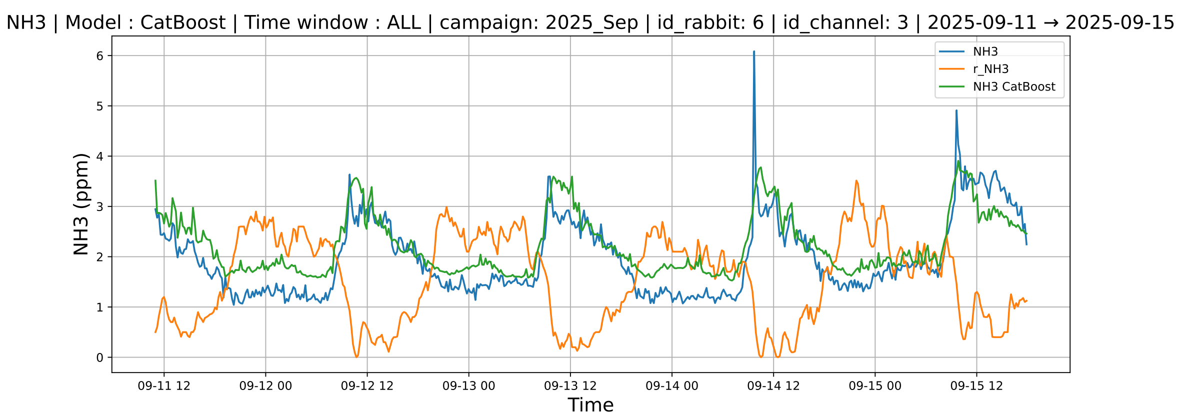
NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15



4.1.8.3 Time window : ALL





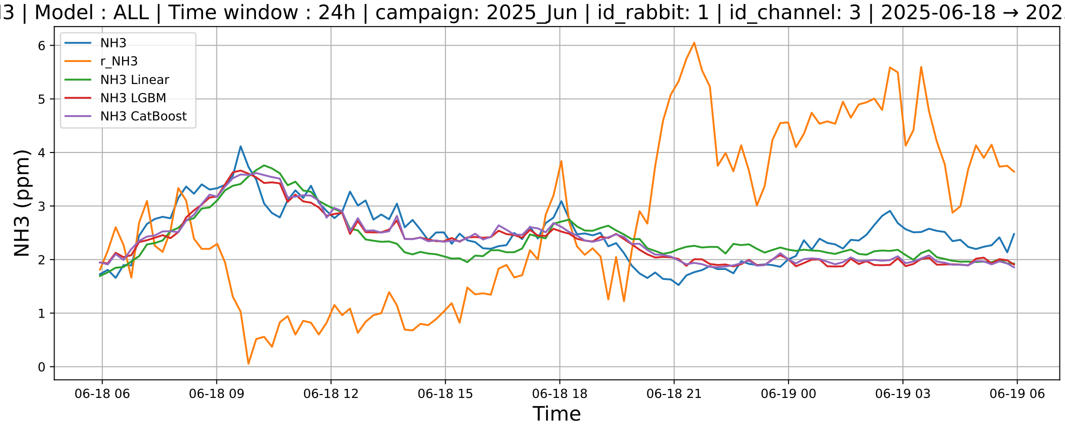


4.2 Compare plots

4.2.1 Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3

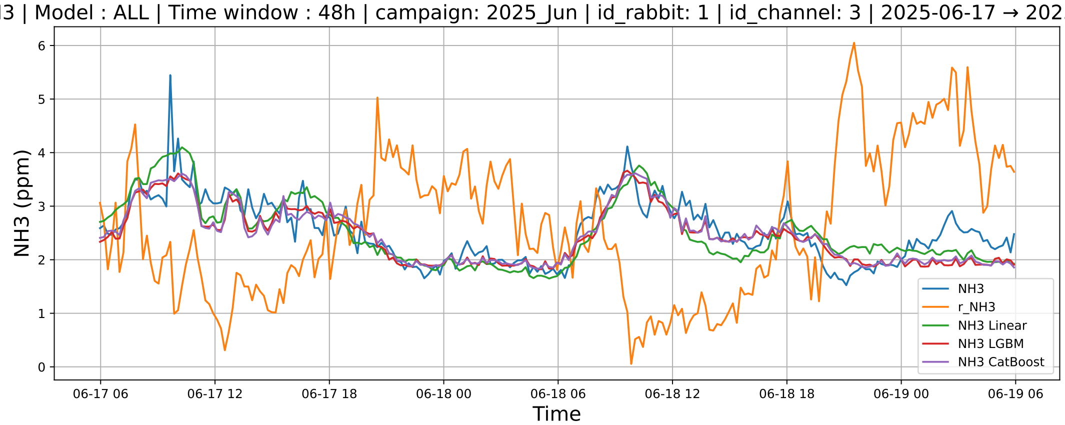
4.2.1.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19



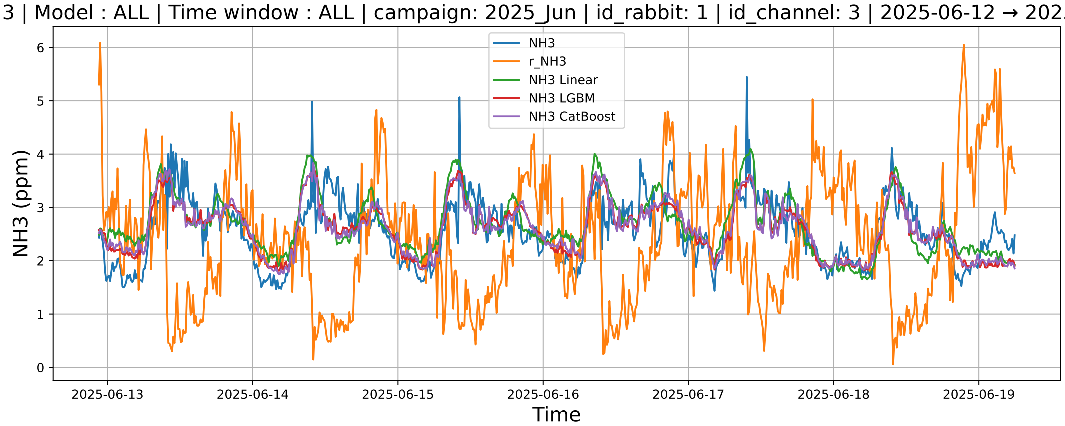
4.2.1.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19



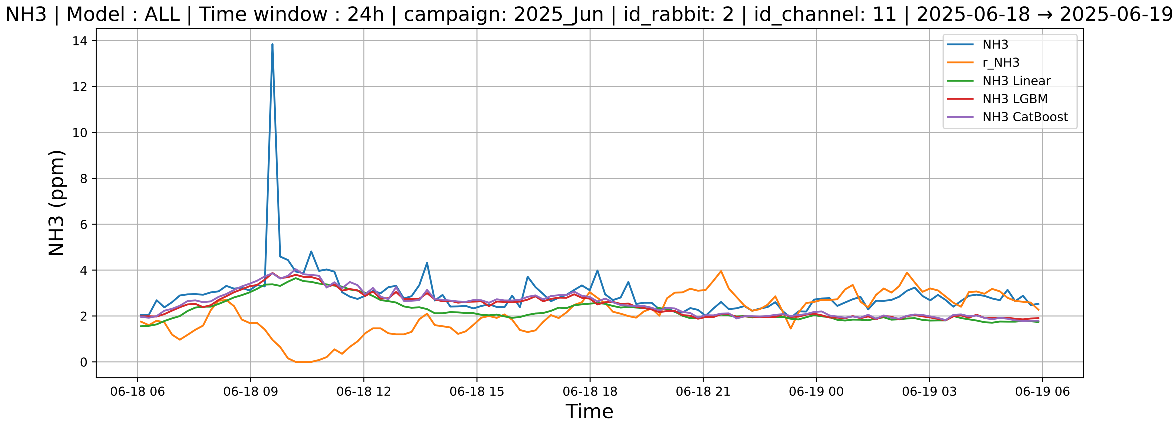
4.2.1.3 Time window : ALL

NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19

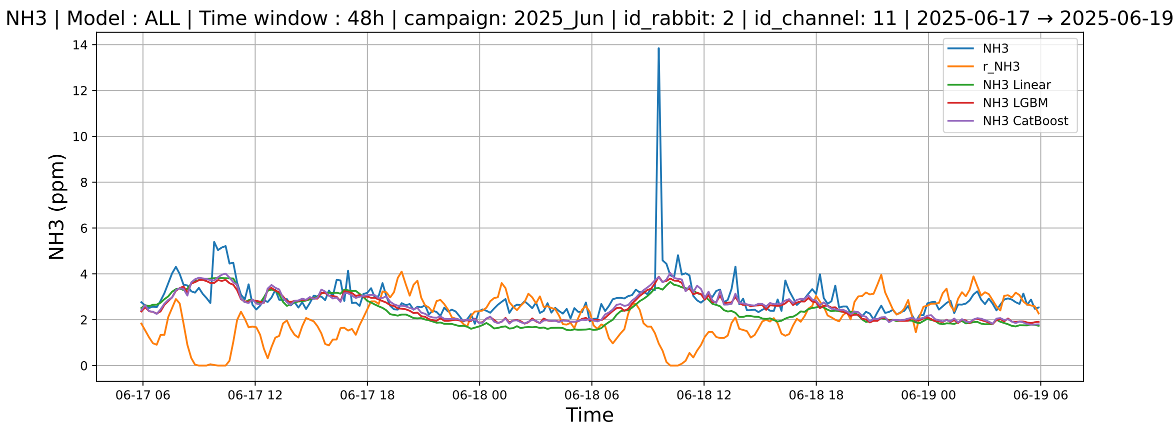


4.2.2 Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11

4.2.2.1 Time window : 24

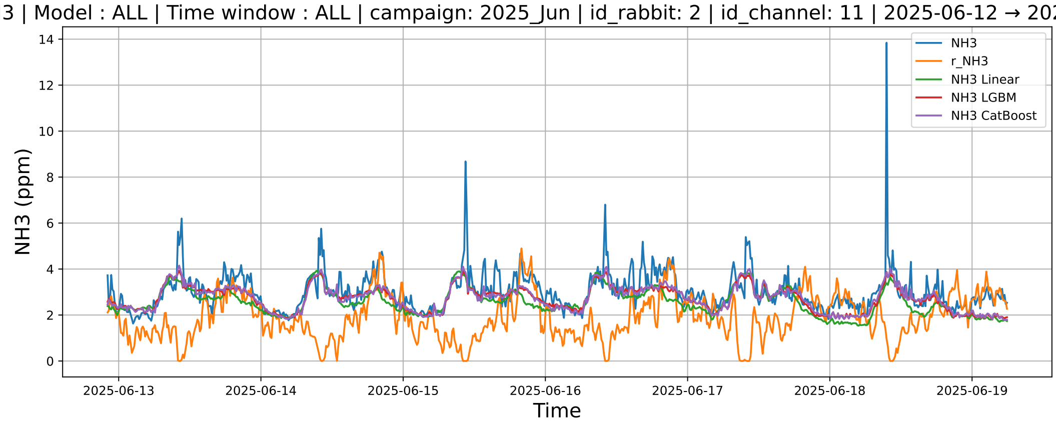


4.2.2.2 Time window : 48



4.2.2.3 Time window : ALL

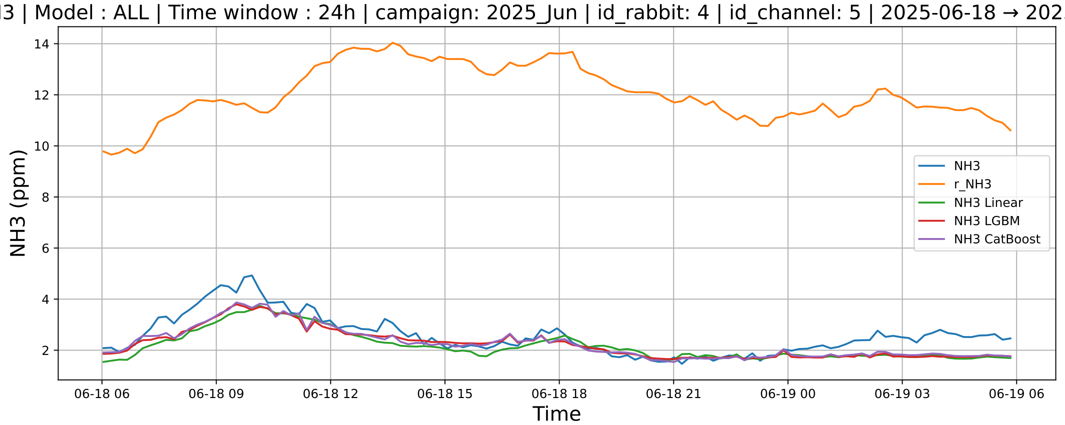
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



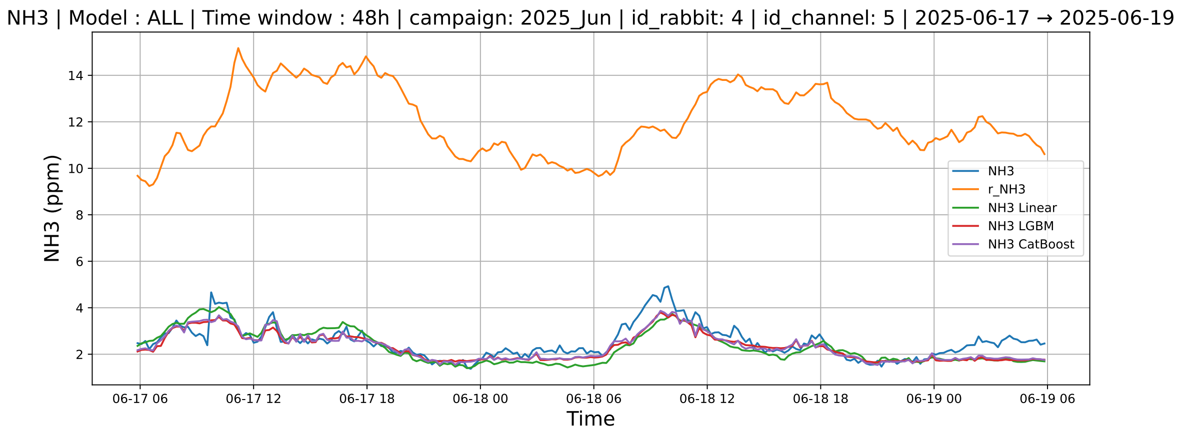
4.2.3 Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5

4.2.3.1 Time window : 24

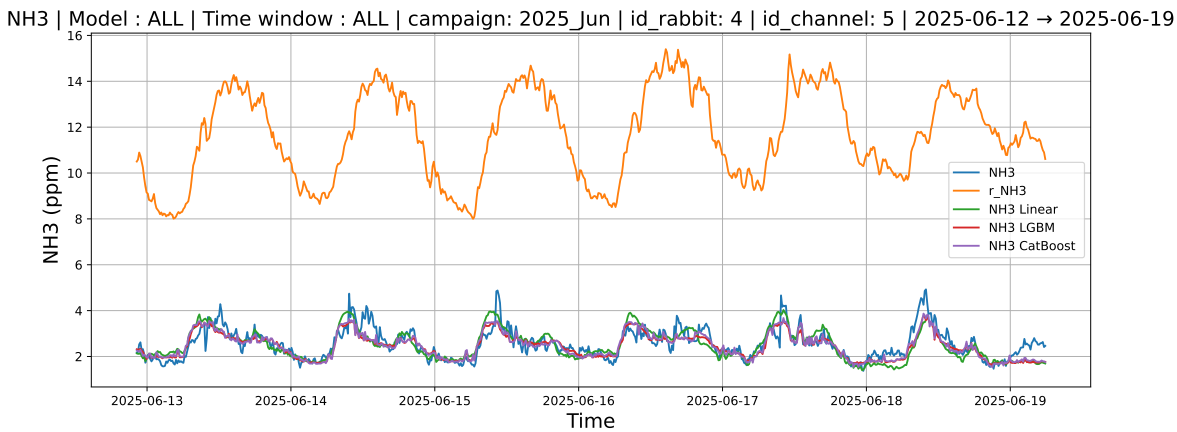
NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19



4.2.3.2 Time window : 48

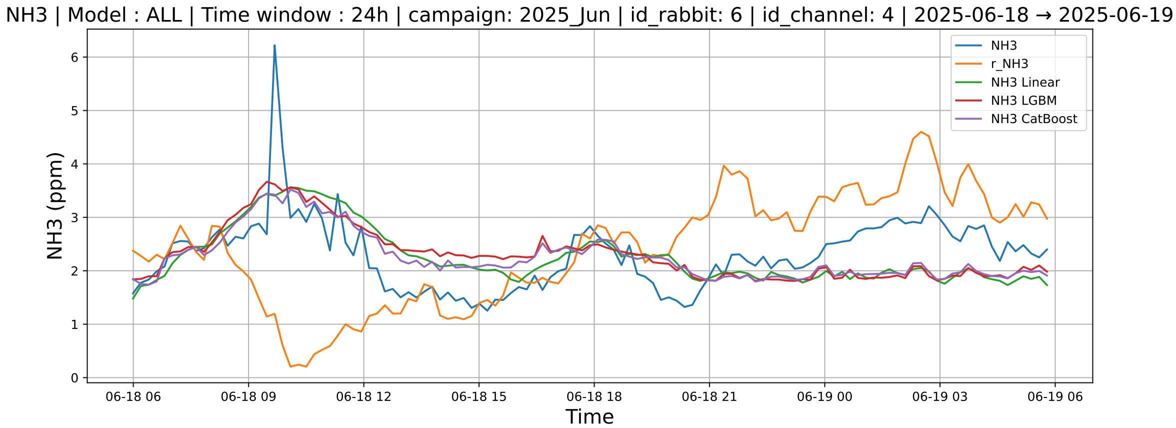


4.2.3.3 Time window : ALL

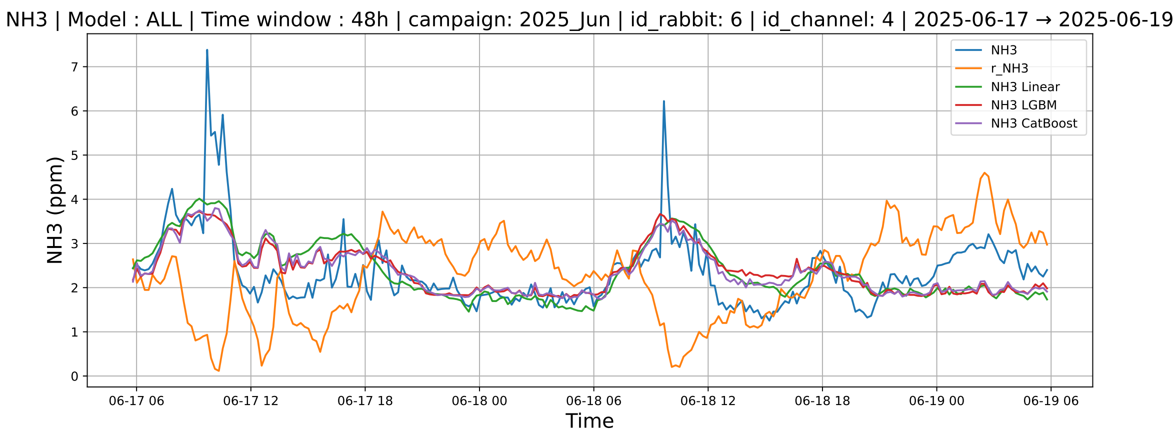


4.2.4 Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4

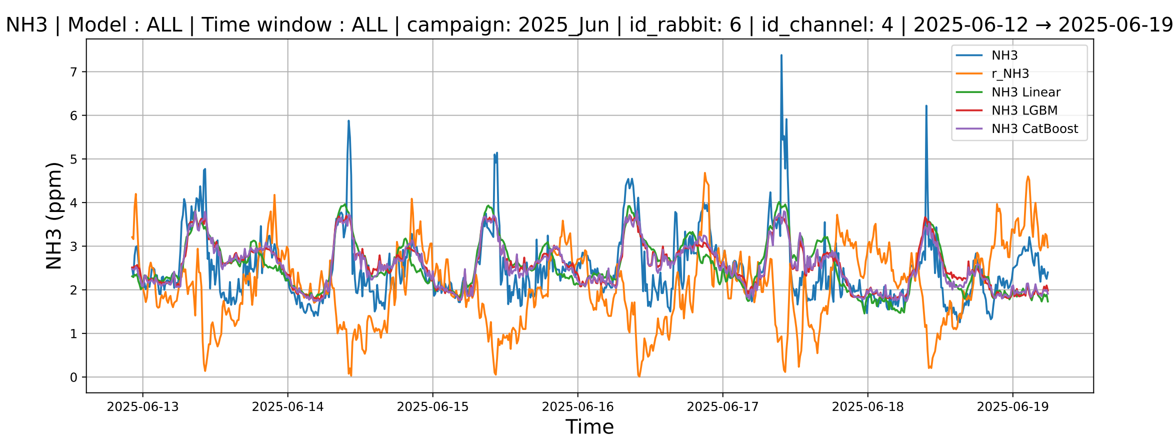
4.2.4.1 Time window : 24



4.2.4.2 Time window : 48



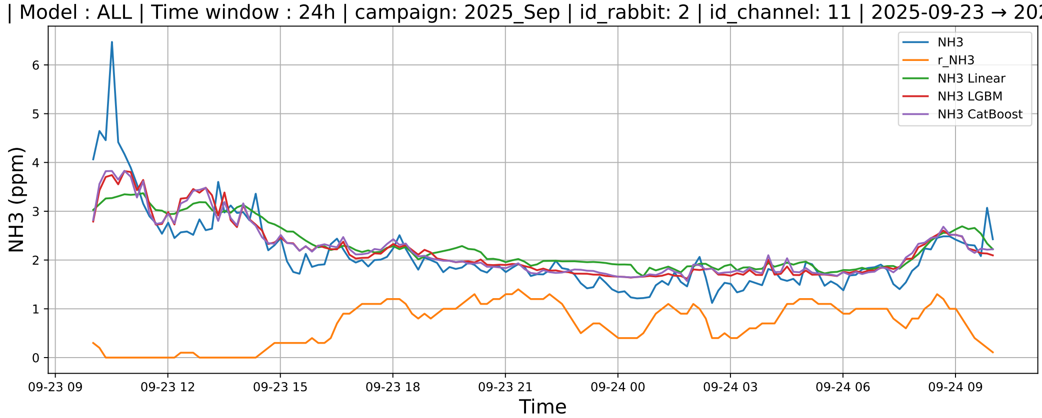
4.2.4.3 Time window : ALL



4.2.5 Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11

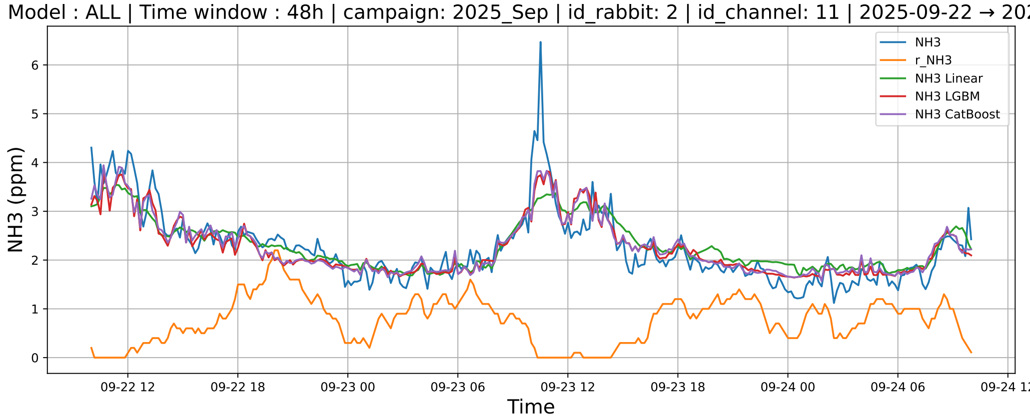
4.2.5.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24

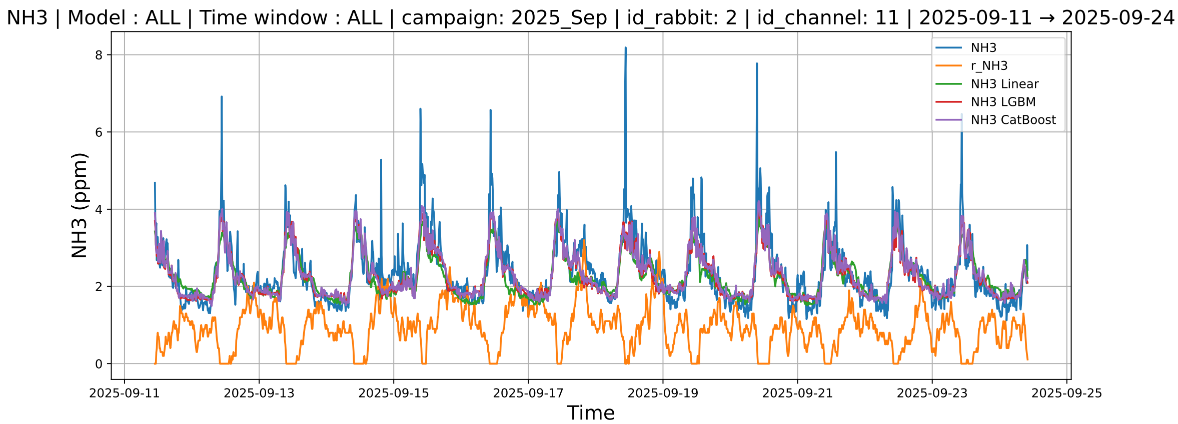


4.2.5.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24

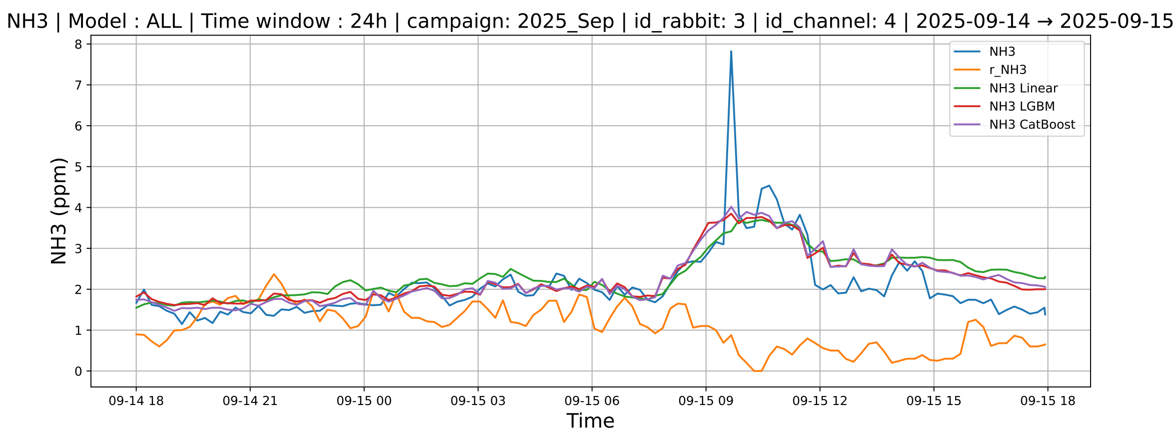


4.2.5.3 Time window : ALL

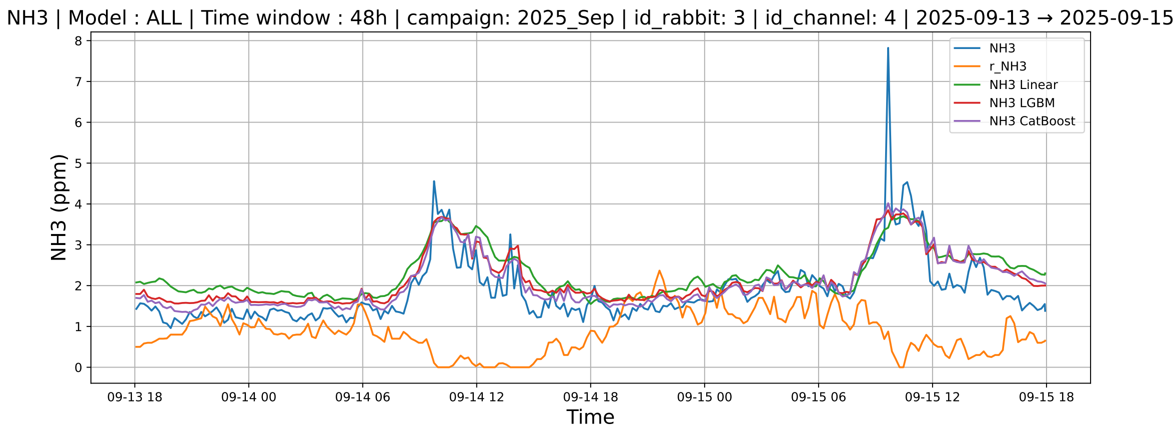


4.2.6 Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4

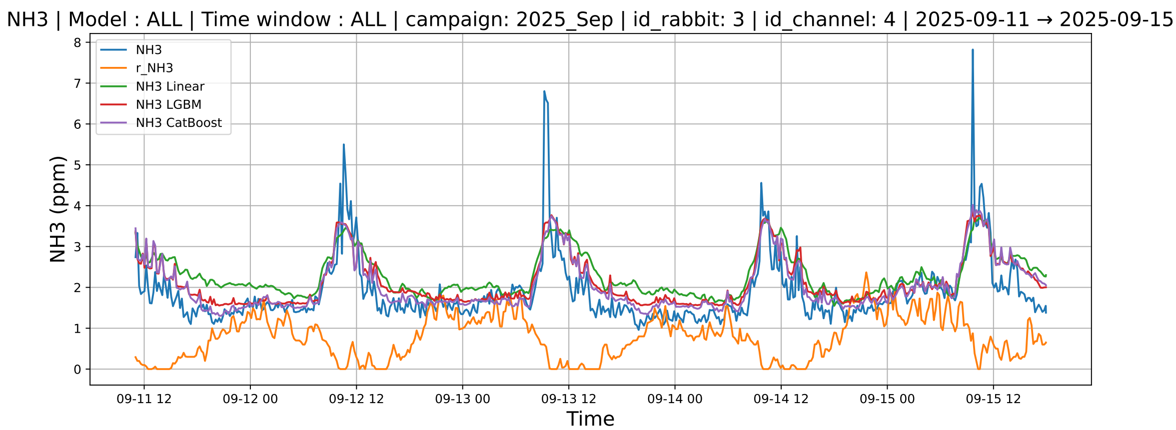
4.2.6.1 Time window : 24



4.2.6.2 Time window : 48

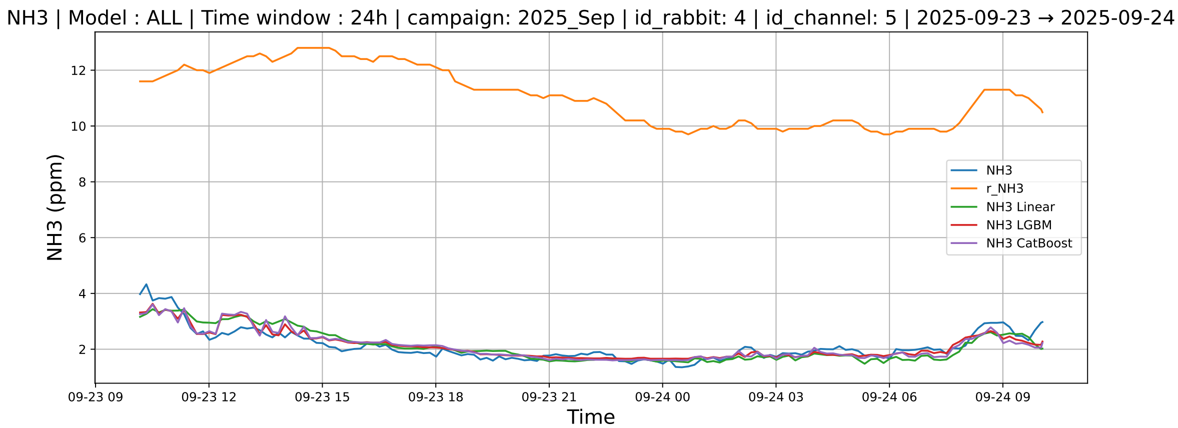


4.2.6.3 Time window : ALL

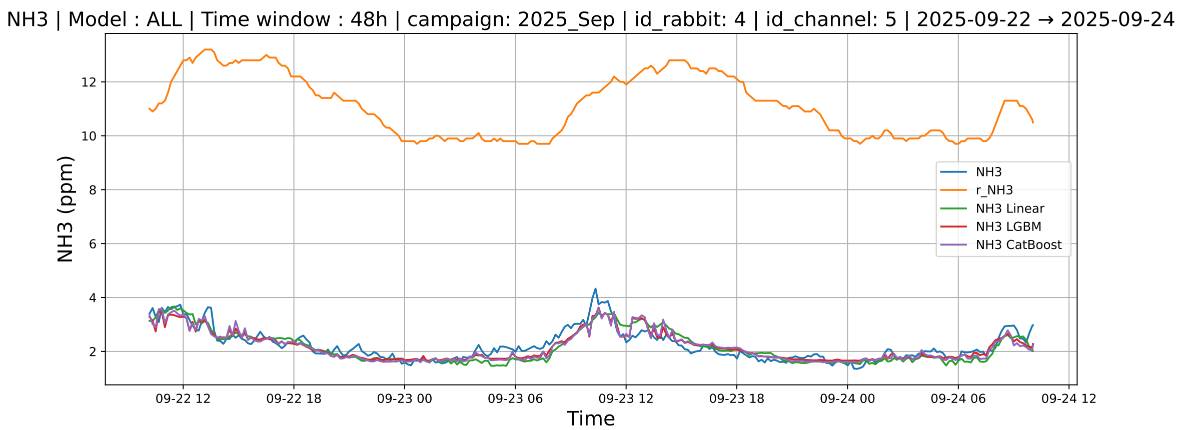


4.2.7 Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5

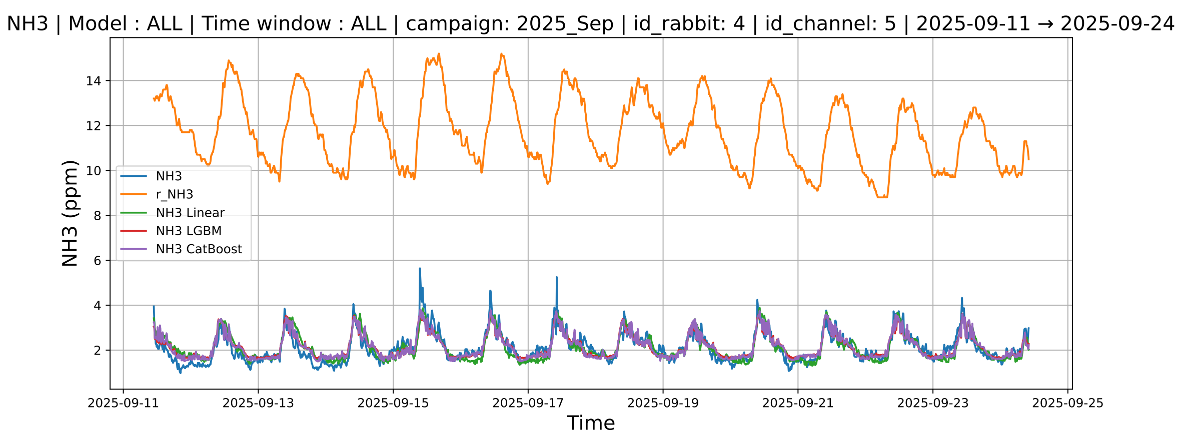
4.2.7.1 Time window : 24



4.2.7.2 Time window : 48



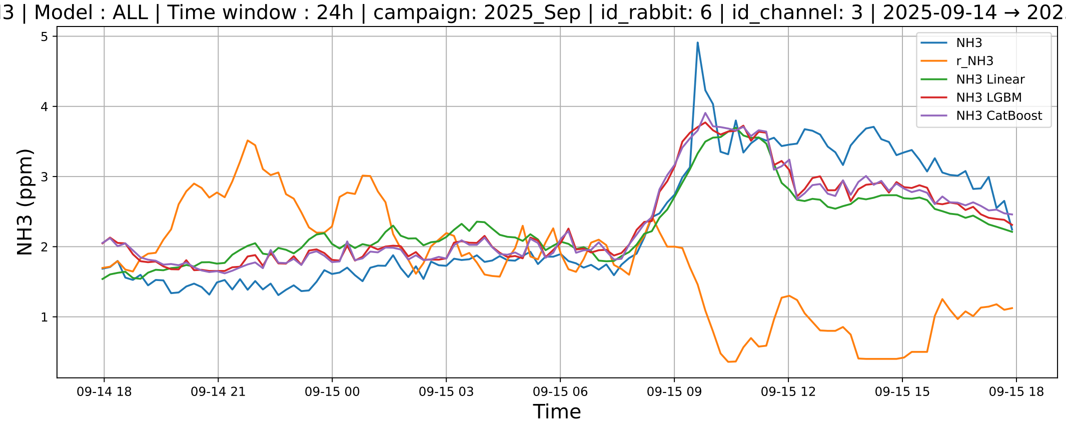
4.2.7.3 Time window : ALL



4.2.8 Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3

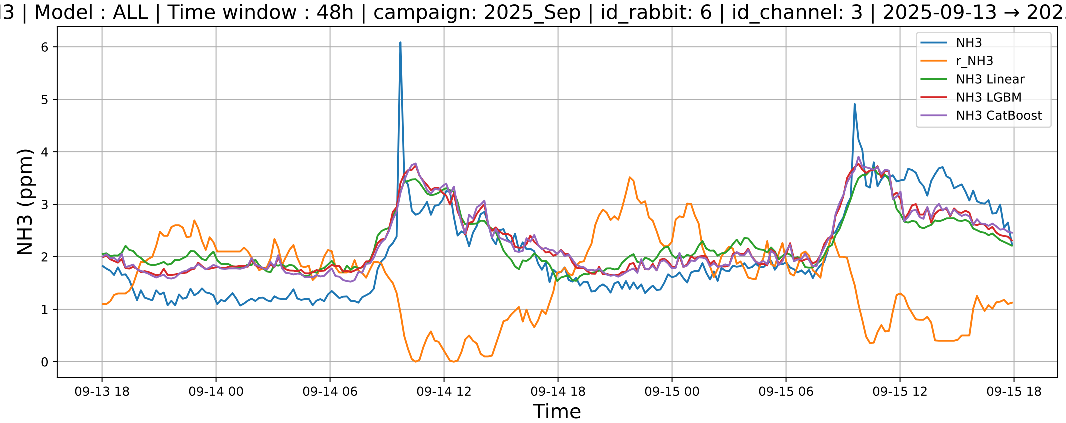
4.2.8.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15

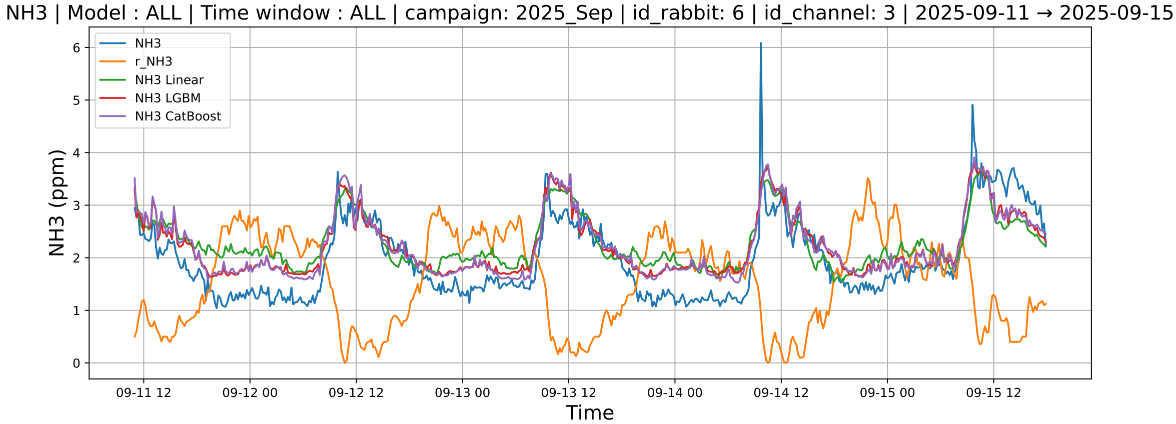


4.2.8.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15



4.2.8.3 Time window : ALL


NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15
NH3
r_NH3
NH3 Linear
NH3 LGBM
NH3 CatBoost
NH3 (ppm)
Time
09-11 12
09-12 00
09-12 12
09-13 00
09-13 12
09-14 00
09-14 12
09-15 00
09-15 12
0
1
2
3
4
5
6